00:11:56.000 --> 00:11:59.000
Okay. We will starting…

00:11:59.000 --> 00:12:29.000
Five minutes.

00:17:48.000 --> 00:17:54.000
Okay.

00:17:54.000 --> 00:17:55.000
remote participants here.

00:17:55.000 --> 00:18:24.000
vulnerable participant to you.

00:18:24.000 --> 00:18:33.000
Oh, uh.

00:18:33.000 --> 00:18:35.000
Hello? Hi. Okay.

00:18:35.000 --> 00:18:39.000
Yes, let's start.

00:18:39.000 --> 00:18:43.000
can remote participants hear me?

00:18:43.000 --> 00:18:45.000
if you can hear me.

00:18:45.000 --> 00:18:50.000
Prese them up.

00:18:50.000 --> 00:18:53.000
Okay.

00:18:53.000 --> 00:18:58.000
All right.

00:18:58.000 --> 00:19:07.000
Let's start it.

00:19:07.000 --> 00:19:13.000
I hope more people are coming in later, but yeah, let's start the opening session

00:19:13.000 --> 00:19:16.000
Okay, so…

00:19:16.000 --> 00:19:20.000
Thank you for coming.

00:19:20.000 --> 00:19:23.000
Okay.

00:19:23.000 --> 00:19:29.000
Thank you very much for coming to Layer 2 meetup as that's San Francisco

00:19:29.000 --> 00:19:36.000
I'm Takaya working in an admin team of begin and also a research fellow at Georgetown University.

00:19:36.000 --> 00:19:41.000
So before starting the opening session, I would like to

00:19:41.000 --> 00:19:43.000
Thank you.

00:19:43.000 --> 00:19:48.000
the event sponsor, DGUS, for providing this fantastic place

00:19:48.000 --> 00:19:50.000
And this is our first time to have an event.

00:19:50.000 --> 00:19:55.000
In San Francisco. So it is a pleasure to find

00:19:55.000 --> 00:19:58.000
such a fantastic partner here.

00:19:58.000 --> 00:20:01.000
Okay, so we have…

00:20:01.000 --> 00:20:06.000
50 minutes, 15 minutes for the opening session

00:20:06.000 --> 00:20:12.000
Where I would like to give you all a brief introduction of what bigging is.

00:20:12.000 --> 00:20:16.000
And so have you ever had a

00:20:16.000 --> 00:20:19.000
I think you have already heard of

00:20:19.000 --> 00:20:22.000
you know what it what begin is but

00:20:22.000 --> 00:20:28.000
begin blockchain governance initiative network was founded in 2020

00:20:28.000 --> 00:20:30.000
Before then, like…

00:20:30.000 --> 00:20:36.000
G20 summit was held in Tokyo, Japan that year.

00:20:36.000 --> 00:20:38.000
And there was…

00:20:38.000 --> 00:20:39.000
Round?

00:20:39.000 --> 00:20:44.000
roundtable where Adam Beck, which is a Bitcoin core developer

00:20:44.000 --> 00:20:47.000
who is here now.

00:20:47.000 --> 00:20:59.000
class not the chair of the Financial Stability Board of the time at the time, and Jim Rye, so-called the Japanese father of the internet, joined to propose the necessity of

00:20:59.000 --> 00:21:02.000
multi-stakeholder discussion.

00:21:02.000 --> 00:21:06.000
or sound governance, sound blockchain governance

00:21:06.000 --> 00:21:09.000
So based on that discussion

00:21:09.000 --> 00:21:12.000
begin was…

00:21:12.000 --> 00:21:22.000
found it as a nonprofit by Dr. Matsuo with other initial contributors, including regulators, business operators

00:21:22.000 --> 00:21:26.000
on developers, including Bitcoin core developers

00:21:26.000 --> 00:21:30.000
and key persons from academia.

00:21:30.000 --> 00:21:38.000
So I will go back to the previous slide. The mission of begin is to provide a place where

00:21:38.000 --> 00:21:42.000
different stakeholders discuss blockchain governance challenges

00:21:42.000 --> 00:21:46.000
And in the end, to create standards

00:21:46.000 --> 00:21:49.000
And best practices in technical

00:21:49.000 --> 00:21:53.000
operational and the regulatory spheres.

00:21:53.000 --> 00:22:01.000
So we already had 10 conferences in a variety of regions in the world for the last four years.

00:22:01.000 --> 00:22:05.000
The last one happened in Tokyo this March.

00:22:05.000 --> 00:22:11.000
And the next one will be in Washington, D.C. next week on October 21st and 22nd.

00:22:11.000 --> 00:22:17.000
So please feel free to register from the QR code here if you haven't.

00:22:17.000 --> 00:22:23.000
I will wait for like 10 seconds now

00:22:23.000 --> 00:22:26.000
So we're going to talk about a bunch of

00:22:26.000 --> 00:22:30.000
topics in washington dc which will be like on

00:22:30.000 --> 00:22:35.000
Well, two days before right before the DC FinTech Week.

00:22:35.000 --> 00:22:39.000
Okay, I will move on.

00:22:39.000 --> 00:22:41.000
So…

00:22:41.000 --> 00:22:44.000
It's challenging for

00:22:44.000 --> 00:22:53.000
different stakeholders to gather in the same place since we often have conflict with many other blockchain related conferences

00:22:53.000 --> 00:22:58.000
So when we had the last conference, it actually conflicted with Isa Demba, which was

00:22:58.000 --> 00:23:03.000
a little bit unfortunate. But like this, there was um

00:23:03.000 --> 00:23:16.000
This was a challenge we had. So we started having small meetup events co-located with other blockchain events, including the ones organized by the Bitcoin community.

00:23:16.000 --> 00:23:25.000
the SREM community and identity, internet identity workshop, and ISO TC307, etc.

00:23:25.000 --> 00:23:30.000
But last month, we co-located our layer two meetup event

00:23:30.000 --> 00:23:38.000
With Bitcoin Tokyo 2024, where we had a great conversation with Bitcoin developers about

00:23:38.000 --> 00:23:42.000
Regulatory uncertainty for Lightning Network.

00:23:42.000 --> 00:23:46.000
wallet governance, cybersecurity information sharing.

00:23:46.000 --> 00:23:48.000
And now we are in ESF.

00:23:48.000 --> 00:23:57.000
in co-locating with ESA Global and pragama to have a conversation with people in the East RM community.

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So…

00:23:59.000 --> 00:24:04.000
The reason why we call our small meetup event, Layer 2 meetup.

00:24:04.000 --> 00:24:06.000
is that we

00:24:06.000 --> 00:24:10.000
would like to roll up discussion points made in these events

00:24:10.000 --> 00:24:14.000
to our main blockchain block conference

00:24:14.000 --> 00:24:20.000
And it's Matapha and it's not like we will discuss only layer two

00:24:20.000 --> 00:24:23.000
So our purpose for these conferences is

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an event, it's not just to have a conversation with like

00:24:28.000 --> 00:24:32.000
different communities, but to create documents based on the discussion and

00:24:32.000 --> 00:24:40.000
innovate them to standards and best practices in the blockchain industry.

00:24:40.000 --> 00:24:45.000
Okay, this is a list of the documents we created so far.

00:24:45.000 --> 00:24:50.000
What are we going to publish a bunch of other documents tc next year

00:24:50.000 --> 00:24:55.000
And I would like to make sure that

00:24:55.000 --> 00:25:01.000
all of the session all over the session will be on the Chatham house rule which means that

00:25:01.000 --> 00:25:05.000
all of their remarks in remarks in the session will be anonymized.

00:25:05.000 --> 00:25:12.000
In our meeting reports and the recordings will be not will not be disclosed

00:25:12.000 --> 00:25:14.000
to the public so

00:25:14.000 --> 00:25:16.000
you're not allowed to

00:25:16.000 --> 00:25:18.000
associate remarks made here.

00:25:18.000 --> 00:25:21.000
with any specific person later.

00:25:21.000 --> 00:25:25.000
Okay.

00:25:25.000 --> 00:25:28.000
So this is end of the opening

00:25:28.000 --> 00:25:30.000
and if you

00:25:30.000 --> 00:25:33.000
Anyone has any question

00:25:33.000 --> 00:25:36.000
So far, dude.

00:25:36.000 --> 00:25:39.000
Raise your hands like

00:25:39.000 --> 00:25:42.000
Feel free to speak up, but…

00:25:42.000 --> 00:25:45.000
Okay.

00:25:45.000 --> 00:25:48.000
If there isn't any other question

00:25:48.000 --> 00:25:51.000
I think we can move on to the

00:25:51.000 --> 00:25:58.000
First session, Layer 2 governance.

00:25:58.000 --> 00:26:04.000
Okay.

00:26:04.000 --> 00:26:06.000
I also wanted to make sure that

00:26:06.000 --> 00:26:08.000
Again.

00:26:08.000 --> 00:26:11.000
This isn't Chad Hammerhauser rule.

00:26:11.000 --> 00:26:13.000
under the Chatham house rule so

00:26:13.000 --> 00:26:21.000
We are recording this session now, but it's not going to be public, disclosed publicly.

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And we're going to use it only for the purpose of like reading, creating a meeting request.

00:26:27.000 --> 00:26:32.000
Okay.

00:26:32.000 --> 00:26:38.000
So let's move on to the first session.

00:26:38.000 --> 00:26:41.000
This is…

00:26:41.000 --> 00:26:47.000
to ask a opinion on the Instagram community about governance challenges

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that L2s, that layer twos have

00:26:50.000 --> 00:26:53.000
and discuss solutions being developed.

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So I'm not an expert

00:26:57.000 --> 00:26:58.000
on East RM.

00:26:58.000 --> 00:27:01.000
While trying to catch up with the latest trends.

00:27:01.000 --> 00:27:05.000
But it would be fantastic if we could learn from you

00:27:05.000 --> 00:27:08.000
you all about the latest trends.

00:27:08.000 --> 00:27:11.000
solutions discussed in the community

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And we will create a document of what we discuss.

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here and we can later make it easier for regulators and different stakeholders

00:27:20.000 --> 00:27:23.000
and circulate the document to them.

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And we also have the same session in washington dc

00:27:28.000 --> 00:27:35.000
where there is expected to be many regulators so we can have a conversation about this with regulators as well.

00:27:35.000 --> 00:27:39.000
And so this is an introduction page

00:27:39.000 --> 00:27:44.000
And layer two is now, you know, being developed

00:27:44.000 --> 00:27:49.000
as a way to address the scalability and cost efficiency challenges.

00:27:49.000 --> 00:27:52.000
But we need to…

00:27:52.000 --> 00:27:54.000
face the reality that

00:27:54.000 --> 00:27:57.000
You know, there is a trade-off

00:27:57.000 --> 00:27:59.000
um they

00:27:59.000 --> 00:28:02.000
I use support for achieving higher scalability.

00:28:02.000 --> 00:28:07.000
But it's sometimes introduced new security and governance challenges.

00:28:07.000 --> 00:28:09.000
And…

00:28:09.000 --> 00:28:17.000
From regulator perspectives, there isn't any research that objectively analyzes the trust points created by layer twos

00:28:17.000 --> 00:28:20.000
And I think this is going to be a

00:28:20.000 --> 00:28:29.000

you know serving as a valuable resource for business and also these regulatory authorities to understand the technical limitations

00:28:29.000 --> 00:28:32.000
And what are the current

00:28:32.000 --> 00:28:36.000
solutions that are being discussed in the username community.

00:28:36.000 --> 00:28:41.000
And I'm not saying that regulatory intervention is necessary

00:28:41.000 --> 00:28:43.000
But we have to

00:28:43.000 --> 00:28:48.000
regulators also regulators and other like business operators need to

00:28:48.000 --> 00:28:51.000
have a good understanding of, you know.

00:28:51.000 --> 00:28:55.000
the challenges, the L2s currently have

00:28:55.000 --> 00:28:58.000
And maybe we can consider the regulations.

00:28:58.000 --> 00:29:04.000
or if there is any appropriate technical solutions currently underdeveloped.

00:29:04.000 --> 00:29:13.000
under development, I think this could be also a solutions for these challenges. But I think I believe that it is going to become a topic of buzzer

00:29:13.000 --> 00:29:16.000
like future discussion.

00:29:16.000 --> 00:29:18.000
And…

00:29:18.000 --> 00:29:23.000
Yes. So we're going to create a document where

00:29:23.000 --> 00:29:30.000
we can aware we identify the governance challenges and security challenges of l2s

00:29:30.000 --> 00:29:35.000
And the scaling solution, the solutions to solve these problems

00:29:35.000 --> 00:29:39.000
In that document we

00:29:39.000 --> 00:29:43.000
publish at the beginning. So this is the purpose of this this

00:29:43.000 --> 00:29:46.000
session and um

00:29:46.000 --> 00:29:50.000
You know, we wanted to talk about this because you know we

00:29:50.000 --> 00:29:54.000
believe that it's important to have a conversation with the Syrian community first.

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And later, we can…

00:29:56.000 --> 00:30:00.000
talk about the same thing with regulators in Washington, DC so

00:30:00.000 --> 00:30:10.000
Yes, this is the whole purpose of this session.

00:30:10.000 --> 00:30:17.000
Okay.

00:30:17.000 --> 00:30:22.000
So this is a very…

00:30:22.000 --> 00:30:25.000
concise um

00:30:25.000 --> 00:30:28.000
Slide, but…

00:30:28.000 --> 00:30:30.000
When it comes to discussing L2U,

00:30:30.000 --> 00:30:34.000
I feel like the definition is often ambiguous.

00:30:34.000 --> 00:30:37.000
People talk about L2 blockchains such as like base

00:30:37.000 --> 00:30:40.000
optimism like recently like uniswap

00:30:40.000 --> 00:30:42.000
launched new

00:30:42.000 --> 00:30:45.000
L2, like uni chain or something

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But I hear people talk about like roll up body doom

00:30:50.000 --> 00:30:53.000
like plasma and etc which are not

00:30:53.000 --> 00:30:57.000
like L2, it's more like the operation between L2 and L1.

00:30:57.000 --> 00:31:01.000
And we discussed this in a working group.

00:31:01.000 --> 00:31:05.000
And the first question we could ask when you hear

00:31:05.000 --> 00:31:09.000
L2 is whether it is blockchain based or not, I think.

00:31:09.000 --> 00:31:13.000
So one of the good examples of no

00:31:13.000 --> 00:31:15.000
is state channel.

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So state channel is

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So the lightning network

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is a great example of state channel.

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Pick is not a blockchain, but it's working as a L2. It's like working as a scalability solution.

00:31:28.000 --> 00:31:30.000
For the Bitcoin.

00:31:30.000 --> 00:31:32.000
So…

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Yeah, like Lightning Network is huge in a bit bitcoin ecosystem and I'm not much familiar with state channel or project on Ethereum, but, you know, state channel is not the blockchain based l2

00:31:45.000 --> 00:31:50.000
But if the answer to that question is yes, yeah, it is blockchain based.

00:31:50.000 --> 00:31:54.000
there are different types of blockchain-based L2s of course

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But it's often a difference in the operations between L1 and L2.

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So the L2,

00:32:02.000 --> 00:32:04.000
itself is just a blockchain

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But the difference between

00:32:07.000 --> 00:32:10.000
difference across these l2s

00:32:10.000 --> 00:32:14.000
is the difference

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in the operation between like L1 and L2.

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So how they interoperate with

00:32:21.000 --> 00:32:23.000
L1. Okay.

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So…

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Assuming…

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we can achieve better scalability by making transactional L2 rather than L1.

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we have different solutions for blockchain-based L2s to ensure that it doesn't

00:32:44.000 --> 00:32:51.000
harm to scale the security of L2 series of operation between L1 and L2.

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So these are plasma, roll-up, volidium, and et cetera.

00:32:57.000 --> 00:32:59.000
But the term plasma

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could be used to revert to the chains themselves, but it could also refer to the operations among

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these chains and O1 so it's not

00:33:07.000 --> 00:33:10.000
it's a bit more ambiguous term so

00:33:10.000 --> 00:33:14.000
it's important to be careful about using this term.

00:33:14.000 --> 00:33:16.000
But anyway, um.

00:33:16.000 --> 00:33:19.000
So I talked about the definition of l2

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So I want to also talk about

00:33:21.000 --> 00:33:23.000
the definition of governance.

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Okay, so this is leading to the recap of big game block 10.

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which happened last

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next this March in Tokyo, Japan.

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So, yeah.

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was a session about L2 governance in

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Tokyo as well. And all the speakers of this session is Justin Drake, who is a researcher

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in the Easter Yaman community.

00:33:51.000 --> 00:34:02.000
And he gave us the definition of governance. He said the definition of a governance system is all about changing the rules of the game.

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So we can change

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the rules or L2s either through a governance multi-sync

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Which is a set of like semi-trusted parties

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or governance tokens

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for more decentralized version of multisix.

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In governance

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can be sometimes seen as a magic tool

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to solve any problem, to fix bags, to add features.

00:34:29.000 --> 00:34:32.000
To add incentives and et cetera.

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But we have to face the reality that it also comes with one major

00:34:38.000 --> 00:34:42.000
downside. The governance can be an attacker vector.

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So if you are interested in longevity

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building systems that are lost

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for like decades and centuries for example

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or if you are

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like interested in maximizing

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a credible neutrality of the system

00:35:01.000 --> 00:35:07.000
that could be a bad thing. The governance introducing new governance is a bad thing.

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So, but…

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From…

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like other perspectives, I think like the ethereum sometimes like

00:35:18.000 --> 00:35:21.000
believe that governance is always

00:35:21.000 --> 00:35:24.000
like a magic tool to solve any problem.

00:35:24.000 --> 00:35:27.000
So we have to face the reality.

00:35:27.000 --> 00:35:31.000
that it could sometimes like

00:35:31.000 --> 00:35:36.000
Another major downside to attack vector.

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So…

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So the whole point of L2s, Layer 2s, is to try and scale l1

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and detain the vast majority of the security properties of layer one.

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So this is an important statement.

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that Justin Drake made in the last conference.

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So he also

00:36:04.000 --> 00:36:08.000
use this slide to talk about, you know.

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different kinds of L2s.

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And L2 is a piece of infrastructure with three layers

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data availability layer.

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sequencing layer

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which chooses the order of transactions

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And also on an execution layer, which defines a virtual machine and execution environment

00:36:32.000 --> 00:36:36.000
So it might be misleading to use the term leia

00:36:36.000 --> 00:36:45.000
But the key probably wanted to mention that there are three things to consider for L2s, like data availability.

00:36:45.000 --> 00:36:47.000
sequencing and execution.

00:36:47.000 --> 00:36:51.000
So I would like to

00:36:51.000 --> 00:36:55.000
talk about talk about each by eight each by each

00:36:55.000 --> 00:36:59.000
And for execution with

00:36:59.000 --> 00:37:01.000
Custom execution

00:37:01.000 --> 00:37:03.000
the layer two

00:37:03.000 --> 00:37:08.000
has its own execution environment which operates under different rules

00:37:08.000 --> 00:37:11.000
and mechanisms than ECRM's EBM.

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So this can include

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Using a unique virtual machine

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or a specific way

00:37:19.000 --> 00:37:23.000
of executing smart contracts.

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And if SARM undergoes a change, for example.

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The L2 also needs to adjust its execution environment accordingly.

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which can increase governance burdens on the l2

00:37:37.000 --> 00:37:40.000
So in negative

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execution on the right

00:37:42.000 --> 00:37:50.000
The L2 relies directly on ECRM's EVM and runs transaction in the same way.

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And as a result, if Ethereum's rule changes, for example, the L2 automatically aligns with those changes without requiring adjustments.

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This allows the L2 to use ESRM's functionalities as internal resources.

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But on the other hand…

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If you use custom execution

00:38:16.000 --> 00:38:19.000
it means that you have like your own

00:38:19.000 --> 00:38:26.000
EVM. So that means like if, you know, the rules of CRM changes, you have to also adjust accordingly

00:38:26.000 --> 00:38:29.000
By noticing

00:38:29.000 --> 00:38:35.000
changes that are made in the E3M and then have to update the you know

00:38:35.000 --> 00:38:37.000
environment accordingly.

00:38:37.000 --> 00:38:42.000
So it's kind of a little bit burden for the L2 governance.

00:38:42.000 --> 00:38:48.000
So I would like to go to data next.

00:38:48.000 --> 00:38:52.000
So if you are consuming

00:38:52.000 --> 00:38:56.000
External data availability, which is different to Isrium.

00:38:56.000 --> 00:38:58.000
there is a possibility

00:38:58.000 --> 00:39:03.000
that data availability layer will break down right

00:39:03.000 --> 00:39:06.000

But you need to invoke governance.

00:39:06.000 --> 00:39:10.000
to choose the new data availability layer.

00:39:10.000 --> 00:39:12.000
If you choose to opt into a roll wrap.

00:39:12.000 --> 00:39:15.000
On the other hand.

00:39:15.000 --> 00:39:18.000
you are considering security assumptions and

00:39:18.000 --> 00:39:25.000
You don't have to use data availability layer and all of the transaction data will be

00:39:25.000 --> 00:39:28.000
on chain on ethereum.

00:39:28.000 --> 00:39:33.000
So there are the differences. If you go to the left end

00:39:33.000 --> 00:39:35.000
you could use body doom you're going to use

00:39:35.000 --> 00:39:42.000
external data availability layer, which means that you have to store

00:39:42.000 --> 00:39:46.000
you're going to store all of the transaction data on l2

00:39:46.000 --> 00:39:50.000
externally but if you use row app

00:39:50.000 --> 00:39:53.000
all of the transaction data that

00:39:53.000 --> 00:39:55.000
on L2 will be

00:39:55.000 --> 00:39:58.000
stored on L1, on the CRM.

00:39:58.000 --> 00:40:03.000
So this is a big difference.

00:40:03.000 --> 00:40:10.000

So in terms of sequencing, so most L2 scaling solutions, most L2 solutions

00:40:10.000 --> 00:40:12.000
use centralized

00:40:12.000 --> 00:40:17.000
sequences responsible for determining the order of transactions.

00:40:17.000 --> 00:40:20.000
So however, this idea suggests

00:40:20.000 --> 00:40:26.000
L2s could potentially use sequencing services from L1 itself.

00:40:26.000 --> 00:40:28.000
So in other words.

00:40:28.000 --> 00:40:32.000
The proposals on Ethereum's mainnet

00:40:32.000 --> 00:40:36.000
who sequence transactions at l1

00:40:36.000 --> 00:40:39.000
could also be used for L2 sequencing.

00:40:39.000 --> 00:40:41.000
So Wyom is a must.

00:40:41.000 --> 00:40:45.000
benefits of this approach is that it would enable

00:40:45.000 --> 00:40:50.000
synchronous composability between different l2s

00:40:50.000 --> 00:40:53.000
In country, like East RM.

00:40:53.000 --> 00:40:55.000
faces fragmentation.

00:40:55.000 --> 00:40:59.000
Where each L2 functions as a

00:40:59.000 --> 00:41:03.000
independent silo leading to fragmented

00:41:03.000 --> 00:41:07.000

liquidity applications and user bases.

00:41:07.000 --> 00:41:10.000
But with this new sequencing mechanism

00:41:10.000 --> 00:41:12.000
A2s could collaborate.

00:41:12.000 --> 00:41:20.000
more smoothly reducing fragmentation and creating a more unified ecosystem. So ESRM has us

00:41:20.000 --> 00:41:22.000
So Ethereum is currently

00:41:22.000 --> 00:41:28.000
creating a super chain. I think this is also relevant to this discussion.

00:41:28.000 --> 00:41:35.000
But anyway, these three are the kind of like categories of l2s

00:41:35.000 --> 00:41:40.000
that was which was suggested by Justin Drake in a CRM.

00:41:40.000 --> 00:41:42.000
Foundation.

00:41:42.000 --> 00:41:46.000
Okay.

00:41:46.000 --> 00:41:50.000
Any questions about this?

00:41:50.000 --> 00:41:52.000
Okay.

00:41:52.000 --> 00:41:57.000
So what I started doing, working on

00:41:57.000 --> 00:42:00.000
based on discussion is that based on

00:42:00.000 --> 00:42:05.000
I would have liked, I wanted to classify

00:42:05.000 --> 00:42:09.000
You know, L2s and L1, L2 operations

00:42:09.000 --> 00:42:13.000
So, you know, I talked about a state channel

00:42:13.000 --> 00:42:16.000
And there are also

00:42:16.000 --> 00:42:18.000
L2L1 operations called

00:42:18.000 --> 00:42:22.000
Plasma, optimistic, TK Roarab.

00:42:22.000 --> 00:42:26.000
Bali doom and…

00:42:26.000 --> 00:42:28.000
They are…

00:42:28.000 --> 00:42:33.000
used in a different way. And there are a lot of like use cases and

00:42:33.000 --> 00:42:37.000
in each operation.

00:42:37.000 --> 00:42:40.000
is tailored for like specific

00:42:40.000 --> 00:42:42.000
like use use cases

00:42:42.000 --> 00:42:46.000
And also the way of

00:42:46.000 --> 00:42:48.000
storing data is different.

00:42:48.000 --> 00:42:52.000
across these operation methods.

00:42:52.000 --> 00:42:55.000
in sequences.

00:42:55.000 --> 00:42:58.000
So, for example.

00:42:58.000 --> 00:43:00.000
It's say…

00:43:00.000 --> 00:43:03.000
If you use roll-ups.

00:43:03.000 --> 00:43:05.000
an optimistic roller up ticker roll-ups.

00:43:05.000 --> 00:43:08.000
all of the data

00:43:08.000 --> 00:43:11.000
transaction data on l2

00:43:11.000 --> 00:43:13.000
will be stored in

00:43:13.000 --> 00:43:15.000
L1, basically.

00:43:15.000 --> 00:43:18.000
There are a lot of some exceptions.

00:43:18.000 --> 00:43:28.000
But if you use Belladium, you're going to use layer data availability, data availability layers, which means that you're going to use external data

00:43:28.000 --> 00:43:34.000
like data

00:43:34.000 --> 00:43:36.000
each platform

00:43:36.000 --> 00:43:41.000
So it means that the transaction on L2 will be stored off chain.

00:43:41.000 --> 00:43:45.000
Not on Ethereum itself.

00:43:45.000 --> 00:43:48.000
So also…

00:43:48.000 --> 00:43:51.000
I don't think I should talk about stage channel.

00:43:51.000 --> 00:43:54.000
this time but also

00:43:54.000 --> 00:43:56.000
Plasma is a little bit

00:43:56.000 --> 00:44:00.000
obsolete to compare with like Europe and the value so

00:44:00.000 --> 00:44:04.000
I don't know if we should talk about the details of like basma

00:44:04.000 --> 00:44:08.000
But yeah, in terms of data storage

00:44:08.000 --> 00:44:13.000
it also stores all of the L2 transaction data off chain

00:44:13.000 --> 00:44:15.000
And…

00:44:15.000 --> 00:44:17.000
Yes.

00:44:17.000 --> 00:44:21.000
And in terms of sequencing.

00:44:21.000 --> 00:44:23.000
There should be some people

00:44:23.000 --> 00:44:25.000
called sequencers

00:44:25.000 --> 00:44:28.000
who has the responsibility of

00:44:28.000 --> 00:44:32.000
storing

00:44:32.000 --> 00:44:38.000
data, transaction data on L2s to L1, or like say sometimes like

00:44:38.000 --> 00:44:41.000
work on the

00:44:41.000 --> 00:44:46.000
you know security solutions

00:44:46.000 --> 00:44:50.000
For example, like if you use Optimistic roll-up

00:44:50.000 --> 00:44:58.000
you sequencer is supposed to

00:44:58.000 --> 00:45:02.000
put crowd blue. No, no, no. Sorry.

00:45:02.000 --> 00:45:04.000
If you use ZKR app.

00:45:04.000 --> 00:45:10.000
Sequence is responsible for

00:45:10.000 --> 00:45:12.000
putting quality proof.

00:45:12.000 --> 00:45:14.000
to the L1.

00:45:14.000 --> 00:45:17.000
And I think this is…

00:45:17.000 --> 00:45:21.000
are responsible for and if they

00:45:21.000 --> 00:45:23.000
make some mistakes

00:45:23.000 --> 00:45:27.000
for on these orgs, they will be slashed

00:45:27.000 --> 00:45:29.000
Because like they basically deposit

00:45:29.000 --> 00:45:31.000
um fans

00:45:31.000 --> 00:45:34.000
or yeah so

00:45:34.000 --> 00:45:40.000
And yeah, there are also like security solutions are different across these

00:45:40.000 --> 00:45:42.000
L2, L1 operations.

00:45:42.000 --> 00:45:47.000
And there are also security challenges in each

00:45:47.000 --> 00:45:49.000
operation methods

00:45:49.000 --> 00:45:54.000
And yeah, I would like to talk about the details of this but

00:45:54.000 --> 00:45:58.000
Is everybody familiar with

00:45:58.000 --> 00:46:00.000
these how these

00:46:00.000 --> 00:46:03.000
like operation works like

00:46:03.000 --> 00:46:05.000
optimistic plasma

00:46:05.000 --> 00:46:07.000
um

00:46:07.000 --> 00:46:09.000
optimistic roll up, ticky roll-up.

00:46:09.000 --> 00:46:11.000
volume, etc.

00:46:11.000 --> 00:46:15.000
I don't know, like, how about the remote participants if

00:46:15.000 --> 00:46:26.000
Are you all familiar with this and

00:46:26.000 --> 00:46:34.000
So I created these slides because I think it is a good education contents for regulators and business operators.

00:46:34.000 --> 00:46:39.000
But I think it's…

00:46:39.000 --> 00:46:45.000
it could be very busy, easy for like ECRM developers, I guess. So I don't know if i should

00:46:45.000 --> 00:46:58.000
talk about this.

00:46:58.000 --> 00:47:03.000
Okay.

00:47:03.000 --> 00:47:05.000

So…

00:47:05.000 --> 00:47:12.000
If there isn't any question, I would like to go to

00:47:12.000 --> 00:47:42.000
the data storage challenges

00:47:56.000 --> 00:48:00.000
store transaction data of L2 that happens or l2

00:48:00.000 --> 00:48:03.000
And the one of the…

00:48:03.000 --> 00:48:05.000
solutions, it's cold data

00:48:05.000 --> 00:48:08.000
So…

00:48:08.000 --> 00:48:12.000
cold data refers to

00:48:12.000 --> 00:48:24.000
Yeah, cold data is the most cost efficient method it was said it was said that, you know, it was most cost efficient method for

00:48:24.000 --> 00:48:27.000
storing data on the L1 blockchain

00:48:27.000 --> 00:48:30.000
And…

00:48:30.000 --> 00:48:33.000
So all of the data, transaction data on l2

00:48:33.000 --> 00:48:39.000
will be stored as cool data to l1

00:48:39.000 --> 00:48:42.000
by…

00:48:42.000 --> 00:48:50.000
Because it is necessary like the storing transaction data on L1 is necessary because everyone needs to see

00:48:50.000 --> 00:48:52.000
all of the transaction data

00:48:52.000 --> 00:48:54.000
on L2 eventually

00:48:54.000 --> 00:48:57.000
Because…

00:48:57.000 --> 00:49:00.000
Someone needs to submit fraud proofs.

00:49:00.000 --> 00:49:04.000
by checking the transaction data

00:49:04.000 --> 00:49:06.000
all of the transaction data on l2

00:49:06.000 --> 00:49:08.000
on l1

00:49:08.000 --> 00:49:15.000
And if you see some wrong transactions data on l1

00:49:15.000 --> 00:49:21.000
So you can see the right transaction on l1

00:49:21.000 --> 00:49:25.000
And if there is someone who wants to exit

00:49:25.000 --> 00:49:30.000
And who wants to make a transaction from the smart contract

00:49:30.000 --> 00:49:34.000
that has his fans

00:49:34.000 --> 00:49:36.000
to he's

00:49:36.000 --> 00:49:39.000
address only one

00:49:39.000 --> 00:49:43.000
Okay.

00:49:43.000 --> 00:49:46.000
Wait.

00:49:46.000 --> 00:49:49.000
I think.

00:49:49.000 --> 00:50:19.000
Wait, wait, wait, wait. Let me check.

00:50:53.000 --> 00:50:56.000
Sorry, I'm back.

00:50:56.000 --> 00:51:05.000
Can you all hear me?

00:51:05.000 --> 00:51:07.000
Okay, okay.

00:51:07.000 --> 00:51:12.000
So I explained a little bit about like cold data so

00:51:12.000 --> 00:51:16.000
Cold data is a method to

00:51:16.000 --> 00:51:20.000
But all of the transaction data on l1

00:51:20.000 --> 00:51:23.000
And if you store

00:51:23.000 --> 00:51:27.000
the transaction data as cold data on

00:51:27.000 --> 00:51:29.000
L1, the ESRM.

00:51:29.000 --> 00:51:31.000
it's very cost efficient.

00:51:31.000 --> 00:51:35.000
So that's why they used cold data

00:51:35.000 --> 00:51:37.000
as a data format.

00:51:37.000 --> 00:51:41.000
of to to store

00:51:41.000 --> 00:51:46.000
all of this transaction on L1, L2.

00:51:46.000 --> 00:51:49.000
And, um.

00:51:49.000 --> 00:51:52.000
Yes. But…

00:51:52.000 --> 00:51:58.000
They started creating the Etrium foundation eastern

00:51:58.000 --> 00:52:04.000
He started creating new methods to data to store data off chain

00:52:04.000 --> 00:52:08.000
So a new data format called bulb bulb

00:52:08.000 --> 00:52:13.000
It was introduced by EIP45.

00:52:13.000 --> 00:52:17.000
844 in um

00:52:17.000 --> 00:52:24.000
bulbs are not stored directly on chain and they are transmitted and stored off chain.

00:52:24.000 --> 00:52:26.000
But…

00:52:26.000 --> 00:52:30.000
Yeah, they are increasingly

00:52:30.000 --> 00:52:34.000
in optimistic roll-ups where it is necessary to publish layer two

00:52:34.000 --> 00:52:38.000
transaction data to enable the submission of route pollution.

00:52:38.000 --> 00:52:41.000
and yeah

00:52:41.000 --> 00:52:46.000
um

00:52:46.000 --> 00:52:50.000
So…

00:52:50.000 --> 00:52:53.000
It's very similar to cold data, but…

00:52:53.000 --> 00:52:56.000
they allow it allows…

00:52:56.000 --> 00:53:01.000
use us to store all of the transaction data on

00:53:01.000 --> 00:53:07.000
off-chain instead of like on-chain on E3M. So it is more cost of efficiency

00:53:07.000 --> 00:53:12.000
cost coefficient. So that's why they started using this instead of um

00:53:12.000 --> 00:53:18.000
cold data and, um.

00:53:18.000 --> 00:53:20.000
But…

00:53:20.000 --> 00:53:29.000
The Polydium uses data availability to be the layoff, which is an external layout to only just to store

00:53:29.000 --> 00:53:34.000
data of transaction data, data of transaction on l2

00:53:34.000 --> 00:53:40.000
And so if you use data availability data layer

00:53:40.000 --> 00:53:46.000
the data will not be stored on on chain, on ESRM.

00:53:46.000 --> 00:53:47.000
So all of the transaction data should be

00:53:47.000 --> 00:53:52.000
and lay all the data availability layer

00:53:52.000 --> 00:53:54.000
And in order to

00:53:54.000 --> 00:54:03.000
ensure that all of the transaction data on data availability layer is not wrong is correct

00:54:03.000 --> 00:54:06.000
And there should be some people who

00:54:06.000 --> 00:54:11.000
monitor the data availability day layer

00:54:11.000 --> 00:54:15.000

And the data availability committee is used to secure

00:54:15.000 --> 00:54:22.000
to ensure that the data on data availability layer is not

00:54:22.000 --> 00:54:25.000
is correct.

00:54:25.000 --> 00:54:28.000
Okay, so…

00:54:28.000 --> 00:54:33.000
But in order to ensure the security of data availability layer.

00:54:33.000 --> 00:54:37.000
there should be other governance mechanism.

00:54:37.000 --> 00:54:40.000
So, um.

00:54:40.000 --> 00:54:48.000
Several people, like multiple people needs to be in the needs to be monitoring the

00:54:48.000 --> 00:54:55.000
security of data availability layer, which means that sometimes like they issue governance tokens to

00:54:55.000 --> 00:55:02.000
and create a security mechanism only for the data availability layer.

00:55:02.000 --> 00:55:04.000
And…

00:55:04.000 --> 00:55:09.000
But if they introduce new governance systems like this.

00:55:09.000 --> 00:55:14.000
It's sometimes possible that there could be some like collusion

00:55:14.000 --> 00:55:20.000
among the operators of data availability layer.

00:55:20.000 --> 00:55:23.000
And you have to

00:55:23.000 --> 00:55:26.000
be careful about

00:55:26.000 --> 00:55:29.000
how to establish security.

00:55:29.000 --> 00:55:35.000
mechanisms of the data availability data. So this is

00:55:35.000 --> 00:55:40.000
One of the good examples of like

00:55:40.000 --> 00:55:42.000
new governance.

00:55:42.000 --> 00:55:47.000
of the security challenges that are introduced by

00:55:47.000 --> 00:55:56.000
new governance system, new governance mechanism

00:55:56.000 --> 00:55:58.000
so also

00:55:58.000 --> 00:56:01.000
There are challenges as well.

00:56:01.000 --> 00:56:05.000
So there should be sequences

00:56:05.000 --> 00:56:10.000
who is responsible for

00:56:10.000 --> 00:56:15.000
storing all of the transaction data on l1

00:56:15.000 --> 00:56:19.000
transaction data on L2s to l1

00:56:19.000 --> 00:56:22.000
But, you know.

00:56:22.000 --> 00:56:26.000
it's some it's possible that they…

00:56:26.000 --> 00:56:29.000
submit wrong transaction data on l1

00:56:29.000 --> 00:56:32.000
And it's possible that they

00:56:32.000 --> 00:56:36.000
you know i intend to

00:56:36.000 --> 00:56:44.000
cast or wrong transaction data only once so that you know they can get they can make some kind of like profit out of it

00:56:44.000 --> 00:56:52.000
So it's possible that there are some like collusions among sequences as well

00:56:52.000 --> 00:56:55.000
So, um.

00:56:55.000 --> 00:57:02.000
Yeah, if it's a sequences are too few, the number of sequences are too few

00:57:02.000 --> 00:57:07.000
it will lead to the insufficient decentralization and

00:57:07.000 --> 00:57:12.000
Also, there could be higher possibility of collusions among

00:57:12.000 --> 00:57:24.000
sequences. So this is same problem across all of the L1, L2 operations like state channel plasma, optimistic or opt and volume.

00:57:24.000 --> 00:57:28.000
And also they use a lot of like smart contracts to

00:57:28.000 --> 00:57:32.000
realize the operation of L1 and L2, L2 interoperability.

00:57:32.000 --> 00:57:38.000
So if there are any like vulnerabilities in smart contracts they use

00:57:38.000 --> 00:57:45.000
On Ethereum, this is definitely going to be attacker vector.

00:57:45.000 --> 00:57:47.000
Yes.

00:57:47.000 --> 00:57:50.000
Good.

00:57:50.000 --> 00:57:53.000
security mechanism

00:57:53.000 --> 00:57:55.000

of each operation

00:57:55.000 --> 00:58:01.000
they issue tokens, governance tokens, and say

00:58:01.000 --> 00:58:05.000
ask the sequences to ask

00:58:05.000 --> 00:58:07.000
deposit funds

00:58:07.000 --> 00:58:10.000
saw that they could be slashed deep sea

00:58:10.000 --> 00:58:13.000
make something bad. They do something bad.

00:58:13.000 --> 00:58:17.000
But we have to ensure that

00:58:17.000 --> 00:58:21.000
The proof of commitment outweighs collusion of commitment.

00:58:21.000 --> 00:58:25.000
And this kind of calculation is going to be very complicated.

00:58:25.000 --> 00:58:30.000
And we have to make sure that the equation is

00:58:30.000 --> 00:58:32.000
Always.

00:58:32.000 --> 00:58:35.000
satisfied so

00:58:35.000 --> 00:58:41.000
But it's very complicated operation. It's going to be a very complicated operation that I'm not really sure.

00:58:41.000 --> 00:58:44.000
If all of the operations

00:58:44.000 --> 00:58:47.000
make sure that these equations

00:58:47.000 --> 00:58:51.000
are satisfied all of the time.

00:58:51.000 --> 00:58:55.000
So this is, I think, one of the challenges level twos have.

00:58:55.000 --> 00:58:57.000
As of now.

00:58:57.000 --> 00:59:00.000
So yeah, this is

00:59:00.000 --> 00:59:03.000
the overview of sequencing challenges, and I talked about

00:59:03.000 --> 00:59:07.000
the data challenges as well in here

00:59:07.000 --> 00:59:10.000
Like…

00:59:10.000 --> 00:59:17.000
Yeah, as I mentioned, data availability, if you use data availability layer and if you use any methods to

00:59:17.000 --> 00:59:21.000
store data off chain instead of on-chain

00:59:21.000 --> 00:59:28.000
there should be some other governance problems.

00:59:28.000 --> 00:59:32.000
So yeah, so…

00:59:32.000 --> 00:59:34.000
about this, I would like to ask

00:59:34.000 --> 00:59:36.000
all of the audience

00:59:36.000 --> 00:59:39.000
many questions like

00:59:39.000 --> 00:59:45.000
What are the current solutions that are being developed in an ethereum community

00:59:45.000 --> 00:59:47.000
to…

00:59:47.000 --> 00:59:50.000
you know, what are the solutions

00:59:50.000 --> 00:59:55.000
to like address the problems of sequencing

00:59:55.000 --> 00:59:58.000
the governance sequencing and data storage

00:59:58.000 --> 01:00:00.000
In the Ethereum community

01:00:00.000 --> 01:00:04.000
So this is a question.

01:00:04.000 --> 01:00:06.000
that I wanted to ask and like

01:00:06.000 --> 01:00:10.000
So if there isn't any good solutions of now probably

01:00:10.000 --> 01:00:14.000
There should be some regulatory intervention but which is

01:00:14.000 --> 01:00:16.000
Definitely not.

01:00:16.000 --> 01:00:21.000
what the Ethereum community do not want, does not want so

01:00:21.000 --> 01:00:23.000
So…

01:00:23.000 --> 01:00:25.000
I think…

01:00:25.000 --> 01:00:28.000
We would like to discuss

01:00:28.000 --> 01:00:32.000
Okay.

01:00:32.000 --> 01:00:35.000
And I think we can create a document

01:00:35.000 --> 01:00:38.000
that

01:00:38.000 --> 01:00:41.000
where there is a list of the solutions

01:00:41.000 --> 01:00:45.000
And yes, um.

01:00:45.000 --> 01:00:47.000
I can bring this

01:00:47.000 --> 01:00:52.000
documented to the, you know, conference we're going to have in Washington, D.C.

01:00:52.000 --> 01:00:58.000
So, okay.

01:00:58.000 --> 01:01:02.000
So let me start with a pretty

01:01:02.000 --> 01:01:04.000
Basic question.

01:01:04.000 --> 01:01:08.000
So when we discuss about lightning

01:01:08.000 --> 01:01:10.000
things in Tokyo.

01:01:10.000 --> 01:01:12.000
a few weeks ago.

01:01:12.000 --> 01:01:14.000
One of the discussion points either

01:01:14.000 --> 01:01:17.000
if that you know if that

01:01:17.000 --> 01:01:19.000
Lightening a node.

01:01:19.000 --> 01:01:21.000
could be that target double rate.

01:01:21.000 --> 01:01:23.000
for the money transmission.

01:01:23.000 --> 01:01:26.000
So when, you know, before…

01:01:26.000 --> 01:01:29.000
doing some comparison among these five types of

01:01:29.000 --> 01:01:38.000
Thank you.

01:01:38.000 --> 01:01:41.000
Barbara's two record off-chain data.

01:01:41.000 --> 01:01:44.000
could be the target of regret.

01:01:44.000 --> 01:01:47.000
From a negative point of view for the many

01:01:47.000 --> 01:01:50.000
so different many

01:01:50.000 --> 01:01:52.000
different types of transaction over ethereum.

01:01:52.000 --> 01:02:00.000
This is my basic question.

01:02:00.000 --> 01:02:05.000
I think I can partly answer that question like

01:02:05.000 --> 01:02:08.000
So…

01:02:08.000 --> 01:02:10.000
you know.

01:02:10.000 --> 01:02:14.000
I think we should, what do we need to start from

01:02:14.000 --> 01:02:17.000
is I think the question I…

01:02:17.000 --> 01:02:19.000
Probably need to ask first is

01:02:19.000 --> 01:02:21.000
like what are the risks involved with

01:02:21.000 --> 01:02:23.000
the, you know.

01:02:23.000 --> 01:02:27.000
regulatory risks.

01:02:27.000 --> 01:02:30.000
um

01:02:30.000 --> 01:02:36.000
involved with that are involved with like the different methods of like storing data

01:02:36.000 --> 01:02:40.000
So…

01:02:40.000 --> 01:02:44.000
I think it's important to start from talking about why

01:02:44.000 --> 01:02:46.000
it is better to

01:02:46.000 --> 01:02:48.000
store data.

01:02:48.000 --> 01:02:51.000
Off-chain.

01:02:51.000 --> 01:02:54.000
Because if you

01:02:54.000 --> 01:02:56.000
put all of the data.

01:02:56.000 --> 01:02:58.000
transaction data on l2s

01:02:58.000 --> 01:03:01.000
to L1, which means that it's

01:03:01.000 --> 01:03:05.000
going to be stored permanently, permanently.

01:03:05.000 --> 01:03:08.000
But…

01:03:08.000 --> 01:03:11.000
If you store

01:03:11.000 --> 01:03:14.000
transaction data on l2s

01:03:14.000 --> 01:03:19.000
to someone else, that means that you can't guarantee that

01:03:19.000 --> 01:03:21.000
it's going, it's wrong.

01:03:21.000 --> 01:03:26.000
Yeah, it's cracked, right? So somebody manipulate the data

01:03:26.000 --> 01:03:31.000
And like someone made the wrong transaction data on

01:03:31.000 --> 01:03:34.000
the on the place where

01:03:34.000 --> 01:03:37.000
the transaction data is supposed to be.

01:03:37.000 --> 01:03:40.000
right if they do that, if they do it.

01:03:40.000 --> 01:03:43.000
That means that

01:03:43.000 --> 01:03:53.000
When you want to exit the L1, L2 and want to make a real transaction on L1,

01:03:53.000 --> 01:03:55.000
you can…

01:03:55.000 --> 01:03:58.000
submit the wrong transaction on L1.

01:03:58.000 --> 01:04:01.000
So that is a risk.

01:04:01.000 --> 01:04:04.000
In terms of

01:04:04.000 --> 01:04:09.000
like the financial risk, I think. Financial risk.

01:04:09.000 --> 01:04:11.000
So…

01:04:11.000 --> 01:04:16.000
So the question, going back to the question about like if they should regulate

01:04:16.000 --> 01:04:18.000
the data storage

01:04:18.000 --> 01:04:21.000
project or like

01:04:21.000 --> 01:04:24.000
should be regulated or not i think

01:04:24.000 --> 01:04:29.000
if this is, if they

01:04:29.000 --> 01:04:32.000
are the cause of

01:04:32.000 --> 01:04:34.000
manipulating

01:04:34.000 --> 01:04:41.000
something and

01:04:41.000 --> 01:04:43.000
Allowing…

01:04:43.000 --> 01:04:47.000
people to make wrong transaction that

01:04:47.000 --> 01:04:50.000
harm other people

01:04:50.000 --> 01:04:56.000
I think it's going to be the discussion of like customer protection

01:04:56.000 --> 01:04:58.000
And…

01:04:58.000 --> 01:05:03.000
Yeah, I think this is where regulators should also think about. But I don't know if it's

01:05:03.000 --> 01:05:07.000
A good answer to that question the professor made.

01:05:07.000 --> 01:05:21.000
And…

01:05:21.000 --> 01:05:25.000
I'm not the expert on that real estate.

01:05:25.000 --> 01:05:31.000
So when we began beginning the activities.

01:05:31.000 --> 01:05:37.000
So I ran that from, you know, regulator guides about that three

01:05:37.000 --> 01:05:39.000
basic categories of

01:05:39.000 --> 01:05:43.000
why the regret and is regulated cryptocurrencies?

01:05:43.000 --> 01:05:46.000
It's a general thing for, you know.

01:05:46.000 --> 01:05:51.000
for the decentralized finance and also for traditional finance

01:05:51.000 --> 01:05:53.000
Yeah, three reasons. First.

01:05:53.000 --> 01:05:56.000
is financial stability.

01:05:56.000 --> 01:06:00.000
The second one is consumer protection or investor protection

01:06:00.000 --> 01:06:03.000
The third one is preventing financial crime.

01:06:03.000 --> 01:06:09.000
this is a country so they related to KYC, CFT or something like that.

01:06:09.000 --> 01:06:12.000
So there are three different

01:06:12.000 --> 01:06:16.000
reasons why we need to regulate something.

01:06:16.000 --> 01:06:20.000
So, you know, for consumer protection or investor protection

01:06:20.000 --> 01:06:23.000
you know, losing, losing…

01:06:23.000 --> 01:06:27.000
losing off-schain data from their

01:06:27.000 --> 01:06:30.000
you know that bookchain computers

01:06:30.000 --> 01:06:34.000
So that might be some.

01:06:34.000 --> 01:06:36.000
some you know reasonable

01:06:36.000 --> 01:06:42.000
That's how, you know, losing money from some users of the cryptocurrency happened

01:06:42.000 --> 01:06:47.000
So at that moment, the chain

01:06:47.000 --> 01:06:52.000
storage it could be that custody or something like that custody of assets or something like that.

01:06:52.000 --> 01:06:57.000
Anyway, that is our so

01:06:57.000 --> 01:07:01.000
That is the kind of that concern from the consumer protection point of view.

01:07:01.000 --> 01:07:11.000
Therefore, preliminary.

01:07:11.000 --> 01:07:15.000
You know, FBI do

01:07:15.000 --> 01:07:20.000
That's all.

01:07:20.000 --> 01:07:25.000
That could be that other concern from the FBI point of view

01:07:25.000 --> 01:07:29.000
So this is the reason why that's so

01:07:29.000 --> 01:07:32.000
I ask that question that so that off-chain

01:07:32.000 --> 01:07:35.000
computers octane spreadsheet

01:07:35.000 --> 01:07:40.000
could be the target of regret or something like that.

01:07:40.000 --> 01:07:44.000

And so I think before this begins work

01:07:44.000 --> 01:07:47.000
we need to rest up that

01:07:47.000 --> 01:07:51.000
So the target to regulate

01:07:51.000 --> 01:07:56.000
the discussion. So we need to restart that. So why Legreta need to care about

01:07:56.000 --> 01:07:59.000
that off-tain data then so

01:07:59.000 --> 01:08:02.000
If that sum of you know we have

01:08:02.000 --> 01:08:05.000
five typical

01:08:05.000 --> 01:08:07.000
you know protocols for their

01:08:07.000 --> 01:08:10.000
So if, you know.

01:08:10.000 --> 01:08:15.000
Scheme B is better than scheme A from that that

01:08:15.000 --> 01:08:20.000
Not to really get some point of view. That could be that good, you know.

01:08:20.000 --> 01:08:26.000
work for the comparing different types of data to solution.

01:08:26.000 --> 01:08:28.000
This is my opinion.

01:08:28.000 --> 01:08:31.000
Yeah, that is true.

01:08:31.000 --> 01:08:41.000
So we also had a conversation about the regulatory uncertainty for Lightning Network, which is one of the L2 scaling solutions for Bitcoin.

01:08:41.000 --> 01:08:44.000
And what we discussed was

01:08:44.000 --> 01:08:47.000
you know what what are the regulatory leaks

01:08:47.000 --> 01:08:49.000
involved with um

01:08:49.000 --> 01:08:54.000
blinding network. Lightning Network itself is a very complicated technical

01:08:54.000 --> 01:08:56.000
You know, protocol

01:08:56.000 --> 01:09:01.000
So most of the regulators probably do not have a good understanding of it, but we

01:09:01.000 --> 01:09:10.000
kind of like learning about it and, you know, made a list of like regulatory uncertainties and regulatory risks involved with Lightning Network.

01:09:10.000 --> 01:09:15.000
And we discussed it with Bitcoin core developers in Tokyo.

01:09:15.000 --> 01:09:18.000
And yeah, that is true so that is true

01:09:18.000 --> 01:09:21.000
I also stated some challenges

01:09:21.000 --> 01:09:25.000
The L2s have on Ethereum have

01:09:26.000 --> 01:09:29.000
But it's not great. It's not a good resource still for

01:09:29.000 --> 01:09:34.000
regulators because like it's not written as from the regulatory perspective so

01:09:34.000 --> 01:09:40.000
I think it's important to make a rest of regulatory risks involved with these

01:09:40.000 --> 01:09:43.000
L2s and

01:09:43.000 --> 01:09:52.000
I think it's important it's better to clearly state what are the challenges we should like regulators should look at.

01:09:52.000 --> 01:09:56.000

And, you know, what are the technical solutions

01:09:56.000 --> 01:09:58.000
the username community.

01:09:58.000 --> 01:10:03.000
is trying to, you know, create as of now.

01:10:03.000 --> 01:10:09.000
in yes um

01:10:09.000 --> 01:10:17.000
I think this is a future work of begin.

01:10:17.000 --> 01:10:21.000
Okay.

01:10:21.000 --> 01:10:23.000
So…

01:10:23.000 --> 01:10:27.000
I also wanted to ask this question. I don't know if, you know…

01:10:27.000 --> 01:10:31.000
How many, how many people are like, you know.

01:10:31.000 --> 01:10:35.000
much involved with the eastern community and discuss this matter

01:10:35.000 --> 01:10:37.000
but um

01:10:37.000 --> 01:10:41.000
I think it's important to have a good understanding of like how

01:10:41.000 --> 01:10:44.000
G's um

01:10:44.000 --> 01:10:49.000
L2 and operation between L1 and L2 work.

01:10:49.000 --> 01:10:55.000
And when we have the session about this in Washington, D.C,

01:10:55.000 --> 01:10:58.000
Should we talk about…

01:10:58.000 --> 01:11:01.000

Should we make an explanation about these things?

01:11:01.000 --> 01:11:06.000
in this session, because I think if i talk about

01:11:06.000 --> 01:11:10.000
this thing immediately probably people will not understand.

01:11:10.000 --> 01:11:13.000
this. So I think it's better to have a

01:11:13.000 --> 01:11:16.000
good like brief description of

01:11:16.000 --> 01:11:19.000
What is road ramp what is

01:11:19.000 --> 01:11:25.000
Polydium, what is plasma etc because

01:11:25.000 --> 01:11:30.000
I think it's…

01:11:30.000 --> 01:11:39.000
Because most of the participants in Washington, D.C. will be probably regulators and policymakers and business operators instead of developers so

01:11:39.000 --> 01:11:42.000
I think it's important to

01:11:42.000 --> 01:11:44.000
Have an excellent.

01:11:44.000 --> 01:11:48.000
to make an explanation about this first.

01:11:48.000 --> 01:11:52.000
Before talking about the details of like the challenges.

01:11:52.000 --> 01:11:56.000
they have.

01:11:56.000 --> 01:11:59.000
In…

01:11:59.000 --> 01:12:03.000
Also, I need to make sure that

01:12:03.000 --> 01:12:07.000
um…

01:12:07.000 --> 01:12:12.000
Yeah, so…

01:12:12.000 --> 01:12:15.000
the document will be

01:12:15.000 --> 01:12:19.000
easier to understand for

01:12:19.000 --> 01:12:23.000
regulators, I guess.

01:12:23.000 --> 01:12:27.000
Okay.

01:12:27.000 --> 01:12:29.000
Any question?

01:12:29.000 --> 01:12:31.000
about this?

01:12:31.000 --> 01:12:38.000
So far.

01:12:38.000 --> 01:12:42.000
I'm asking many questions hello about that.

01:12:42.000 --> 01:12:59.000
for this comparison

01:12:59.000 --> 01:13:25.000
not in this table.

01:13:25.000 --> 01:13:29.000
less efficiency efficient than

01:13:29.000 --> 01:13:31.000
Okay.

01:13:31.000 --> 01:13:37.000
IAP, EIP, 44, 48, 44 and like data availability layer

01:13:37.000 --> 01:13:39.000
But, you know, if you use cold data

01:13:39.000 --> 01:13:46.000
I think it's more secure, but if you use data availability layer, it's less secure.

01:13:46.000 --> 01:13:51.000
And yeah, I can also put this perspective as well on this table

01:13:51.000 --> 01:13:55.000
you know, um.

01:13:55.000 --> 01:14:18.000
Okay.

01:14:18.000 --> 01:14:20.000
I just wanted to make some comments.

01:14:20.000 --> 01:14:23.000
a presentation so

01:14:23.000 --> 01:14:28.000
I totally remember that there's some controversy that the uh

01:14:28.000 --> 01:14:37.000
Not operator of the lightning network could be regarded as a money transfer servicer.

01:14:37.000 --> 01:14:41.000
the in traditional finance deep

01:14:41.000 --> 01:14:44.000
base.

01:14:44.000 --> 01:14:47.000
They, I mean the money transfer service needs to be regulated

01:14:47.000 --> 01:14:49.000
is that the…

01:14:49.000 --> 01:14:52.000
there's some chance to embezzle the

01:14:52.000 --> 01:14:55.000
the customer's money

01:14:55.000 --> 01:15:00.000
So as you know, that they technically have the ability to do that

01:15:00.000 --> 01:15:03.000
So in that case that the uh

01:15:03.000 --> 01:15:08.000
The criteria, this scaling system or mechanism

01:15:08.000 --> 01:15:10.000
has to be, I mean there

01:15:10.000 --> 01:15:13.000
in the operator, I mean the author

01:15:13.000 --> 01:15:15.000
It should be called the uh

01:15:15.000 --> 01:15:21.000
sequencer. Sequencer needs to be regulated is that they are

01:15:21.000 --> 01:15:24.000
if days uh

01:15:24.000 --> 01:15:26.000
they have technically uh

01:15:26.000 --> 01:15:29.000
abilities to embezzle their

01:15:29.000 --> 01:15:34.000
Customers, the users crypto

01:15:34.000 --> 01:15:39.000
So I'm not familiar with the scaling system.

01:15:39.000 --> 01:15:45.000
mechanism, but if the sequencer has technically ability

01:15:45.000 --> 01:15:49.000
to embezzle the user's crypto the day has to be

01:15:49.000 --> 01:15:51.000
regulated. I think that's the

01:15:51.000 --> 01:16:03.000
criteria and the maybe, yeah, as you said that the uh we're going to have a meeting next week, but I don't think the regulator could understand all of them

01:16:03.000 --> 01:16:05.000
So maybe…

01:16:05.000 --> 01:16:09.000
But my point is that they are

01:16:09.000 --> 01:16:12.000
I'm not familiar with this, but I'm the uh

01:16:12.000 --> 01:16:17.000
I know that's the some of these has a very centralized sequencer

01:16:17.000 --> 01:16:19.000
But I don't think we

01:16:19.000 --> 01:16:23.000
we need to

01:16:23.000 --> 01:16:25.000
we need to refuse the

01:16:25.000 --> 01:16:29.000
all of these centralized sequencer system because that's it

01:16:29.000 --> 01:16:36.000
I do know that there is some very famous trilemma among the

01:16:36.000 --> 01:16:39.000
this system now they

01:16:39.000 --> 01:16:42.000
boxing so in that case

01:16:42.000 --> 01:16:47.000
This is about the efficiency and the versus the

01:16:47.000 --> 01:16:51.000
cost of the complying the regulation.

01:16:51.000 --> 01:16:53.000
So if the uh

01:16:53.000 --> 01:16:58.000
some of the very centralized, so-called centralized sequence

01:16:58.000 --> 01:17:01.000
sequencing system is very efficient.

01:17:01.000 --> 01:17:03.000
Even if there's some

01:17:03.000 --> 01:17:07.000
regulatory costs. So in that case, they have to be

01:17:07.000 --> 01:17:09.000
there has to be a choice.

01:17:09.000 --> 01:17:12.000
Yeah, to adapting the dutch system.

01:17:12.000 --> 01:17:17.000
So, but yeah, that's the point so

01:17:17.000 --> 01:17:23.000
Madam, but the problem is that the user, the from user perspective the

01:17:23.000 --> 01:17:26.000
they have the they don't have any ability

01:17:26.000 --> 01:17:29.000
to which sequencing system has a

01:17:29.000 --> 01:17:37.000
very centralized system. In that case, the user needs to make sure that the

01:17:37.000 --> 01:17:40.000
this system, this operator

01:17:40.000 --> 01:17:44.000
is taking uh is addressing their

01:17:44.000 --> 01:17:47.000
regulatory program or not so

01:17:47.000 --> 01:17:49.000
maybe there has to be some

01:17:49.000 --> 01:17:54.000
very clear criteria to jet safety it is.

01:17:54.000 --> 01:17:58.000
they are complying with them.

01:17:58.000 --> 01:18:00.000
So…

01:18:00.000 --> 01:18:08.000
I looked up the details of what each element

01:18:08.000 --> 01:18:14.000
Thank you.

01:18:14.000 --> 01:18:21.000
you can see the differences between like these L2s in terms of like cost efficiency, scalability

01:18:21.000 --> 01:18:23.000
And, you know, how…

01:18:23.000 --> 01:18:26.000
SDDs.

01:18:26.000 --> 01:18:28.000
And I think it's important

01:18:28.000 --> 01:18:38.000
just have this kind of like educational content, educational content as well but

01:18:38.000 --> 01:18:44.000
I'm not with like what you as a regulator as a country like discussing and

01:18:44.000 --> 01:18:50.000
Is anyone familiar with, like…

01:18:50.000 --> 01:18:55.000
whether this kind of thing is, you know, kind of like discussed in the

01:18:55.000 --> 01:19:00.000
Washington, D.C, like…

01:19:00.000 --> 01:19:07.000
So this is a question to the audience, I guess.

01:19:07.000 --> 01:19:09.000
Have you ever heard of this?

01:19:09.000 --> 01:19:22.000
I'm sorry, sorry.

01:19:22.000 --> 01:19:25.000
I'm not…

01:19:25.000 --> 01:19:51.000
I'm not entirely sure. I guess…

01:19:51.000 --> 01:19:53.000
So you're sort of just

01:19:53.000 --> 01:19:57.000
discussion of the different sort of categories of regulatory risk.

01:19:57.000 --> 01:20:00.000
I think the thing is to try to map that

01:20:00.000 --> 01:20:03.000
onto these characteristics. And so

01:20:03.000 --> 01:20:12.000
a walkthrough of the different types of L2s could be useful if it's presented in terms of those characteristics.

01:20:12.000 --> 01:20:16.000
So that it's a little easier for the regulator to understand and why do I, you know, what are the

01:20:16.000 --> 01:20:19.000
What are the choices between these different

01:20:19.000 --> 01:20:22.000
protocols is different approaches.

01:20:22.000 --> 01:20:27.000
And why would you be doing that from a commercial perspective, from a technological perspective?

01:20:27.000 --> 01:20:29.000
But what's the consequence?

01:20:29.000 --> 01:20:32.000
And if you start thinking about

01:20:32.000 --> 01:20:35.000
financial stability as a regulatory prerogative.

01:20:35.000 --> 01:20:44.000
what's going to drive that is you're concerned in terms of financial stability

01:20:44.000 --> 01:20:47.000
About everything up and down the stack. You care about

01:20:47.000 --> 01:20:55.000
speed because markets need to be able to move quickly, but you also are concerned about finality. You're concerned about

01:20:55.000 --> 01:20:58.000
Proof.

01:20:58.000 --> 01:21:03.000
At the consumer protection level.

01:21:03.000 --> 01:21:07.000
it's going to be similar, but a little bit different nuance.

01:21:07.000 --> 01:21:14.000
In terms of law enforcement, I think there's a lot of focus on

01:21:14.000 --> 01:21:16.000
money laundering.

01:21:16.000 --> 01:21:18.000
But I think it's important

01:21:18.000 --> 01:21:21.000
I'll say this as a um as a

01:21:21.000 --> 01:21:26.000
tax lawyer that I also what gets left out is tax evasion.

01:21:26.000 --> 01:21:30.000
Which is a very important part of financial crime as well.

01:21:30.000 --> 01:21:34.000
So there's a whole side topic around tax reporting

01:21:34.000 --> 01:21:36.000
And tax compliance.

01:21:36.000 --> 01:21:41.000
And that gets to issues around whether you have a centralized

01:21:41.000 --> 01:21:52.000
sequencer who the regulators are going to want to regulate that sequencer, not just from an AML compliance perspective, but also tax reporting.

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Right now, there are regulations in place

01:21:56.000 --> 01:21:59.000
to force

01:21:59.000 --> 01:22:01.000
centralized exchanges

01:22:01.000 --> 01:22:08.000
to rules will be going into effect at the end of this year in the US to

01:22:08.000 --> 01:22:13.000

For centralized exchanges to report to the IRS about transactions from a tax reporting perspective.

01:22:13.000 --> 01:22:19.000
But there's proposed rules that would apply that to creators of

01:22:19.000 --> 01:22:24.000
DeFi protocols. And so just software developers could be

01:22:24.000 --> 01:22:28.000
Subject to these same sorts of rules. So I think

01:22:28.000 --> 01:22:33.000
each of the different regulatory regimes has its own

01:22:33.000 --> 01:22:38.000
goal that it's trying to achieve. And when you were talking about the data layer.

01:22:38.000 --> 01:22:50.000
I think what's interesting there is what is it you care about the data? You were proposing if I have off-chain data, that could be a problem from a regulator's perspective.

01:22:50.000 --> 01:22:55.000
But is the regulator concerned about

01:22:55.000 --> 01:22:59.000
The provable content of that data or the mere absence of it.

01:22:59.000 --> 01:23:04.000
So, right. So, but those are two sort of separate issues, right? Because…

01:23:04.000 --> 01:23:20.000
It is the case that you could prove the validity of that off-chain data, but the real concern is what if the data simply disappears? And now the records that say tax authorities might want to have access to or law enforcement might want to have access to

01:23:20.000 --> 01:23:28.000
It's just gone. And it's no longer a question, oh, fine, I've got a hash that I've got recorded on an L1, so what if I don't know what that hashed?

01:23:28.000 --> 01:23:32.000
If I don't have some sort of those off-chain records.

01:23:32.000 --> 01:23:35.000
then that creates a problem. So from a regulatory perspective.

01:23:35.000 --> 01:23:41.000
you need some sort of persistent data store if that's the information that they want.

01:23:41.000 --> 01:23:43.000
So…

01:23:43.000 --> 01:23:49.000
I don't know, it's not an answer to your question, but that was sort of my reaction to what you've been talking about so far.

01:23:49.000 --> 01:23:51.000
So if you're talking about trying to present to

01:23:51.000 --> 01:23:55.000
regulator folks in dc

01:23:55.000 --> 01:23:58.000
on this, sort of focusing on

01:23:58.000 --> 01:24:01.000
what are the key differences between these different

01:24:01.000 --> 01:24:04.000
systems that

01:24:04.000 --> 01:24:08.000
have impacts or implications for

01:24:08.000 --> 01:24:12.000
the kinds of things that feed into regulatory regimes, whether it is speed

01:24:12.000 --> 01:24:18.000
validity, certainty, or just availability of the information.

01:24:18.000 --> 01:24:20.000
And…

01:24:20.000 --> 01:24:23.000
Boiling that down so that you can see, well.

01:24:23.000 --> 01:24:31.000
You know, I've got the data is completely on chain, so therefore you know it's going to be there versus it's off-chain and maybe you don't unless we build a system that

01:24:31.000 --> 01:24:36.000
persists that off-chain data.

01:24:36.000 --> 01:24:39.000
Anyway, some random thoughts.

01:24:39.000 --> 01:24:42.000
That needs a song for Spectator, but then I definitely…

01:24:42.000 --> 01:24:47.000
Yeah, I understand. I have to make a change to the, you know

01:24:47.000 --> 01:24:50.000
how to how how i can

01:24:50.000 --> 01:24:52.000
present this so

01:24:52.000 --> 01:24:54.000
Yeah, thank you, Hodis.

01:24:54.000 --> 01:24:57.000
Bringing this perspective. Okay.

01:24:57.000 --> 01:25:01.000
I have a comment. So yeah, question to you

01:25:01.000 --> 01:25:09.000
So you're doing a lot of job for that you know communicating you know the blockchain guys and you know government guys who

01:25:09.000 --> 01:25:13.000
It was something.

01:25:13.000 --> 01:25:15.000
You know, so…

01:25:15.000 --> 01:25:20.000
I don't know that detail about the United States, but in Japan, other countries

01:25:20.000 --> 01:25:28.000
The most current crypto regulation is concentrated on the layer one, you know, Bitcoin, Ethereum, something like that.

01:25:28.000 --> 01:25:34.000
Very few people care about Alea 2. What happened to the DeFi areas, Alea 2 protocols? What is the meaning of

01:25:34.000 --> 01:25:37.000
or implication to regulations.

01:25:37.000 --> 01:25:42.000
So it's very complicated for the government people or you know

01:25:42.000 --> 01:25:45.000
you know politicians to understand that each each

01:25:45.000 --> 01:25:48.000
time, I think, you know, what is the roll-up

01:25:48.000 --> 01:25:51.000
Well, something like that, so…

01:25:51.000 --> 01:25:53.000
it's not easy for

01:25:53.000 --> 01:25:57.000
you know to have to have command understanding for that.

01:25:57.000 --> 01:26:00.000
every once or every time to explain that.

01:26:00.000 --> 01:26:02.000
the detail about the layout protocols.

01:26:02.000 --> 01:26:07.000
So what we should do to solve that problem to

01:26:07.000 --> 01:26:10.000
having command understanding.

01:26:10.000 --> 01:26:14.000
Well, I think one way to think about that is

01:26:14.000 --> 01:26:19.000
think about presenting the information with the layer one as your baseline.

01:26:19.000 --> 01:26:21.000
You know, we tend

01:26:21.000 --> 01:26:24.000
folks who are not as close to the technology

01:26:24.000 --> 01:26:27.000
the sound bites that you get

01:26:27.000 --> 01:26:32.000
that they may be familiar with is this idea that a blockchain is an immutable record.

01:26:32.000 --> 01:26:37.000
of information right and oh okay so i've got this immutable record. That means I just have to go look it up

01:26:37.000 --> 01:26:39.000
And I have the transaction.

01:26:39.000 --> 01:26:44.000
Well, the nuance here, for example, with like an optimistic roll-up is that

01:26:44.000 --> 01:26:51.000
No, I don't have necessarily the information from the layer two isn't

01:26:51.000 --> 01:27:00.000
itself being written onto layer one. I'm leveraging the security of layer one in order to secure my transactions that I'm writing on layer two. That's a nuance.

01:27:00.000 --> 01:27:06.000
But it has implications in terms of

01:27:06.000 --> 01:27:09.000
various of those factors that they might concern about about the

01:27:09.000 --> 01:27:14.000
the security, the validity of the information, the availability of the information.

01:27:14.000 --> 01:27:26.000
But you can also point out that on layer one, well, fine, we should all just stay on layer one then, right? But the problem there is, of course, scalability and execution speed. And so if you want

01:27:26.000 --> 01:27:31.000
this to function as a stable financial system

01:27:31.000 --> 01:27:33.000
there needs to be there

01:27:33.000 --> 01:27:39.000
innovation that allows us to get greater execution speed. And so you're making different choices.

01:27:39.000 --> 01:27:44.000
There is one other thing that I think

01:27:44.000 --> 01:27:49.000
is an important topic to think about across these various regulatory regimes.

01:27:49.000 --> 01:27:53.000
which is interoperability because

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as we're moving into a multi-chain world

01:27:56.000 --> 01:28:02.000
the ways in which we operate across chains become

01:28:02.000 --> 01:28:08.000
extremely important from a security and a sort of regulatory perspective.

01:28:08.000 --> 01:28:20.000
Regulators are focused on regulating transactions that are happening in one sense and then they move off into some other chain or more importantly, I think a lot of the hacking and the thefts that have been going on have to do with

01:28:20.000 --> 01:28:24.000
moving value between chains and the insecurity of bridges.

01:28:24.000 --> 01:28:26.000
So I think

01:28:26.000 --> 01:28:37.000
if from a consumer protection perspective, you're saying we want to make this tool available to the consumer to be able to use crypto for their day-to-day transactions.

01:28:37.000 --> 01:28:44.000
Well, but now the consumer's not sure, but is my crypto

01:28:44.000 --> 01:28:46.000
On…

01:28:46.000 --> 01:28:49.000
Optimism? Is it on base? Is it on…

01:28:49.000 --> 01:28:56.000
you know what what network am I on and how do I, it's not just now the issue of on-ramping and off ramping from the TradFi system.

01:28:56.000 --> 01:28:59.000
It's now how does my value move

01:28:59.000 --> 01:29:01.000
across this multi-chain environment.

01:29:01.000 --> 01:29:05.000
I would think that

01:29:05.000 --> 01:29:13.000
It's a complicated and difficult topic to talk about with regulators, but I think it's in the long run something they will care about.

01:29:13.000 --> 01:29:15.000
Because it also has things like even antitrust

01:29:15.000 --> 01:29:20.000
implications. If you start using chains and the

01:29:20.000 --> 01:29:37.000
interoperability or non-interoperability of chains as a way to create moats in a business environment.

01:29:37.000 --> 01:29:51.000
Yeah, I don't think it comes up. That's one that's just sort of occurring to me, but it's not something that I've heard come brought up as an issue.

01:29:51.000 --> 01:29:54.000
Thank you very much.

01:29:54.000 --> 01:29:57.000
I…

01:29:57.000 --> 01:30:01.000
So I have like…

01:30:01.000 --> 01:30:06.000
25 minutes for the session.

01:30:06.000 --> 01:30:11.000
Yes. Any questions from…

01:30:11.000 --> 01:30:26.000
remote participants.

01:30:26.000 --> 01:30:36.000
Yeah, I…

01:30:36.000 --> 01:30:47.000
So maybe I can also…

01:30:47.000 --> 01:30:49.000
I don't know if like…

01:30:49.000 --> 01:30:52.000
people in here are familiar with this.

01:30:52.000 --> 01:30:57.000
But have you ever heard of like leakage sticking

01:30:57.000 --> 01:31:00.000
liquid staking, liquid staking. Okay.

01:31:00.000 --> 01:31:03.000
So leakage staking is basically like you know you

01:31:03.000 --> 01:31:08.000
can delegate your ether to someone else.

01:31:08.000 --> 01:31:10.000
through this protocol and you can get

01:31:10.000 --> 01:31:13.000
liquid staking tokens.

01:31:13.000 --> 01:31:17.000
out of the protocol. So it's like a derivative a product

01:31:17.000 --> 01:31:19.000
And you can also

01:31:19.000 --> 01:31:24.000
deposited into some other like DeFi protocol and you can get resticking

01:31:24.000 --> 01:31:30.000
rewards from the, you know, DeFi protocol by depositing

01:31:30.000 --> 01:31:32.000
the leaked staking token.

01:31:32.000 --> 01:31:34.000
So…

01:31:34.000 --> 01:31:36.000
Have you ever heard of restaking?

01:31:36.000 --> 01:31:39.000
So this is a very different

01:31:39.000 --> 01:31:41.000
concept actually

01:31:41.000 --> 01:31:48.000
So when we talk about like data availability layer or like sequencing sequencer

01:31:48.000 --> 01:31:51.000
In order to ensure

01:31:51.000 --> 01:31:56.000
the security mechanism of something else

01:31:56.000 --> 01:32:00.000
They sometimes issue governance tokens

01:32:00.000 --> 01:32:04.000
And based on the governance tokens say

01:32:04.000 --> 01:32:07.000
you know the

01:32:07.000 --> 01:32:09.000
ask them to

01:32:09.000 --> 01:32:14.000
They ask them to deposit the governance tokens and the new tokens

01:32:14.000 --> 01:32:17.000
to someone somewhere else, somewhere.

01:32:17.000 --> 01:32:20.000
And if…

01:32:20.000 --> 01:32:26.000
people like sequencers or like data ability availability team managers

01:32:26.000 --> 01:32:30.000
engage in bad activities.

01:32:30.000 --> 01:32:34.000
the fans deposited into the smart contract or something will be slashed.

01:32:34.000 --> 01:32:37.000
So this is how the governance works.

01:32:37.000 --> 01:32:40.000
based in the East Arian community.

01:32:40.000 --> 01:32:43.000
Right. So

01:32:43.000 --> 01:32:51.000
So actively validating these services is like, you know, the good example of AVS is like the data availability layer.

01:32:51.000 --> 01:32:54.000
So, um.

01:32:54.000 --> 01:32:59.000
So when you want to launch a data availability layer.

01:32:59.000 --> 01:33:01.000
Like for the first time.

01:33:01.000 --> 01:33:03.000
you have to

01:33:03.000 --> 01:33:06.000
can't like collect many

01:33:06.000 --> 01:33:10.000
people who will be responsible for

01:33:10.000 --> 01:33:13.000
You know, sequencing work right

01:33:13.000 --> 01:33:17.000
Because if there isn't any any person

01:33:17.000 --> 01:33:20.000
who is going to be responsible for the

01:33:20.000 --> 01:33:23.000
a sequencing work.

01:33:23.000 --> 01:33:28.000
That data availability data will not work.

01:33:28.000 --> 01:33:30.000
Well.

01:33:30.000 --> 01:33:34.000
Okay. So…

01:33:34.000 --> 01:33:36.000
there is a challenge.

01:33:36.000 --> 01:33:39.000
So it's challenging to

01:33:39.000 --> 01:33:41.000
um

01:33:41.000 --> 01:33:43.000
collect

01:33:43.000 --> 01:33:47.000
many people

01:33:47.000 --> 01:33:49.000
to ensure the

01:33:49.000 --> 01:33:52.000
security mechanisms of like

01:33:52.000 --> 01:33:56.000
new service like data availability layer.

01:33:56.000 --> 01:33:58.000
Because…

01:33:58.000 --> 01:34:03.000
the person who work as a data availability manager, for example.

01:34:03.000 --> 01:34:08.000
needs to work on some kind of like, you know, work.

01:34:08.000 --> 01:34:12.000
But it's very technical it's

01:34:12.000 --> 01:34:15.000
it takes time. It also takes time

01:34:15.000 --> 01:34:18.000
consumes a lot of data sometimes.

01:34:18.000 --> 01:34:19.000
Recording in progress.

01:34:19.000 --> 01:34:20.000
you know to operate the

01:34:20.000 --> 01:34:29.000
accordingly progress, man.

01:34:29.000 --> 01:34:33.000
Okay, so…

01:34:33.000 --> 01:34:35.000
What did risk taking do?

01:34:35.000 --> 01:34:37.000
is that… So…

01:34:37.000 --> 01:34:40.000
these new

01:34:40.000 --> 01:34:45.000
protocol doesn't have to

01:34:45.000 --> 01:34:52.000
issue governance tokens or like new tokens to secure to have the security mechanisms their own security mechanism

01:34:52.000 --> 01:34:58.000
what they do is that they can also they can reuse

01:34:58.000 --> 01:35:01.000
the staked ETHA on ESRAM.

01:35:01.000 --> 01:35:03.000
So…

01:35:03.000 --> 01:35:07.000
If you restake ESA,

01:35:07.000 --> 01:35:10.000
Through restaking protocol

01:35:10.000 --> 01:35:12.000
what you can do

01:35:12.000 --> 01:35:15.000
is that you can also

01:35:15.000 --> 01:35:22.000
engaging work to ensure the security of the activity validated services.

01:35:22.000 --> 01:35:26.000
In addition to Ethereum itself.

01:35:26.000 --> 01:35:29.000
Does that make sense? Like, so…

01:35:29.000 --> 01:35:32.000
And if you do something wrong.

01:35:32.000 --> 01:35:40.000
For the work you are supposed to do for the actively validated services like data availability layer

01:35:40.000 --> 01:35:45.000
you will get slashed through the staked ESA.

01:35:45.000 --> 01:35:50.000
through the ether staked on ethereum.

01:35:50.000 --> 01:35:56.000
So you are also a staker, the body data on East RM.

01:35:56.000 --> 01:36:00.000
And you deposit ETHA on Ethereum, right?

01:36:00.000 --> 01:36:07.000
But you can also engaging work to ensure the security of

01:36:07.000 --> 01:36:09.000
actively validated services.

01:36:09.000 --> 01:36:12.000
And if you do something wrong

01:36:12.000 --> 01:36:18.000
All the work you were supposed to do for the activity validity services

01:36:18.000 --> 01:36:24.000
you will get slash through the ether you've deposited on Ethereum.

01:36:24.000 --> 01:36:26.000
Does he make sense? Like…

01:36:26.000 --> 01:36:30.000
it's very weird.

01:36:30.000 --> 01:36:37.000
So…

01:36:37.000 --> 01:36:39.000
So this is a risk taking.

01:36:39.000 --> 01:36:41.000
In…

01:36:41.000 --> 01:36:45.000
Also, you can restake

01:36:45.000 --> 01:36:47.000
leaked staking token

01:36:47.000 --> 01:36:49.000
like you

01:36:49.000 --> 01:36:51.000
stake, Issa.

01:36:51.000 --> 01:36:58.000
soup, the kids taking protocol and you delegate your ESA onto some other validators

01:36:58.000 --> 01:37:05.000
to stake ESA on Ethereum and you can get leaked staking tokens out of leakage staking protocol

01:37:05.000 --> 01:37:16.000
And you can restake staking tokens to restaking protocol and then you can provide security to actively validated services

01:37:16.000 --> 01:37:18.000
And…

01:37:18.000 --> 01:37:21.000
the rewards you will be able to get

01:37:21.000 --> 01:37:24.000
is the rewards you can get

01:37:24.000 --> 01:37:27.000
a part of rewards

01:37:27.000 --> 01:37:30.000
from the body data work

01:37:30.000 --> 01:37:33.000
of the East Arian validated work

01:37:33.000 --> 01:37:38.000
And also, you can also get a part of the rewards you can

01:37:38.000 --> 01:37:42.000
reward from the risk taking, you know, activity.

01:37:42.000 --> 01:37:46.000
Like, um.

01:37:46.000 --> 01:37:49.000
So the delegated validators

01:37:49.000 --> 01:37:56.000

behind a leaked sticking protocols also engage in the security work of activity validity services

01:37:56.000 --> 01:37:59.000
In…

01:37:59.000 --> 01:38:02.000
the users, you who deposit

01:38:02.000 --> 01:38:09.000
ricket staking token to restaking protocol can also get a product rewards

01:38:09.000 --> 01:38:11.000
from the security work

01:38:11.000 --> 01:38:17.000
the delegated validators do for the actively validated services.

01:38:17.000 --> 01:38:20.000
So this is um

01:38:20.000 --> 01:38:23.000
Yeah.

01:38:23.000 --> 01:38:27.000
This is a complicated workflow to

01:38:27.000 --> 01:38:29.000
enable users to get

01:38:29.000 --> 01:38:35.000
additional yield affirming opportunities.

01:38:35.000 --> 01:38:37.000
In…

01:38:37.000 --> 01:38:44.000
the benefit of restaking for actively borrowed services is that they do not

01:38:44.000 --> 01:38:49.000
to establish their own security mechanisms themselves from scratch.

01:38:49.000 --> 01:38:52.000
And they can rely on the security

01:38:52.000 --> 01:38:56.000
of Ethereum because like, you know, there are a lot of people who already

01:38:56.000 --> 01:38:58.000
stake ESA on Ethereum.

01:38:58.000 --> 01:39:03.000
And they can take advantage of the security that those people provide.

01:39:03.000 --> 01:39:06.000
to the E3M but

01:39:06.000 --> 01:39:11.000
if they do something wrong for the active lead body, they do services work

01:39:11.000 --> 01:39:14.000
they will get scratched.

01:39:14.000 --> 01:39:16.000
through the ESA deposited on Ethereum.

01:39:16.000 --> 01:39:18.000
So…

01:39:18.000 --> 01:39:24.000
And there is also liquidity restaking protocol these days

01:39:24.000 --> 01:39:28.000
And you can so as a user

01:39:28.000 --> 01:39:33.000
Once you get rickety sticking tokens from ricket staking protocol

01:39:33.000 --> 01:39:39.000
you can deposit the kids taking token to restaking protocol and you can get

01:39:39.000 --> 01:39:41.000
restaking token out of it.

01:39:41.000 --> 01:39:44.000
And this is how you can get

01:39:44.000 --> 01:39:47.000
additional yield farming opportunities.

01:39:47.000 --> 01:39:49.000
In…

01:39:49.000 --> 01:39:52.000
So it's becoming complicated.

01:39:52.000 --> 01:39:56.000
But so what I wanted to talk about

01:39:56.000 --> 01:40:03.000
In terms of this is that

01:40:03.000 --> 01:40:07.000
So…

01:40:07.000 --> 01:40:13.000
You know, I talked about the governance problems of data availability layer and sequencer, right?

01:40:13.000 --> 01:40:24.000
But the reason why there is a challenge, it is challenging is that it's very hard. It takes time to establish the security mechanism from scratch.

01:40:24.000 --> 01:40:29.000
But if they use restaking, it's technically possible

01:40:29.000 --> 01:40:36.000
to immediately establish the security mechanisms that are as secure as the ECRM.

01:40:36.000 --> 01:40:41.000
Not as scary secure as Ethereum, but they are like

01:40:41.000 --> 01:40:45.000
more secure than, you know, like establishing it from scratch.

01:40:45.000 --> 01:40:50.000
Right.

01:40:50.000 --> 01:40:54.000
That's why the Ethereum community is

01:40:54.000 --> 01:40:59.000
saying that risk-taking is kind of like a future

01:40:59.000 --> 01:41:05.000
for developers to establish new security mechanisms without

01:41:05.000 --> 01:41:09.000
Without taking much time.

01:41:09.000 --> 01:41:13.000
Okay. But I believe…

01:41:13.000 --> 01:41:16.000

that there is a lot of challenges

01:41:16.000 --> 01:41:18.000
for restaking itself.

01:41:18.000 --> 01:41:23.000
definitely leads to the financial stability risk.

01:41:23.000 --> 01:41:26.000
Okay, so…

01:41:26.000 --> 01:41:30.000
The financial stability risk

01:41:30.000 --> 01:41:33.000
within decentralized finance

01:41:33.000 --> 01:41:36.000
has not been like focused yet by the

01:41:36.000 --> 01:41:37.000
regulatory standards.

01:41:37.000 --> 01:41:38.000
and regulations.

01:41:38.000 --> 01:41:45.000
regulatory standards setting bodies that watch financial stability risks

01:41:45.000 --> 01:41:47.000
um within

01:41:47.000 --> 01:41:49.000
traditional finance like finance

01:41:49.000 --> 01:41:52.000
Financial Stability Board.

01:41:52.000 --> 01:41:59.000
And they say that the spillover effects of stability risk within decentralized finance

01:41:59.000 --> 01:42:03.000
are too limited to be seriously discussed yet.

01:42:03.000 --> 01:42:09.000
but so they only talk about discussed, you know, risks

01:42:09.000 --> 01:42:12.000

spill over effects of like stable coins and like

01:42:12.000 --> 01:42:17.000
EDF and et cetera, but they do not really talk about SWT.

01:42:17.000 --> 01:42:20.000
risks within decentralized finance.

01:42:20.000 --> 01:42:22.000
that much yet but yet

01:42:22.000 --> 01:42:26.000
it's already discussed in the ceramic community

01:42:26.000 --> 01:42:33.000
But yeah, I wanted to just to exchange ideas around like what are the challenges and what are the solutions being discussed to

01:42:33.000 --> 01:42:39.000
you know, prevent the risks, the financial stability risks among around

01:42:39.000 --> 01:42:41.000
restaking.

01:42:41.000 --> 01:42:44.000
In…

01:42:44.000 --> 01:42:50.000
I analyze the characteristics of these yield affirming activities like staking

01:42:50.000 --> 01:42:54.000
We keep staking, native staking, restaking wicked

01:42:54.000 --> 01:42:59.000
staking liquid staking token restaking and leakage restaking and so on

01:42:59.000 --> 01:43:02.000
So, um.

01:43:02.000 --> 01:43:04.000
Yeah. So…

01:43:04.000 --> 01:43:07.000
In this table, I basically…

01:43:07.000 --> 01:43:09.000
you know make a list of like

01:43:09.000 --> 01:43:15.000
what kind of rewards they can users can get out of each activity

01:43:15.000 --> 01:43:21.000
And if you engage, if a user engage in that activity

01:43:21.000 --> 01:43:25.000
That means…

01:43:25.000 --> 01:43:29.000
Who are you going to trust?

01:43:29.000 --> 01:43:34.000
through these activities. For example, like, I think it's lead staking is a good example

01:43:34.000 --> 01:43:37.000
Like if you

01:43:37.000 --> 01:43:40.000
engaging in ricket staking

01:43:40.000 --> 01:43:43.000
you get the rewards

01:43:43.000 --> 01:43:47.000
You receive a part of rewards from rich staking protocol

01:43:47.000 --> 01:43:52.000
dedicated body data is working the body dedicated body data is work right

01:43:52.000 --> 01:43:54.000
And…

01:43:54.000 --> 01:44:01.000
if you deposit ricket staking token you received from ricket staking protocol

01:44:01.000 --> 01:44:07.000
you can also get additional rewards from DeFi that you support with the leakage staking tokens.

01:44:07.000 --> 01:44:09.000
that means that

01:44:09.000 --> 01:44:17.000
you are trusting the delegated body leaders while properly dealing with these validated work, right?

01:44:17.000 --> 01:44:23.000

And you also rely on the security of ricket staking protocol and ESRM.

01:44:23.000 --> 01:44:26.000
I think this is a good example and

01:44:26.000 --> 01:44:28.000
So I also…

01:44:28.000 --> 01:44:31.000
make a list of, you know.

01:44:31.000 --> 01:44:36.000
what kind of rewards a user will be able to get to each activity and

01:44:36.000 --> 01:44:38.000
Hmm, they are

01:44:38.000 --> 01:44:42.000
trusting through these activities.

01:44:42.000 --> 01:44:46.000
And also when you want to

01:44:46.000 --> 01:44:50.000
If you want to

01:44:50.000 --> 01:44:55.000
and get redemption from like, for example, leakage staking protocol

01:44:55.000 --> 01:44:57.000
So like you when when you

01:44:57.000 --> 01:45:10.000
deposit your ESA to the leakage sticking token this leakage staking protocol and the leakage staking protocol delegate your isa onto

01:45:10.000 --> 01:45:13.000
dedicated body data is

01:45:13.000 --> 01:45:15.000
But he…

01:45:15.000 --> 01:45:18.000
Yeah, usually you can immediately

01:45:18.000 --> 01:45:21.000
get

01:45:21.000 --> 01:45:23.000
your ether back from the

01:45:23.000 --> 01:45:27.000
digit staking protocol, but it's sometimes

01:45:27.000 --> 01:45:30.000
takes time for you to

01:45:30.000 --> 01:45:33.000
get redemption from liquid staking protocol because you know

01:45:33.000 --> 01:45:36.000
if you know a number of people

01:45:36.000 --> 01:45:40.000
try to get redemption from one naked sticky

01:45:40.000 --> 01:45:43.000
rigid sticking protocol at the same time probably it's

01:45:43.000 --> 01:45:45.000
difficult for the protocol to

01:45:45.000 --> 01:45:47.000
you know um

01:45:47.000 --> 01:45:49.000
um

01:45:49.000 --> 01:45:51.000
provider fans to all of the users, right?

01:45:51.000 --> 01:45:55.000
So, um.

01:45:55.000 --> 01:45:58.000
these kind of like the limitations of redemption

01:45:58.000 --> 01:46:00.000
could be decided by

01:46:00.000 --> 01:46:03.000
some entities.

01:46:03.000 --> 01:46:07.000
And for example, like if you engage in rickets taking

01:46:07.000 --> 01:46:09.000
We get staking.

01:46:09.000 --> 01:46:14.000
the redemption could be remitted by the kids taking protocol.

01:46:14.000 --> 01:46:18.000
It's fundamentally, you know, by Eastern wicked sticking protocol

01:46:18.000 --> 01:46:24.000
And also like in terms of like restaking as well, we can also

01:46:24.000 --> 01:46:27.000
do the same analysis for

01:46:27.000 --> 01:46:30.000
in terms yeah

01:46:30.000 --> 01:46:32.000
In the same way, yeah. So…

01:46:32.000 --> 01:46:35.000
This is a table I created.

01:46:35.000 --> 01:46:37.000
and so

01:46:37.000 --> 01:46:43.000
When it comes to talking about the financial stability risk in the financial stability board.

01:46:43.000 --> 01:46:47.000
we have like they have like two definitions, the bounding ability

01:46:47.000 --> 01:46:57.000
and shock. The balmy ability is an underlining state within a decentralized financial system that has the potential to amplify that state or trigger

01:46:57.000 --> 01:47:01.000
a new state in the event of a shock or stressed event.

01:47:01.000 --> 01:47:04.000
And shock is a sudden disturbance

01:47:04.000 --> 01:47:13.000
with the decentralized financial system that triggers an amplification of vulnerabilities and or leads to new vulnerabilities emerging.

01:47:13.000 --> 01:47:15.000

So this is um

01:47:15.000 --> 01:47:22.000
definition of the terms that are often used in the financial stability board.

01:47:22.000 --> 01:47:25.000
And I will reuse this terminology

01:47:25.000 --> 01:47:28.000
to analyze the vulnerability and shocks

01:47:28.000 --> 01:47:32.000
around E3M restaking.

01:47:32.000 --> 01:47:36.000
And this is the taxonomy of vulnerabilities.

01:47:36.000 --> 01:47:40.000
And I create that i created

01:47:40.000 --> 01:47:45.000
And I would not like to talk about the details of this taxonomy.

01:47:45.000 --> 01:47:47.000
But I think it's important to

01:47:47.000 --> 01:47:52.000
analyze the vulnerabilities that could lead to the financial stability

01:47:52.000 --> 01:47:57.000
Challenge, financial stability.

01:47:57.000 --> 01:48:00.000
unstable by now.

01:48:00.000 --> 01:48:02.000
Financial stability risks.

01:48:02.000 --> 01:48:06.000
in um

01:48:06.000 --> 01:48:10.000
So this is one of the examples of

01:48:10.000 --> 01:48:14.000
how

01:48:14.000 --> 01:48:18.000

one vulnerability could lead to

01:48:18.000 --> 01:48:25.000
like different vulnerability. I could amplify different vulnerabilities through shock

01:48:25.000 --> 01:48:27.000
So for example.

01:48:27.000 --> 01:48:34.000
disproportionate amount of fancy it's concentrated into specific liquid restaking protocols

01:48:34.000 --> 01:48:37.000
that means that

01:48:37.000 --> 01:48:48.000
So Ricky Tari staking protocols have delegated

01:48:48.000 --> 01:48:50.000
It's taking…

01:48:50.000 --> 01:48:52.000
protocols

01:48:52.000 --> 01:49:00.000
you can body leaders

01:49:00.000 --> 01:49:05.000
Okay, so you can, when, when you

01:49:05.000 --> 01:49:07.000
um

01:49:07.000 --> 01:49:12.000
deposit your, for example, like leakage staking tokens or like isa

01:49:12.000 --> 01:49:15.000
to ricket restaking protocol

01:49:15.000 --> 01:49:20.000
that means that you will delegate your ESA

01:49:20.000 --> 01:49:23.000
to a few AVSs.

01:49:23.000 --> 01:49:27.000
that restaking protocols support.

01:49:27.000 --> 01:49:35.000

But, you know, if a disproportionate amount of fans is concentrated into specific leakage staking protocols

01:49:35.000 --> 01:49:38.000
Which means that the

01:49:38.000 --> 01:49:43.000
you know your isa and people'sa will be

01:49:43.000 --> 01:49:48.000
And Rick is taking tokens will be concentrated into

01:49:48.000 --> 01:49:55.000
a few AVSs that are supported by the specific resticking protocol.

01:49:55.000 --> 01:49:57.000
So…

01:49:57.000 --> 01:50:06.000
This means that a disproportionate amount of staked ESI will be concentrated with very few AVS.

01:50:06.000 --> 01:50:09.000
And for example.

01:50:09.000 --> 01:50:12.000
if there is a shock like

01:50:12.000 --> 01:50:14.000
AVS.

01:50:14.000 --> 01:50:16.000
gets hacked or like has errors

01:50:16.000 --> 01:50:21.000
That means a large amount of ESA gets flushed.

01:50:21.000 --> 01:50:23.000
Because…

01:50:23.000 --> 01:50:25.000
you know.

01:50:25.000 --> 01:50:27.000
A lot of these are

01:50:27.000 --> 01:50:29.000
state restaked

01:50:29.000 --> 01:50:33.000
to very few

01:50:33.000 --> 01:50:34.000
Only a few AVS.

01:50:34.000 --> 01:50:39.000
And a large amount of Ether gets slashed at the same time.

01:50:39.000 --> 01:50:41.000
Because of the heart.

01:50:41.000 --> 01:50:44.000
And, um.

01:50:44.000 --> 01:50:49.000
AVS and Ethereum lose body data a lot of validators

01:50:49.000 --> 01:50:52.000
pick your vector security of ECRM, the IVS.

01:50:52.000 --> 01:50:55.000
In…

01:50:55.000 --> 01:50:58.000
Also…

01:50:58.000 --> 01:51:04.000
Yes, so this is one of the examples of like how vulnerabilities in

01:51:04.000 --> 01:51:07.000
within the decentralized financial system

01:51:07.000 --> 01:51:09.000
get amplified

01:51:09.000 --> 01:51:12.000
And…

01:51:12.000 --> 01:51:15.000
Yes. Also…

01:51:15.000 --> 01:51:18.000
Ricky three staking tokens.

01:51:18.000 --> 01:51:24.000
are circulated within decentralized financial system. And in DeFi protocol

01:51:24.000 --> 01:51:26.000
There are some like…

01:51:26.000 --> 01:51:29.000
How do you say?

01:51:29.000 --> 01:51:35.000
pair of ricketry staking token with like some other

01:51:35.000 --> 01:51:38.000
tokens.

01:51:38.000 --> 01:51:42.000
But would it not be like…

01:51:42.000 --> 01:51:49.000
adequate market liquidity of characteristic tokens within each in each

01:51:49.000 --> 01:51:51.000
DeFi protocol

01:51:51.000 --> 01:51:55.000
And it's easy to…

01:51:55.000 --> 01:51:59.000
For the Deepak wicket restaking token to happen

01:51:59.000 --> 01:52:05.000
the deep echo of leaky resticking token from East RM.

01:52:05.000 --> 01:52:07.000
Issa.

01:52:07.000 --> 01:52:10.000
value and data staking token value

01:52:10.000 --> 01:52:13.000
to be happening. It's easy to

01:52:13.000 --> 01:52:16.000
for it to be happening.

01:52:16.000 --> 01:52:19.000
And if it happens.

01:52:19.000 --> 01:52:24.000
that the pack, it will lead to the Deepak of VT3 staking tokens from ESRM.

01:52:24.000 --> 01:52:27.000
ether as well in

01:52:27.000 --> 01:52:29.000
It's like…

01:52:29.000 --> 01:52:33.000
Spiral. It's like negative spiral of

01:52:33.000 --> 01:52:38.000
Deepaks around restaking tokens and Ricketry staking tokens.

01:52:38.000 --> 01:52:40.000
And I think it's…

01:52:40.000 --> 01:52:45.000
Barry similar to what could happen, what happened in

01:52:45.000 --> 01:52:48.000
the um

01:52:48.000 --> 01:52:50.000
Duff.

01:52:50.000 --> 01:52:55.000
in the shock in 2008 and 2009.

01:52:55.000 --> 01:53:00.000
The same thing could happen within the decentralized financial

01:53:00.000 --> 01:53:02.000
system because like

01:53:02.000 --> 01:53:10.000
staking tokens and wiki3 staking tokens are basically like the dairy Buddy product product within decentralized financial system

01:53:10.000 --> 01:53:13.000
So…

01:53:13.000 --> 01:53:18.000
Yeah, there are a lot of lessons we could learn from these

01:53:18.000 --> 01:53:20.000
you know, past.

01:53:20.000 --> 01:53:22.000

shocks in past.

01:53:22.000 --> 01:53:24.000
financial stability.

01:53:24.000 --> 01:53:26.000
Challenges.

01:53:26.000 --> 01:53:29.000
And this is what

01:53:29.000 --> 01:53:31.000
begin i um

01:53:31.000 --> 01:53:34.000
analyzing right now?

01:53:34.000 --> 01:53:36.000
And yeah.

01:53:36.000 --> 01:53:39.000
So I wanted to ask if

01:53:39.000 --> 01:53:47.000
The Ethereum community is seriously thinking about the financial stability risk within the decentralized financial system through restaking.

01:53:47.000 --> 01:53:49.000
In this event.

01:53:49.000 --> 01:53:52.000
and once

01:53:52.000 --> 01:53:57.000
I published this paper. I would like to circulate it to the financial stability.

01:53:57.000 --> 01:53:59.000
bored when I got to…

01:53:59.000 --> 01:54:03.000
Regulatory authorities that are interested in

01:54:03.000 --> 01:54:06.000
the financial stability challenges within decentralized finance.

01:54:06.000 --> 01:54:09.000
system so

01:54:09.000 --> 01:54:14.000
But I think I wanted to make this presentation because

01:54:14.000 --> 01:54:18.000
And this is definitely related to the governance challenges

01:54:18.000 --> 01:54:20.000
of layer two.

01:54:20.000 --> 01:54:29.000
Because as I said, like data layer twos, scaling solutions use data availability layers and like

01:54:29.000 --> 01:54:34.000
you know sequences and like these kind of like actively validated services

01:54:34.000 --> 01:54:40.000
that use new governance mechanisms.

01:54:40.000 --> 01:54:43.000
And they started using restaking to

01:54:43.000 --> 01:54:47.000
have these

01:54:47.000 --> 01:54:52.000
security mechanisms because they don't want you to take a lot of spend a lot of time, you know

01:54:52.000 --> 01:54:56.000
creating the security mechanism from scratch.

01:54:56.000 --> 01:55:03.000
And they started relying on restaking. But if you rely on restaking a lot.

01:55:03.000 --> 01:55:07.000
that will read to the financial stability stability risks, as I mentioned.

01:55:07.000 --> 01:55:09.000
So…

01:55:09.000 --> 01:55:14.000
this is related.

01:55:14.000 --> 01:55:27.000
Okay. Lots of comments.

01:55:27.000 --> 01:55:34.000
What if measure entities, investors restate to multiple protocols from the same source?

01:55:34.000 --> 01:55:39.000
And then having one of them being slashed.

01:55:39.000 --> 01:55:41.000
Yes. So if…

01:55:41.000 --> 01:55:44.000
This is a good question.

01:55:44.000 --> 01:55:47.000
If…

01:55:47.000 --> 01:55:53.000
one of them being gets slashed that means that all of the protocols

01:55:53.000 --> 01:55:57.000
will lose the security.

01:55:57.000 --> 01:56:02.000
Because all of the multiple protocols are basically

01:56:02.000 --> 01:56:04.000
I rely on rely on

01:56:04.000 --> 01:56:06.000
the same is deposited on Ethereum.

01:56:06.000 --> 01:56:12.000
So why was the protocols get hashed?

01:56:12.000 --> 01:56:17.000
the value that gets lost through one of the protocols

01:56:17.000 --> 01:56:20.000
That's the day support.

01:56:20.000 --> 01:56:24.000
all of the security will be

01:56:24.000 --> 01:56:26.000
lost because of this.

01:56:26.000 --> 01:56:28.000
So that is a problem that is uh

01:56:28.000 --> 01:56:30.000
you know.

01:56:30.000 --> 01:56:42.000
reason why it has a lot of financial stability risk.

01:56:42.000 --> 01:56:46.000
Okay.

01:56:46.000 --> 01:56:50.000
I think.

01:56:50.000 --> 01:56:54.000
That's it?

01:56:54.000 --> 01:56:58.000
Okay, so if there isn't any comment, probably…

01:56:58.000 --> 01:57:01.000
Let's finish it up.

01:57:01.000 --> 01:57:05.000
Thank you. Okay, so it's time for a break.

01:57:05.000 --> 01:57:07.000
And we will

01:57:07.000 --> 01:57:09.000
Yeah, Kofi Blake, and we will come back.

01:57:09.000 --> 01:57:12.000
within 15 minutes.

01:57:12.000 --> 01:57:16.000
Okay, thank you. Thank you for hearing, listening to you.

01:57:16.000 --> 01:57:18.000
This.

01:57:18.000 --> 01:57:48.000
Okay.

02:14:40.000 --> 02:14:42.000
Okay, so let's…

02:14:42.000 --> 02:14:54.000
led you up to the second session.

02:14:54.000 --> 02:14:57.000
So we started his presentation.

02:14:57.000 --> 02:15:27.000
on the warrant governance.

02:15:30.000 --> 02:15:48.000
I don't think I need to use this one. I can share my screen.

02:15:48.000 --> 02:16:01.000
Hello.

02:16:01.000 --> 02:16:04.000
Hello, Mitchell.

02:16:04.000 --> 02:16:05.000
Hi, joining. Hello.

02:16:05.000 --> 02:16:07.000
Hello. Yep.

02:16:07.000 --> 02:16:08.000
Yeah.

02:16:08.000 --> 02:16:10.000
How are you going?

02:16:10.000 --> 02:16:14.000
I've heard that you want me to you wanted to

02:16:14.000 --> 02:16:17.000
You want me to do it first?

02:16:17.000 --> 02:16:22.000
Yeah, that will give you the best amount of time to discuss with the

02:16:22.000 --> 02:16:23.000
Okay. Okay.

02:16:23.000 --> 02:16:26.000
participants from San Francisco?

02:16:26.000 --> 02:16:35.000
Yes, San Francisco.

02:16:35.000 --> 02:16:40.000
So I aim to finish my presentation within 60 minutes.

02:16:40.000 --> 02:16:43.000
So you can use the list.

02:16:43.000 --> 02:16:48.000
30 minutes. Okay, so, okay.

02:16:48.000 --> 02:16:51.000
there's some…

02:16:51.000 --> 02:16:53.000
System trouble, it seems like

02:16:53.000 --> 02:17:15.000
So, okay.

02:17:15.000 --> 02:17:18.000
Good afternoon, everyone. I'm Masaru Yamanaka.

02:17:18.000 --> 02:17:30.000
Today, I'd rather have to make a presentation about an overview of the accountable work concept, which is ongoing discussion topic within the IKP working group or bidding.

02:17:30.000 --> 02:17:33.000
So the…

02:17:33.000 --> 02:17:36.000
I think I can use the 60

02:17:36.000 --> 02:17:41.000
minutes so hydra but i do have to allocate the more time

02:17:41.000 --> 02:17:47.000
for a discussion down my presentation so I keep my presentation short

02:17:47.000 --> 02:17:50.000
I'm not going to dive into the technical detail today.

02:17:50.000 --> 02:17:55.000
I'm going to focus on explaining the idea itself

02:17:55.000 --> 02:18:04.000
Okay, so we define the accountable word as a wallet that enables users to prove their legitimacy

02:18:04.000 --> 02:18:08.000
to their transaction count buddies.

02:18:08.000 --> 02:18:11.000
I'm going to explain about it more.

02:18:11.000 --> 02:18:14.000
Anyway, let me give him my

02:18:14.000 --> 02:18:20.000
presentation, the research background first

02:18:20.000 --> 02:18:28.000
so for silver, the crypt itself is not created for money laundering. However, in reality.

02:18:28.000 --> 02:18:30.000
It's being used for it.

02:18:30.000 --> 02:18:35.000
Agencies like FBI pointed out that so-called

02:18:35.000 --> 02:18:40.000
grip mixers, especially its tornado cache, are being used for money laundering

02:18:40.000 --> 02:18:50.000
The money laundering illustrated here is not limited to crypto process obtained from the crypto hacking hacking

02:18:50.000 --> 02:18:56.000
It can be anything such as the proceeds from illegal drug sales

02:18:56.000 --> 02:19:06.000
If the process is not in crypto yet, it can be converted from fiat to crypto through anonymous transaction on tour

02:19:06.000 --> 02:19:14.000
P2P transaction or crypto ATMs, that doesn't need to KYC or whatever.

02:19:14.000 --> 02:19:23.000
So however, the if money launderer send a crypto to Vaspus virtual asset service providers.

02:19:23.000 --> 02:19:26.000
from illegal transaction platform directory

02:19:26.000 --> 02:19:31.000
it may be detected as an illegal source by them.

02:19:31.000 --> 02:19:34.000
So they, money launderer

02:19:34.000 --> 02:19:43.000
Typically released through crypt mixers like Tornado Cash before depositing into Vaspus

02:19:43.000 --> 02:19:45.000

and then uh they

02:19:45.000 --> 02:19:51.000
declare it as a list limit investment profits from legitimate crypto

02:19:51.000 --> 02:19:56.000
investments and pay the taxes accordingly

02:19:56.000 --> 02:20:03.000
It can be looked like a Leopold prophet.

02:20:03.000 --> 02:20:05.000
So here's the dilemma for

02:20:05.000 --> 02:20:14.000
of this problem the due to these programs, the use of the tornado cache is officially prohibited in the United States.

02:20:14.000 --> 02:20:24.000
Vaspers in the United States are also prohibited from receiving crypto from the wallets that have used Tornado Cash.

02:20:24.000 --> 02:20:28.000
So, of course, the money launderers knows it

02:20:28.000 --> 02:20:39.000
So instead of the depositing directly from their wallet, they have received the crypto from the Tornado Cash, they will try to make a detour.

02:20:39.000 --> 02:20:44.000
through several wallets before depositing into wasps

02:20:44.000 --> 02:20:47.000
So here's an additional program for that.

02:20:47.000 --> 02:20:54.000
Even if the bus can detect the warrant has indirectly used the tornado cache

02:20:54.000 --> 02:20:59.000
the user themselves may not have used the Tornado Cache themselves.

02:20:59.000 --> 02:21:04.000
the user may have simply received the crypto that went through the

02:21:04.000 --> 02:21:09.000
A tornado cash as a payment for the legitimate sales of goods or services.

02:21:09.000 --> 02:21:12.000

that from the on-chain data perspective.

02:21:12.000 --> 02:21:16.000
this transaction data will look exactly the same.

02:21:16.000 --> 02:21:22.000
as the crypto was coming out of the tornado cache going through several wallets

02:21:22.000 --> 02:21:27.000
And then attempting to be deposited into a PASP.

02:21:27.000 --> 02:21:29.000
In other words, it's

02:21:29.000 --> 02:21:36.000
it's impossible to determine whether the wallets belongs to the malicious users or

02:21:36.000 --> 02:21:40.000
users who acquired their crypto in good faith.

02:21:40.000 --> 02:21:46.000
Just by using the on-chain data alone.

02:21:46.000 --> 02:21:56.000
So on the previous page, I elastated Tornado Cash for the harmful tour for society

02:21:56.000 --> 02:21:59.000
that MD privacy preserving perspective the

02:21:59.000 --> 02:22:02.000
tornado cash uh is

02:22:02.000 --> 02:22:05.000
actually necessary in reality.

02:22:05.000 --> 02:22:11.000
So as you know that the uh without them, the crypto transaction can be easily traced

02:22:11.000 --> 02:22:19.000
However, if you use the tornado cache and one of the counterparties that you mix the crypt with.

02:22:19.000 --> 02:22:26.000
Turned out to be a money launder. You are technically aiding the laundering the criminal process.

02:22:26.000 --> 02:22:33.000

Moreover, as elastylated on previous page, even if you have never used a tornado cache yourself.

02:22:33.000 --> 02:22:36.000
You cannot reduce this risk to zero.

02:22:36.000 --> 02:22:39.000
In most countries, jurisdictions.

02:22:39.000 --> 02:22:43.000
acquiring the criminal forces in good faith i mean the

02:22:43.000 --> 02:22:46.000
receiving the

02:22:46.000 --> 02:22:48.000
that's a process without knowing

02:22:48.000 --> 02:22:52.000
Often that does not result in criminal charges.

02:22:52.000 --> 02:22:57.000
It's called a, you know, good phase acquisition

02:22:57.000 --> 02:23:02.000
However, the proving the good first acquisition in crypto is not easy at all.

02:23:02.000 --> 02:23:15.000
Because someone might falsely claim that the ignorance, often knowingly buying crypto from the criminal proceeds at a slightly lower price than market price.

02:23:15.000 --> 02:23:20.000
In other words, there's inevitable trade-off between the privacy and the compliance.

02:23:20.000 --> 02:23:25.000
So it's up to users to decide which one to prioritize.

02:23:25.000 --> 02:23:29.000
Sorry. However, what I

02:23:29.000 --> 02:23:36.000
What I see as a program is that the users who want to abide by the law

02:23:36.000 --> 02:23:39.000
not giving them option to do so.

02:23:39.000 --> 02:23:41.000
I mean, if

02:23:41.000 --> 02:23:46.000
When users transact and receive the crypto from the

02:23:46.000 --> 02:23:52.000
their account properties, there's no way to verify or prove to each other that the

02:23:52.000 --> 02:23:59.000
Crypto was not derived from the criminal process that is the program.

02:23:59.000 --> 02:24:04.000
So that's why we need a technologies like that that

02:24:04.000 --> 02:24:08.000
That allowed users to comply with the regulations on their own.

02:24:08.000 --> 02:24:14.000
So as I just write it on the top right the

02:24:14.000 --> 02:24:25.000
it is difficult for a recipient of crypto to verify whether the sender, I mean the originated obtained the risk symmetry by checking on-chain data

02:24:25.000 --> 02:24:31.000
But if it could be proven to a counterparty in the form of this.

02:24:31.000 --> 02:24:34.000
verifiable proofs of knowing uh checking

02:24:34.000 --> 02:24:38.000
I'm sorry, without checking the on-chain data

02:24:38.000 --> 02:24:43.000
the process would be great and greatly simplified.

02:24:43.000 --> 02:24:46.000
So our working group.

02:24:46.000 --> 02:24:49.000
in begin is now discussing

02:24:49.000 --> 02:24:51.000
what the proof should prove

02:24:51.000 --> 02:24:56.000
And the insurance

02:24:56.000 --> 02:24:58.000

insurance procedures.

02:24:58.000 --> 02:25:03.000
And how to reduce the risk of the personal information being identified.

02:25:03.000 --> 02:25:09.000
Yeah. That's where we discussing about at moment.

02:25:09.000 --> 02:25:12.000
And the default.

02:25:12.000 --> 02:25:14.000
defining the list demosy.

02:25:14.000 --> 02:25:25.000
So even if the card is considered a less meat for you, the definition of the list mercy itself is totally dependent on the jurisdictions and the entities.

02:25:25.000 --> 02:25:28.000
So it cannot be universally defined.

02:25:28.000 --> 02:25:34.000
In other words, this should be defined freely by

02:25:34.000 --> 02:25:36.000
each user.

02:25:36.000 --> 02:25:44.000
For example, there's some crypto watch list published by the fbi

02:25:44.000 --> 02:25:46.000
that list uh

02:25:46.000 --> 02:25:56.000
But it's up to each individual to decide whether to conduct with wallets on that list or

02:25:56.000 --> 02:26:01.000
wallets that have transact with that wallet on that list

02:26:01.000 --> 02:26:15.000
So what matter is that being able to know the crypto, you are about to receive from the counterparty is derived from these addresses prior to the transaction.

02:26:15.000 --> 02:26:18.000
and if

02:26:18.000 --> 02:26:21.000
everyone decides to exclude the wallets.

02:26:21.000 --> 02:26:23.000
On the FBI list.

02:26:23.000 --> 02:26:25.000
the bitcoin on that

02:26:25.000 --> 02:26:29.000
on the warrants will effectively be unusable.

02:26:29.000 --> 02:26:35.000
Technically, this is the consensus of compliance our idea

02:26:35.000 --> 02:26:40.000
So this is the core concept of the accountable wallet.

02:26:40.000 --> 02:26:46.000
Last year, Beta Rick ended his co-authors proposed the same

02:26:46.000 --> 02:26:52.000
a similar idea of us, I mean there, which is the it's called their privacy pools uh

02:26:52.000 --> 02:26:54.000
which gained the attention

02:26:54.000 --> 02:26:59.000
Last year, this idea attempts to solve

02:26:59.000 --> 02:27:05.000
Roughly same idea, roughly same program from the different approaches

02:27:05.000 --> 02:27:15.000
But I believe that our research works is very compatible with their uploads.

02:27:15.000 --> 02:27:26.000
Privacy pool mixing protocol that allows users to mix their crypt assets only among the legitimate users. It's called the association sets.

02:27:26.000 --> 02:27:35.000
Which means that the privacy pools allow list limit users to mix their crypto in isolation from the

02:27:35.000 --> 02:27:38.000
Brock Listed users.

02:27:38.000 --> 02:27:49.000
As it is a mixer, you can protect the privacy of transaction and you can also prevent mixing your crypto with some money launder

02:27:49.000 --> 02:27:57.000
Which was an issue with the existing crypto mixers such as tornado cash you know

02:27:57.000 --> 02:27:59.000
As I explained earlier.

02:27:59.000 --> 02:28:03.000
users can receive their membership proof.

02:28:03.000 --> 02:28:06.000
upon withdrawal.

02:28:06.000 --> 02:28:10.000
Proving that their withdrawal

02:28:10.000 --> 02:28:14.000
was mixed only among the association sets.

02:28:14.000 --> 02:28:18.000
which groups that is also the deal with the door of

02:28:18.000 --> 02:28:22.000
association sets constructed as a

02:28:22.000 --> 02:28:30.000
either incursion list, I mean, which means that i mean the white listing

02:28:30.000 --> 02:28:35.000
Or could be constructed as an exclusion list.

02:28:35.000 --> 02:28:38.000
Which means that block listed.

02:28:38.000 --> 02:28:46.000
So, and then the paper said that the trusted third party are expected to

02:28:46.000 --> 02:28:50.000
create those sets first.

02:28:50.000 --> 02:28:59.000
But this creates a separation of whether the law-abiding users can prove the legitimacy of the crypto or while preserving

02:28:59.000 --> 02:29:06.000
privacy while non-compliant users cannot create such proofs.

02:29:06.000 --> 02:29:14.000

Okay, so yeah i'm going to explain there. For example, just write it above T.

02:29:14.000 --> 02:29:19.000
For example, let's say that the uh

02:29:19.000 --> 02:29:25.000
even the bottom the bottom of the uh

02:29:25.000 --> 02:29:33.000
chart is illicit users and the other four knows that the

02:29:33.000 --> 02:29:41.000
He holds the crypto obtained by illegally using the DeFi protocol

02:29:41.000 --> 02:29:45.000
uh like uh

02:29:45.000 --> 02:29:47.000
price manipulation, whatever.

02:29:47.000 --> 02:29:52.000
And the other four decided to exclude if when mixing crypto.

02:29:52.000 --> 02:30:00.000
the four can prove that they did not mix their own deposits with

02:30:00.000 --> 02:30:04.000
it was deposits when this withdrawing the

02:30:04.000 --> 02:30:06.000
them from their privacy pool.

02:30:06.000 --> 02:30:10.000
by receiving the membership proofs, meaning that the

02:30:10.000 --> 02:30:14.000
They are not the source of the illicit funds.

02:30:14.000 --> 02:30:21.000
And did not aid if it's money laundering, that's how it works.

02:30:21.000 --> 02:30:30.000
So one of the key challenges highlighted in their paper is how to form the legitimate association sets fairly

02:30:30.000 --> 02:30:39.000
Accountable warrants can address these challenges by enabling the users to prove their legacy mercy.

02:30:39.000 --> 02:30:47.000
Association sets should be formed among those who have applied the same level of the compliance rules

02:30:47.000 --> 02:30:54.000
And they can't prove it. In the diagram below the association from sets one

02:30:54.000 --> 02:30:56.000
is the is a set

02:30:56.000 --> 02:30:59.000
that can be joined that can be

02:30:59.000 --> 02:31:07.000
be joined only by those who can prove the same level of the legitimacy using the accountable wallets

02:31:07.000 --> 02:31:16.000
So in this scenario, some of the users may also have the additional lures as their own preferences.

02:31:16.000 --> 02:31:20.000
For example, some user may

02:31:20.000 --> 02:31:24.000
prefer to mix only with the US residents.

02:31:24.000 --> 02:31:27.000
which now narwhal sets.

02:31:27.000 --> 02:31:32.000
So these are those with such preference

02:31:32.000 --> 02:31:35.000
cells form the association set to here

02:31:35.000 --> 02:31:38.000
in the diagram below

02:31:38.000 --> 02:31:47.000
Mixing crypto within that those association sets provides the membership proof that he

02:31:47.000 --> 02:31:49.000
mixing was done in

02:31:49.000 --> 02:31:52.000
Only way that uh

02:31:52.000 --> 02:31:56.000
someone who has the same rules they uh

02:31:56.000 --> 02:32:00.000
Which can prove that the crypto assets were not makes it

02:32:00.000 --> 02:32:07.000
not mixed with illegal warrants and the illegal warrants at different reasons that defined on the

02:32:07.000 --> 02:32:09.000
you know uh each

02:32:09.000 --> 02:32:11.000
Association set.

02:32:11.000 --> 02:32:16.000
So that's how it works.

02:32:16.000 --> 02:32:20.000
Okay, so our proposed accountable word framework

02:32:20.000 --> 02:32:31.000
is a new approach to addressing these challenges by enabling the worlds to prove that they are less immersive across multiple dimensions.

02:32:31.000 --> 02:32:33.000
Including the word holder identity

02:32:33.000 --> 02:32:38.000
past conduct and the crypto asset provenance

02:32:38.000 --> 02:32:40.000
This document explores the

02:32:40.000 --> 02:32:44.000
technical implementation of these proofs.

02:32:44.000 --> 02:32:47.000
And we also discuss

02:32:47.000 --> 02:32:57.000
incentive structures to hold it and the privacy considerations as well.

02:32:57.000 --> 02:33:00.000
I think the this

02:33:00.000 --> 02:33:03.000
that's all for the presentation for now.

02:33:03.000 --> 02:33:12.000
lists are just about the appendices.

02:33:12.000 --> 02:33:23.000
I think let me stop here. The eyes are after here the audience's honest thoughts on this approach.

02:33:23.000 --> 02:33:34.000
If I could.

02:33:34.000 --> 02:33:41.000
If not, I can continue to explain my appendice sizes.

02:33:41.000 --> 02:33:44.000
As well?

02:33:44.000 --> 02:33:47.000
Okay, in that case, yeah.

02:33:47.000 --> 02:33:54.000
Michel, do you want to say something?

02:33:54.000 --> 02:33:56.000
Hey.

02:33:56.000 --> 02:34:01.000
No, you can keep going. I guess one of the questions that I would like

02:34:01.000 --> 02:34:03.000
us to discuss is

02:34:03.000 --> 02:34:06.000
Can you hear me? Sorry.

02:34:06.000 --> 02:34:07.000
Here.

02:34:07.000 --> 02:34:10.000
is from the previous

02:34:10.000 --> 02:34:16.000
times that we've spoken about this subject. I think one of the more

02:34:16.000 --> 02:34:20.000
I guess controversial and important things to discuss is this

02:34:20.000 --> 02:34:26.000
potential future where there's two different economies running and

02:34:26.000 --> 02:34:31.000
I had an interesting conversation this evening with someone from the industry.

02:34:31.000 --> 02:34:35.000
Who, when I spoke about

02:34:35.000 --> 02:34:40.000
there being an accountable wallet economy and there being a non-accountable wallet economy

02:34:40.000 --> 02:34:42.000
They simply just said, well.

02:34:42.000 --> 02:34:45.000
it's always been like that.

02:34:45.000 --> 02:34:51.000
So how is it different now that it's on a blockchain? There's always been dark markets, there's always been black economies.

02:34:51.000 --> 02:34:53.000
So what like

02:34:53.000 --> 02:34:57.000
what is different here?

02:34:57.000 --> 02:35:01.000
And I think that that's an interesting scope to take this conversation is

02:35:01.000 --> 02:35:03.000
how when we do get these

02:35:03.000 --> 02:35:07.000
separations between the accountable economy and the non

02:35:07.000 --> 02:35:11.000
accountable or like the fully decentralized permissionless economy

02:35:11.000 --> 02:35:14.000
I think what are the ramifications of that that

02:35:14.000 --> 02:35:18.000
may be different to what we've already seen in the world where

02:35:18.000 --> 02:35:20.000
There's inequality issues and

02:35:20.000 --> 02:35:23.000
Those sorts of things that come up when

02:35:23.000 --> 02:35:26.000
access to

02:35:26.000 --> 02:35:31.000
the ability to spend value becomes

02:35:31.000 --> 02:35:34.000
separated by…

02:35:34.000 --> 02:35:37.000
accountability.

02:35:37.000 --> 02:35:38.000
I hope that makes sense.

02:35:38.000 --> 02:35:41.000
Yes, thank you. Yeah, thank you, Michiel.

02:35:41.000 --> 02:35:49.000
So let me explain before the for those the food did not attend the last meeting, it's heard in tokyo.

02:35:49.000 --> 02:35:51.000
Last week, I guess.

02:35:51.000 --> 02:36:00.000
So there was some of the core developer of the Bitcoin Lightning Network. He pointed out that the

02:36:00.000 --> 02:36:05.000
there's some concern about the functionality of Bitcoin when we

02:36:05.000 --> 02:36:07.000
use the this

02:36:07.000 --> 02:36:10.000
the from this idea

02:36:10.000 --> 02:36:12.000
So…

02:36:12.000 --> 02:36:15.000

Yeah, I understand that our accountable word framework

02:36:15.000 --> 02:36:21.000
my potentially impact fungibility of the

02:36:21.000 --> 02:36:25.000
a bitcoin uh not only about bitcoin crypto assets

02:36:25.000 --> 02:36:30.000
I think it could be a leader

02:36:30.000 --> 02:36:32.000
Hi, Steve.

02:36:32.000 --> 02:36:41.000
price difference could make a price difference between the accountable wallets of the people who uses the accountable audits

02:36:41.000 --> 02:36:45.000
And to the

02:36:45.000 --> 02:36:52.000
that would create this different economy. One is that the economy uh

02:36:52.000 --> 02:36:56.000
who uses the accountable words and the other one is the uh

02:36:56.000 --> 02:37:01.000
economy who everyone doesn't use the accountable for it.

02:37:01.000 --> 02:37:03.000
So…

02:37:03.000 --> 02:37:09.000
Yeah, that concern is understandable.

02:37:09.000 --> 02:37:14.000
But I'm the our essential glory is that the not to create a divide

02:37:14.000 --> 02:37:16.000
I'm in the but

02:37:16.000 --> 02:37:18.000
That's not a purpose.

02:37:18.000 --> 02:37:22.000
That's just a result of the

02:37:22.000 --> 02:37:25.000
each investor's choices

02:37:25.000 --> 02:37:29.000
I guess. So…

02:37:29.000 --> 02:37:34.000
I guess that we need to provide these users the way that

02:37:34.000 --> 02:37:37.000
freedom choices

02:37:37.000 --> 02:37:43.000
to keep the idea compliance, I mean, that they wish to adhere to

02:37:43.000 --> 02:37:46.000
uh uh

02:37:46.000 --> 02:37:51.000
the route the wire engaging in the crypt transactions

02:37:51.000 --> 02:37:57.000
So…

02:37:57.000 --> 02:38:07.000
I don't know how to say, but I guess that the actual program is that the crypto users who desire to maintain the certain level of the

02:38:07.000 --> 02:38:12.000
compliance doesn't have a tours

02:38:12.000 --> 02:38:19.000
concrete tours for verifying the whether the crypto assets they uh they are accountabilities

02:38:19.000 --> 02:38:22.000
At the moment is

02:38:22.000 --> 02:38:26.000
coming from the legal sources or not.

02:38:26.000 --> 02:38:31.000
the moment so

02:38:31.000 --> 02:38:34.000
But the problem is that the uh

02:38:34.000 --> 02:38:38.000
Even when I say the uh

02:38:38.000 --> 02:38:40.000
many people

02:38:40.000 --> 02:38:46.000
users want to judge if their counterparty is legitimate or not but um the

02:38:46.000 --> 02:38:48.000
The listing receipt.

02:38:48.000 --> 02:38:53.000
may vary across the jurisdictions and the users.

02:38:53.000 --> 02:39:01.000
So therefore the compliance rules within this framework should be

02:39:01.000 --> 02:39:04.000
viewed as a model of the um

02:39:04.000 --> 02:39:08.000
personal choices, preferences.

02:39:08.000 --> 02:39:12.000
But the point is that the uh if

02:39:12.000 --> 02:39:17.000
that all users agree to exclude some addresses.

02:39:17.000 --> 02:39:22.000
dots are effectively become unusable.

02:39:22.000 --> 02:39:25.000
That's the consensus of the

02:39:25.000 --> 02:39:27.000
Compliance.

02:39:27.000 --> 02:39:31.000
And that's the core idea of our

02:39:31.000 --> 02:39:33.000
concept.

02:39:33.000 --> 02:39:38.000
So because that the um

02:39:38.000 --> 02:39:45.000
One of the significant challenges in combating the money laundering

02:39:45.000 --> 02:39:48.000
is that the, uh.

02:39:48.000 --> 02:39:53.000
inability to seize the crypt assets

02:39:53.000 --> 02:39:57.000
from the money launder or a terrorist.

02:39:57.000 --> 02:40:02.000
So even if we identify the

02:40:02.000 --> 02:40:04.000
Which wallets?

02:40:04.000 --> 02:40:08.000
I'll be hoarding the criminal process.

02:40:08.000 --> 02:40:13.000
So that's why our approach is focusing on the

02:40:13.000 --> 02:40:16.000
enabling the list limit to users to call

02:40:16.000 --> 02:40:24.000
to refuse the transaction with these addresses a bit prior to the transactions.

02:40:24.000 --> 02:40:26.000
And…

02:40:26.000 --> 02:40:35.000
Yeah, that's a cool idea.

02:40:35.000 --> 02:40:41.000
I really wanted to hear the opinion from someone against this idea.

02:40:41.000 --> 02:40:44.000
Yes, please.

02:40:44.000 --> 02:40:55.000
Because I'm huge.

02:40:55.000 --> 02:40:58.000
So are you considering to you know

02:40:58.000 --> 02:41:02.000
force users to use this type of

02:41:02.000 --> 02:41:04.000
Well, it's…

02:41:04.000 --> 02:41:06.000
like a user who lives in

02:41:06.000 --> 02:41:11.000
jurisdiction or do you just like let people use because

02:41:11.000 --> 02:41:18.000
people has a choice, probably there's no motivation for them to use it. So what's the motivation for that?

02:41:18.000 --> 02:41:20.000
Thank you very much.

02:41:20.000 --> 02:41:28.000
Actually, we are not going to force users to use this accountable warrants because, as I said.

02:41:28.000 --> 02:41:32.000
at the complying with some rules or

02:41:32.000 --> 02:41:38.000
compliance rule of the ruled by the jurisdiction is uh

02:41:38.000 --> 02:41:42.000
totally up to the users. I mean the uh

02:41:42.000 --> 02:41:49.000
Considering the borderless feature optical crypto, we cannot focus on

02:41:49.000 --> 02:41:53.000
We cannot make a rule for the rules for the

02:41:53.000 --> 02:41:57.000
our entire crypt world so

02:41:57.000 --> 02:42:01.000
So that's why we need to discuss about the

02:42:01.000 --> 02:42:04.000
Halloween.

02:42:04.000 --> 02:42:09.000
make users use these words i mean the

02:42:09.000 --> 02:42:16.000

to use the warrants.

02:42:16.000 --> 02:42:20.000
So many things in this paper about the

02:42:20.000 --> 02:42:22.000
how we can make an incentive for that

02:42:22.000 --> 02:42:25.000
For example, that the uh

02:42:25.000 --> 02:42:31.000
Maybe you've heard that there's some uh

02:42:31.000 --> 02:42:34.000
some system is called the uh

02:42:34.000 --> 02:42:36.000
Strawberry root protocol.

02:42:36.000 --> 02:42:40.000
Yeah, it has been already, yeah.

02:42:40.000 --> 02:42:46.000
applied to so many jurisdictions as yeah

02:42:46.000 --> 02:42:49.000
I'm sure the US has already

02:42:49.000 --> 02:42:51.000
Using the

02:42:51.000 --> 02:42:53.000
tribal protocols.

02:42:53.000 --> 02:42:58.000
and embed them the uh i prefer that the

02:42:58.000 --> 02:43:01.000
uh crypt uh

02:43:01.000 --> 02:43:04.000
Vaspus has already suffered.

02:43:04.000 --> 02:43:06.000
from the very

02:43:06.000 --> 02:43:11.000

very costly operation for that the to

02:43:11.000 --> 02:43:13.000
to use the

02:43:13.000 --> 02:43:15.000
Tropical protocols

02:43:15.000 --> 02:43:17.000
But I'm the if they

02:43:17.000 --> 02:43:22.000
If the user can their resumer seat just by

02:43:22.000 --> 02:43:27.000
presenting to some legitimacy proof to vaspus

02:43:27.000 --> 02:43:30.000
And then the bus can reduce their costs

02:43:30.000 --> 02:43:36.000
the maybe there could be a chance that the bus can uh

02:43:36.000 --> 02:43:38.000
more favorite

02:43:38.000 --> 02:43:44.000
more favorite uh transaction term to their customers

02:43:44.000 --> 02:43:47.000
who can prove their legitimacy.

02:43:47.000 --> 02:43:49.000
verifiable way.

02:43:49.000 --> 02:43:54.000
In that case, the dusp can reduce their costs and the

02:43:54.000 --> 02:44:01.000
That's why we would like to name this idea as a decentralized KYC idea.

02:44:01.000 --> 02:44:12.000
framework. So if like we can uh we can share the kyc uh verification system now decentralized way

02:44:12.000 --> 02:44:17.000
So, and in that case, it's it's more it's

02:44:17.000 --> 02:44:22.000
it's really similar to the existing finance. For example, the if

02:44:22.000 --> 02:44:28.000
You want to borrow some money from bank day. You need to prove that you are trusted person to the bank.

02:44:28.000 --> 02:44:34.000
So that you can borrow the money with a low interest rate

02:44:34.000 --> 02:44:41.000
So maybe by using the accountable word, you can prove to you at least mercy to others

02:44:41.000 --> 02:44:44.000
so that you can get more favorite

02:44:44.000 --> 02:44:46.000
transaction term uh with

02:44:46.000 --> 02:44:52.000
someone, not only about the P2P transaction, but also the

02:44:52.000 --> 02:44:55.000
DeFi protocol as well. Maybe DeFi protocol uh

02:44:55.000 --> 02:44:58.000
would give them the summit.

02:44:58.000 --> 02:45:00.000
better transaction term

02:45:00.000 --> 02:45:05.000
to someone who has the uh this emergency proof

02:45:05.000 --> 02:45:08.000
that's the that could be an incentive for using the

02:45:08.000 --> 02:45:11.000
accountable warrits.

02:45:11.000 --> 02:45:16.000
Okay, passing over to Mitchell before right

02:45:16.000 --> 02:45:17.000
Later, but, um.

02:45:17.000 --> 02:45:18.000
Yeah.

02:45:18.000 --> 02:45:22.000
So I wanted to add on this comment because like

02:45:22.000 --> 02:45:25.000
You know, as myself also mentioned.

02:45:25.000 --> 02:45:40.000
You know, crypto exchanges spend a lot of money and time for, you know, being compliant to the travel rule and also they needed to use blockchain analysis tools and they paid a lot of money to you know okay analysis and trm whatever.

02:45:40.000 --> 02:45:46.000
So I think they have a lot of incentive to implement this to be compliant, to

02:45:46.000 --> 02:45:49.000
The same thing, the same level of…

02:45:49.000 --> 02:45:52.000
like, you know, ML, CFT,

02:45:52.000 --> 02:45:58.000
like measures

02:45:58.000 --> 02:46:00.000
Through this. And…

02:46:00.000 --> 02:46:07.000
Yeah, so that is, I think, great motivation, could be a great motivation for crypto exchanges, as he mentioned but

02:46:07.000 --> 02:46:09.000
I think Mitchell raised

02:46:09.000 --> 02:46:12.000
his hand first so i think

02:46:12.000 --> 02:46:13.000
we can pass over to him and later

02:46:13.000 --> 02:46:15.000
Yeah.

02:46:15.000 --> 02:46:16.000
You can, yeah.

02:46:16.000 --> 02:46:20.000
I just had one comment on this line of thought and

02:46:20.000 --> 02:46:23.000
I think it's an important one.

02:46:23.000 --> 02:46:35.000
I thought one way we could describe the accountable wallet ecosystem is more that it's an incentivized selective disclosure ecosystem.

02:46:35.000 --> 02:46:39.000
So there's a bit of a difference between

02:46:39.000 --> 02:46:41.000
like a forced

02:46:41.000 --> 02:46:47.000
disclosure of information like the current KYC system works where

02:46:47.000 --> 02:46:50.000
your um like

02:46:50.000 --> 02:46:59.000
you have to provide a document in order to verify you are who you say you are in order to access particular financial products, whether that be crypto or whatever.

02:46:59.000 --> 02:47:02.000
However, this system

02:47:02.000 --> 02:47:05.000
draws upon

02:47:05.000 --> 02:47:14.000
the idea that, well, in the future everything will move towards being more permissionless, access to financial systems should

02:47:14.000 --> 02:47:23.000
be available to the masses, anyone who has access to the internet in theory. So what we're trying to build here is

02:47:23.000 --> 02:47:26.000
the inverse to a

02:47:26.000 --> 02:47:32.000
permission system. It's more of an incentivized selective disclosure system. Now, the only way to

02:47:32.000 --> 02:47:37.000
actually make that work is you have to really focus on the incentives

02:47:37.000 --> 02:47:41.000
And I think that's part of the problem that we run into when we discuss this research.

02:47:41.000 --> 02:47:46.000
is that people struggle to find the incentives here.

02:47:46.000 --> 02:47:50.000
Takaya mentioned one incentive just now where

02:47:50.000 --> 02:47:53.000
there could be an incentive for the

02:47:53.000 --> 02:48:07.000
analytics companies in order to build a system like this. So then sharing of data is easier. There is some level of standardization around the wallets that are able to interact with this system. I think that that's

02:48:07.000 --> 02:48:12.000
While it is an incentive, I think it's a difficult incentive to work off because

02:48:12.000 --> 02:48:16.000
the on-chain analytics companies

02:48:16.000 --> 02:48:19.000
make money from their private data sets.

02:48:19.000 --> 02:48:22.000
And this in theory is making

02:48:22.000 --> 02:48:24.000
data sets more available.

02:48:24.000 --> 02:48:33.000
for the accountable ecosystem. So I think that might be one difficult thing as it grows

02:48:33.000 --> 02:48:37.000
One thing that I was thinking, well.

02:48:37.000 --> 02:48:41.000
Sato was talking was

02:48:41.000 --> 02:48:48.000
it's almost like you need to get to a critical mass of people being willing to disclose

02:48:48.000 --> 02:48:53.000
that they are a part of this accountable association set or wallet ecosystem.

02:48:53.000 --> 02:48:55.000

for that to be great enough.

02:48:55.000 --> 02:48:57.000
For the privacy benefits.

02:48:57.000 --> 02:49:01.000
of being a part of this accountable ecosystem to start to

02:49:01.000 --> 02:49:07.000
emerge. So if in theory you actually get better privacy

02:49:07.000 --> 02:49:14.000
by providing selective disclosure and being a part of a wallet ecosystem where you know that the wallets that you use

02:49:14.000 --> 02:49:17.000
have this level of accountability in place.

02:49:17.000 --> 02:49:21.000
That then starts to become an incentive.

02:49:21.000 --> 02:49:26.000
But there's a tipping point there. Up until that tipping point.

02:49:26.000 --> 02:49:29.000
building an accountable wallet

02:49:29.000 --> 02:49:43.000
is actually just a full disclosure that I'm regulatory compliant and I've provided my documentation, etc. So you don't get any as like you don't get privacy in that sense until it gets to that

02:49:43.000 --> 02:49:48.000
that level where there's an ability to be synonymous again.

02:49:48.000 --> 02:49:51.000
So there are just a couple ideas that I wanted to throw out there.

02:49:51.000 --> 02:49:54.000
that um

02:49:54.000 --> 02:50:00.000
we might not be able to design an accountable wallet ecosystem

02:50:00.000 --> 02:50:08.000
without allowing for the incentives to design it in of itself

02:50:08.000 --> 02:50:13.000

If that makes sense. So allow for people to find the incentives of

02:50:13.000 --> 02:50:17.000
the selective disclosures that they do in order to access opportunities.

02:50:17.000 --> 02:50:28.000
One thing that Masato said, which I found interesting as one of the incentives that could result is you could get some sort of loyalty discount or preferential treatment.

02:50:28.000 --> 02:50:31.000
In any given economy.

02:50:31.000 --> 02:50:35.000
If you provide a level of accountability or selective disclosure to

02:50:35.000 --> 02:50:41.000
your identity and that you are not a part of like

02:50:41.000 --> 02:50:43.000
a flow of funds which could be

02:50:43.000 --> 02:50:45.000
elicit so

02:50:45.000 --> 02:50:51.000
Yeah, they're just a few ideas I wanted to throw out there and sort of answer.

02:50:51.000 --> 02:51:00.000
some of the research that we've been doing and the direction that we could take.

02:51:00.000 --> 02:51:03.000
Yeah, I…

02:51:03.000 --> 02:51:05.000
Seems to me that that

02:51:05.000 --> 02:51:08.000
your sort of foundational starting point is

02:51:08.000 --> 02:51:11.000
the traditional financial system

02:51:11.000 --> 02:51:15.000
access to fiat and

02:51:15.000 --> 02:51:21.000
Fiat on-ramps and off-ramps are fundamentally going to be the thing that is most regulated and then

02:51:21.000 --> 02:51:32.000
the next sort of step, and this is where the travel rule comes in, is in reality, the majority of the people dealing with crypto today are dealing with it through centralized exchanges.

02:51:32.000 --> 02:51:36.000
And there's a process right now by which those exchanges are

02:51:36.000 --> 02:51:40.000
coming within the regulatory ambit more and more so

02:51:40.000 --> 02:51:50.000
But as you start to think about the application of the travel rule as it relates to exchanges, the proposals essentially say an exchange can't

02:51:50.000 --> 02:51:52.000
and can't allow

02:51:52.000 --> 02:51:55.000
for one of his accounts to be used transferred to or from

02:51:55.000 --> 02:51:57.000
an unhosted wallet.

02:51:57.000 --> 02:51:59.000
Unless they have

02:51:59.000 --> 02:52:03.000
identified the owner of that wallet. And that fundamentally

02:52:03.000 --> 02:52:09.000
If that rule in its most extreme form were to go through, that would essentially wall off

02:52:09.000 --> 02:52:12.000
anything in the way of unhosted wallets

02:52:12.000 --> 02:52:15.000
from the rest of the financial system.

02:52:15.000 --> 02:52:21.000
from the rest of the economy, because you'd not be able to move value between those two worlds.

02:52:21.000 --> 02:52:28.000
And what this proposal does, what the privacy pool proposal does, and when we start thinking this through.

02:52:28.000 --> 02:52:31.000

in combination with

02:52:31.000 --> 02:52:34.000
a different form of KYC.

02:52:34.000 --> 02:52:37.000
is that if you could have a single

02:52:37.000 --> 02:52:42.000
regulated entity or KYC a wallet.

02:52:42.000 --> 02:52:45.000
provide you with the certificate you hold in that wallet.

02:52:45.000 --> 02:52:50.000
that if that becomes a passport into the rest of the financial system.

02:52:50.000 --> 02:52:57.000
So that you're not disclosing your PII to every financial institution you ever touch.

02:52:57.000 --> 02:53:06.000
It's a compromise in terms of not having your private information spread all over the place, but have it be have one touch point into the system.

02:53:06.000 --> 02:53:09.000
And then allow this kind of accountable wallet

02:53:09.000 --> 02:53:12.000
spread the

02:53:12.000 --> 02:53:42.000
of an accountable financial system.

02:53:51.000 --> 02:53:55.000
mainstream economy.

02:53:55.000 --> 02:53:58.000
stream economy more inclusive

02:53:58.000 --> 02:54:03.000
to allow for more participants in that economy than the traditional financial system

02:54:03.000 --> 02:54:12.000
this ossified sort of financial system has allowed.

02:54:12.000 --> 02:54:14.000
Yes, thank you very much.

02:54:14.000 --> 02:54:19.000
So if they are the

02:54:19.000 --> 02:54:22.000
inclusive than the non-countable economy

02:54:22.000 --> 02:54:26.000
Which means that the

02:54:26.000 --> 02:54:32.000
the crypto that that helps in uh that that is hoarding the accountable economy

02:54:32.000 --> 02:54:37.000
will be more demanded than the

02:54:37.000 --> 02:54:42.000
a non-accountable economy, which means that the market price of the crypto

02:54:42.000 --> 02:54:47.000
in the accountable economy is slightly higher than the non-accountable economy.

02:54:47.000 --> 02:54:52.000
Which would make some

02:54:52.000 --> 02:54:56.000
incentive to be in the accountable economy.

02:54:56.000 --> 02:55:01.000
For example, if the youth make a transaction with it uh if

02:55:01.000 --> 02:55:24.000
If you are regarded in the accountable economy at the moment, but if you make a deal with someone in unknown accountable economy, which could lead to the you will be no accountable for it. In that case, if you will be regarded as a non-accountable, no one accountable, which means that your

02:55:24.000 --> 02:55:27.000
you have crypto in the market price will be

02:55:27.000 --> 02:55:29.000
decreased yeah that's

02:55:29.000 --> 02:55:35.000
that could be a disincentive to transact with someone

02:55:35.000 --> 02:55:46.000
who cannot approve the UL is mercy.

02:55:46.000 --> 02:55:52.000
that MD, Isaac, I explained this mechanism to this someone

02:55:52.000 --> 02:55:56.000
from the core developer of the bitcoin they uh

02:55:56.000 --> 02:56:00.000
is really afraid of this idea because that could

02:56:00.000 --> 02:56:05.000
really some non-fungible situation on the

02:56:05.000 --> 02:56:11.000
Bitcoin economy. So I could read that we would make some

02:56:11.000 --> 02:56:15.000
good bitcoin bought bitcoin so bottom

02:56:15.000 --> 02:56:22.000
Yeah, maybe I would avoid to use that word, but…

02:56:22.000 --> 02:56:27.000
Yeah, that's the problem.

02:56:27.000 --> 02:56:30.000
Okay, my…

02:56:30.000 --> 02:56:33.000
After hearing everything.

02:56:33.000 --> 02:56:36.000
for this proposal, my interpretation.

02:56:36.000 --> 02:56:41.000
is okay. So with Bitcoin or is the same

02:56:41.000 --> 02:56:43.000
maximalist so they

02:56:43.000 --> 02:56:46.000
would like to make the things

02:56:46.000 --> 02:56:49.000
extremely fungible.

02:56:49.000 --> 02:56:51.000
That's number one.

02:56:51.000 --> 02:56:54.000
part of a policeman.

02:56:54.000 --> 02:56:56.000
need to trace everything.

02:56:56.000 --> 02:57:03.000
That number two. And so the proposal by Vitalik last year was also that using

02:57:03.000 --> 02:57:06.000
privacy and association set is

02:57:06.000 --> 02:57:08.000
the kind of that pronouncing

02:57:08.000 --> 02:57:12.000
both privacy and fungibility

02:57:12.000 --> 02:57:14.000
And, you know, pricing something.

02:57:14.000 --> 02:57:17.000
That's fine, but as Masato said that

02:57:17.000 --> 02:57:22.000
Bitharic proposed a kind of protocol.

02:57:22.000 --> 02:57:27.000
But the major part, the important part was missing.

02:57:27.000 --> 02:57:31.000
that how we decide that association set is so

02:57:31.000 --> 02:57:34.000
So this proposal is complement to that battery

02:57:34.000 --> 02:57:36.000
proposal. That's number one.

02:57:36.000 --> 02:57:38.000
the

02:57:38.000 --> 02:57:43.000
Number two is that, you know, so this proposal says that distinguishing

02:57:43.000 --> 02:57:45.000
accountable economy endeavor.

02:57:45.000 --> 02:57:47.000
no accountable economy.

02:57:47.000 --> 02:57:52.000
I'm not sure that that's a good expression about that.

02:57:52.000 --> 02:57:55.000
that things happen but

02:57:55.000 --> 02:57:59.000
of…

02:57:59.000 --> 02:58:01.000
is…

02:58:01.000 --> 02:58:03.000
Zelle.

02:58:03.000 --> 02:58:06.000
They're also money laundering.

02:58:06.000 --> 02:58:10.000
I don't think it's adobo, but I don't think it's

02:58:10.000 --> 02:58:14.000
So ultimately.

02:58:14.000 --> 02:58:17.000
Surprisingly.

02:58:17.000 --> 02:58:22.000
transactions.

02:58:22.000 --> 02:58:26.000
I don't think that part of who other guys say that kind of statement

02:58:26.000 --> 02:58:29.000
But, you know, you know.

02:58:29.000 --> 02:58:33.000
non-accountable economic, you know.

02:58:33.000 --> 02:58:35.000
you know.

02:58:35.000 --> 02:58:38.000
exists already.

02:58:38.000 --> 02:58:41.000
So we cannot destroy that kind of thing.

02:58:41.000 --> 02:58:43.000
practically but

02:58:43.000 --> 02:58:49.000
so if you know crypt you know you know cryptocurrency or crypt assets or

02:58:49.000 --> 02:58:51.000
rotting economy.

02:58:51.000 --> 02:58:55.000
do not increase that size of

02:58:55.000 --> 02:59:01.000
and no accountable no accountable economy size of no accountable economy

02:59:01.000 --> 02:59:05.000
You know, blockchain technology.

02:59:05.000 --> 02:59:08.000
don't do busing.

02:59:08.000 --> 02:59:10.000
for the society.

02:59:10.000 --> 02:59:15.000
So if this is a goal, so this proposal on that

02:59:15.000 --> 02:59:19.000
that no cost of their worth, you know, custody warrant or, you know.

02:59:19.000 --> 02:59:21.000
work in the cryptocurrency exchange.

02:59:21.000 --> 02:59:27.000
are already regulated by using tribal protocol. But, you know, unhosted war is out

02:59:27.000 --> 02:59:29.000
out of scope of this.

02:59:29.000 --> 02:59:33.000
So, you know, this…

02:59:33.000 --> 02:59:40.000
proposal is a kind of idea to incentivize people to not to receive money from

02:59:40.000 --> 02:59:42.000
some, you know.

02:59:42.000 --> 02:59:44.000
I'm worth with bad reputation.

02:59:44.000 --> 02:59:49.000
And so repetition is created by a decentralized way

02:59:49.000 --> 02:59:55.000
So I don't know about that so decentral way because

02:59:55.000 --> 02:59:59.000
You know, the negativist or sanctionist

02:59:59.000 --> 03:00:03.000
are created by some trusted party.

03:00:03.000 --> 03:00:05.000
about the you know that this four

03:00:05.000 --> 03:00:08.000
this proposal that

03:00:08.000 --> 03:00:12.000
That's kind of association is created by so truly decentralized way

03:00:12.000 --> 03:00:17.000
So, you know, so a large number, a set of large number of people

03:00:17.000 --> 03:00:20.000
We will create some kind of that so you know

03:00:20.000 --> 03:00:24.000
not score, but it's some kind of repetition.

03:00:24.000 --> 03:00:27.000
And they…

03:00:27.000 --> 03:00:32.000
Okay, that is a, you know, Bitcoin guys or Instagram guys want to make it

03:00:32.000 --> 03:00:34.000
extremely fungible.

03:00:34.000 --> 03:00:37.000
with a claim and they claim

03:00:37.000 --> 03:00:40.000
They hate the idea of the colored coins.

03:00:40.000 --> 03:00:44.000
to add some color for each coin.

03:00:44.000 --> 03:00:46.000
But so this proposal is

03:00:46.000 --> 03:00:50.000
not for the coin. It's a proposal for Wallet.

03:00:50.000 --> 03:00:53.000
So my interpretation

03:00:53.000 --> 03:00:55.000
If I…

03:00:55.000 --> 03:00:58.000
add the subtitle to this protocol.

03:00:58.000 --> 03:01:01.000
This is a decentralized

03:01:01.000 --> 03:01:03.000
the vocational system forward.

03:01:03.000 --> 03:01:06.000
decentralized duplication of the world.

03:01:06.000 --> 03:01:09.000
you know that some you know decentralized some

03:01:09.000 --> 03:01:13.000
reputation information is added to each wallet.

03:01:13.000 --> 03:01:22.000
And it's my decision not to receive some coins or not to receive some coin from that wallet.
It's my decision.

03:01:22.000 --> 03:01:25.000
It's not enforcement from someone.

03:01:25.000 --> 03:01:28.000
So we can refer that's some repetition.

03:01:28.000 --> 03:01:31.000
score or the potential information.

03:01:31.000 --> 03:01:33.000
given by that some large numbers.

03:01:33.000 --> 03:01:36.000
people to decide.

03:01:36.000 --> 03:01:40.000
to make my decision to receive or not receive it.

03:01:40.000 --> 03:01:43.000
This is some kind of decentral relocation system.

03:01:43.000 --> 03:01:52.000
This is my interpretation.

03:01:52.000 --> 03:01:55.000
I don't know that that experiential makes sense for you.

03:01:55.000 --> 03:02:02.000
Yeah, I guess the question I would have on that is what are the practical consequences of a wallet having

03:02:02.000 --> 03:02:11.000
a good reputation or a bad reputation that isn't a function of being, say, KYC'd by a regulated entity. If it's just some sort of

03:02:11.000 --> 03:02:14.000
reputational score that's the product of

03:02:14.000 --> 03:02:26.000
You know, I'm in a particular country or I'm able to do data mining to figure out what kind of transactions that wallet does. What's the

03:02:26.000 --> 03:02:30.000
What do we think is the sort of the practical

03:02:30.000 --> 03:02:33.000
Why would I?

03:02:33.000 --> 03:02:36.000
as a hold herbal wallet choose

03:02:36.000 --> 03:02:50.000
to only do business with wallets with high reputation scores. One might be that I think there's less likely a fraud of being cheated. But frankly, a permissionless system like the blockchain is designed so that I don't have to trust you.

03:02:50.000 --> 03:03:00.000
I don't care who you are. I don't have to trust you. So that issue of trust we have in sort of reputation systems on the internet is not relevant really on the blockchain.

03:03:00.000 --> 03:03:09.000
Why would I care whether your wallet has a good reputation or a bad reputation? Why? Because if your wallet has a bad reputation.

03:03:09.000 --> 03:03:17.000
then I run the risk of myself being excluded from the rest of the financial system.

03:03:17.000 --> 03:03:22.000
The point is if regulators may come up with a set of rules to say we're going to draw a

03:03:22.000 --> 03:03:24.000
a line around

03:03:24.000 --> 03:03:28.000
the legitimate financial system

03:03:28.000 --> 03:03:36.000
We'd like them to draw that line as wide, as encompassing as possible so that we make the system as inclusive as possible.

03:03:36.000 --> 03:03:40.000
But that means whatever system you come up with.

03:03:40.000 --> 03:03:46.000
for identifying wallets that have good reputation versus bad reputation.

03:03:46.000 --> 03:03:48.000
I think has to be appealing

03:03:48.000 --> 03:03:54.000
to regulators so that they'll base their regulatory system around it. Otherwise.

03:03:54.000 --> 03:03:59.000
I don't see what, as an individual consumer or user of the system

03:03:59.000 --> 03:04:01.000
All I really care about is

03:04:01.000 --> 03:04:04.000
am I going to have access to be able to transfer value

03:04:04.000 --> 03:04:07.000
to the rest of the financial system.

03:04:07.000 --> 03:04:16.000
or not. That's my primary concern.

03:04:16.000 --> 03:04:22.000
Yeah, thank you for that. Yeah, that could be an incentive to maintain your reputation

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above some certain degree of the criteria.

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for example the uh

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In our paper that we mentioned that the reputation of the BASPUS warrant

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should be disclosed on chain.

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Yeah, it's not unrealistic assumption because that the uh

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In reality, the odds are

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their wallets is already identified on the blockchain explorer

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So in that case the uh

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it would be a situation that their reputation is disclosed on publicly.

03:05:00.000 --> 03:05:07.000
So on-chain data. So in that case, the VASP has could be an incentive

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to maintain their reputation higher because that the if their reputation will go down the uh

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their user may have concern about that VASP could be

03:05:21.000 --> 03:05:24.000
uh sewer.

03:05:24.000 --> 03:05:27.000
sanctioned by some regulators.

03:05:27.000 --> 03:05:31.000
So in that case, the VASP has incentive

03:05:31.000 --> 03:05:34.000
to maintain their reputation higher.

03:05:34.000 --> 03:05:37.000
So which means that the wasp will

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we apply this some

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screening rules to accept which wallet

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can deposit their wallet, their crypto into that fast

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Because that if the

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the BASP would accept a crypto from some external wallets, which has the low reputation the

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the reputation of the vast power will go down. So maybe there will be a situation that the bus will refuse to receive the crypto from the

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external wallet, which has the raw reputation.

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So in that case, which means that the uh

03:06:20.000 --> 03:06:32.000
crypto from the external wallet to user perspective the uh they will have the incentive to maintain higher reputation as well, to use the busp

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Otherwise, they cannot use their bus anymore.

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That's how it works, I hope.

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Just one comment on that in terms of the design space for this reputation system is I think you'd want to be

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careful that it's not

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such a wide open design space that people could design

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exclusionary mechanisms to wall people out of the financial system for reasons that have nothing to do with law enforcement or what have you. I mean, if people just decide certain type of people we don't want to have be part of the financial system.

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And you now have financial service providers who

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are using these mechanisms to only do business with people they like, you know, in ways that are

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really…

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you know, for the purposes of this is to have a financial system that's more inclusive

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You have to be careful, I think, that you can't misapply these systems

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to intentionally exclude people from the financial system.

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So having a set of criteria that are

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sort of acceptable in a broader social sense of we want to exclude you know

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criminals and bad actors from the system. But other than that.

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We ought to let everybody participate, I think is a general principle that that

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ought to be part of the design.

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Go for it, Mitchell.

03:08:08.000 --> 03:08:11.000
Oh, that's a familiar voice.

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Yeah, so I just want to

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sort of second to the point of whoever was speaking in the room. I think that's a really

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important um

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concept to maintain when building these sorts of systems is that it shouldn't be an exclusionary set. It should be an inclusionary set.

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One thing that I was thinking about when we were discussing practical outcomes, and I think that for this to be technologically

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available we need to

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create some level of account abstraction and multiple

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addresses within one wallet which is why

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the accountable wallet system, I think shouldn't just be seen as one address it should be seen as like an encompassing reputation or identity as a whole

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we'll make it

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far more practical for us to go down that path rather than

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just converting a string of a public key into a reputation and then having lots and lots of those, I think it should be instead.

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a set that we possess which have different vectors and different inferences.

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But one of the ideas that I had, and I just wrote down is one thing that a blockchain

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doesn't, I guess, do well in comparison to a traditional financial system is that you can't choose what funds you receive, right?

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Like if you've got a public key and someone wants to send me whatever meme or scam coin, they can as long as they know my public address, they could litter the entire blockchain. And there's a thing called dusting where that happens regularly, where if you have a certain amount of funds in your address, you'll receive a lot of new tokens on that address and if you ever try to move those tokens out, clean your address, etc.

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Some of them could have nefarious contracts involved that you provided permission to spend your USDC, for example. You didn't know what you were signing and you lose your wallet gets drained. I think that that's one of

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the big issues within the industry that we probably don't talk about enough because

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people in the industry are the victims of that and we're meant to be sophisticated enough with our OPSEC, but that could be an incentive where the accountable wallet ecosystem actually really comes to light. And this is where I said account abstraction will be important.

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Because you would need some level of layering to the wallet right like

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I would have one public key that could accept anything and everything. However, then I would have a filter and for that wallet to then send to my

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more heightened reputation, securitized address

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that doesn't deal with any other

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wallet or smart contract, it only deals with the accountable wallet ecosystem smart contracts that I've approved.

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That would then funnel

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the accounts receivable that I get and I could know that they're not going to bite me in the back later if I sign a transaction to move those funds to wherever I would like them to be. So some sort of

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layered system internally for the individual to create

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security could be that practical

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regulatory outcome of safety and soundness that the accountable wallet ecosystem enables.

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So people who use these wallets who have this account abstraction layer tool in place where

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one wallet is used as almost like a

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a gate to then provide access to my more securitized accountable wallet which

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signs up to these principles that the regulator has stipulated that filters out certain smart contracts that are nefarious and only permissions other smart contracts.

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could actually be a way to get this system off the ground in a positive incentive manner rather than a exclusionary incentive manner?

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But I don't know if that is fully technically possible.

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However, I think that

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it could be something that we think about when

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Yeah, discussing this research.

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Per second.

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To the gentleman, you mentioned around exclusions. How do you avoid having a system that is able to have a level of adverse selection scores so that you don't have ever in anybody joining

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But then also we don't want to kind of lead towards exclusions

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But then when we have like things that are identifiers that may be inferred in two different senses, right? Let's say someone's 18 is old.

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That may be good, that may be bad. Let's say someone is living in a certain address locations, maybe good, maybe bad, depending on the context, the person that's evaluating that.

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how do we create identifiers that

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you know neutral, non-invasive, but equally usable

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How do you do that and what kind of examples do you have of that?

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examples of identifiers you'd use?

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there'd be like offensive, neutral.

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Well, I guess the fundamental question is in designing this sort of a system

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what sorts of purposes are you trying to achieve through the mechanism? And if what you're really saying fundamentally is

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All we want to do is exclude bad actors.

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then I think the number of excluded, the sort of parameters you're allowed to use would be very limited, right? And I think that's where the idea you say, okay, inclusion in the system is I have

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identified myself to some sort of regulated entity so that and now I'm in

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I'm either in or out, right and it's in or out

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you know, my age, my race, my country, none of that's relevant. It's just

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Am I… Am I…

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identified if I KYC'd myself.

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That would be a very simplistic

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approach that may not be skeptical, all that acceptable to those who are sort of still trying to maintain their privacy and not be KYC'd to a financial institution. But that's one in which

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He'd simply say that's the only type of criteria we're going to allow.

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But I think some of the use cases that were identified in a moment ago are very compelling for ones where you'd say, no, I'd like to have a wallet that simply that will not do business with

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anything other than my other wallet so that I have a way to use my other wallet as a filter, right? And say, I'm not going to bring coins into my wallet unless I

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personally put them there.

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That's a great use case for this sort of a technology.

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Right. But then are there in between cases where you only want to accept coins from people that are part of some sort of organization that you're part of, but now does that become a nefarious or impermissible use that is exclusionary? And so I think it's going to be very, very difficult to draw those lines.

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I'm speaking.

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Thank you very much. I think I've already spent all of my time. So maybe…

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I need to pass the mic to the

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Me too.

03:15:35.000 --> 03:15:36.000
Thank you all.

03:15:36.000 --> 03:15:40.000
Sure.

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Thank you.

03:15:43.000 --> 03:15:45.000
I will share my screen.

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I think.

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So I'll make this fairly…

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Great. Let me just go into

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slideshow mode. I think we can see this.

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So I guess in a related subject, a lot of the accountable wallet research

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almost predicated the need for this conversation.

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that we notice in the IKP working group.

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However, over the last 12 months we've been

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Creating a paper

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or a discussion point in the working group around

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Wallet governance, but more the sort of typology and the fundamentals of what we actually mean.

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by wallet governance and

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sort of key management so

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I'll just do a quick run through of the concept behind the paper, but I would really like to sort of open the discussion and

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See if there were interest in

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interesting contributions from the group today that we can

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provide and then bring to the block 11 meeting in fact in in

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Next week.

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a lot of the premise of the wallet governance paper is

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to create, I guess, clear terminology and guidelines for a multi-stakeholder conversation to understand. So something that we

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experienced a lot over the last um

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I think it's almost two years of discussions in Begin is when we do talk about this term wallet

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What does that mean like it's it's no longer like

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the pace of leather in my pocket that holds some paper money and then some cards and some government identification at one stage.

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it now is this sort of

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amalgamation of a whole bunch of different things and

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One other thing that

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I would like to bring up in particular is this convergence of

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wallet meaning both

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non-financial and financial applications

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As well as this spin of governance on top and

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That's a question to the group is.

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Are they the three core categories that wallets really

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provide utility to us in society? Is it always going to be a form of

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Identity government uh identity

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finance or governance.

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is the merging of these two coming together where

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governance tokens have value, therefore it's finance and governance and identity tokens provide access to financial systems therefore it's some sort of merging of the two or opportunities, for example.

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Previously, it's been easy to define that because in the meat space, in the physical world, we have really clear indicators of what is

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money and what is identity like it's it's either a person or it's a

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piece of paper in the wallet or a number on an account. But with blockchain systems and sort of digital economies and tokenized ecosystems like this this is getting increasingly blurred.

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So what this paper tries to do is provide some definitions and provide some insight into

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the different technologies that enable wallets.

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Which we can then use as a rubric for identifying the best use cases.

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for when wallets get used.

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For example, in a staking governance scenario, what is the best type of

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wallet technology to use is a multi-party computation wallet. So then we can have multiple layers of authentication and authorization for accessing any

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buns that are earned through staking

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or in a governance scenario, should it be a multi-seq wallet so then we can have multiple people as a part of an organization voting under one

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conglomerate or conglomerate

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How does that become possible so

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How do we reach a shared understanding with governance with regards to wallets?

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And then this also really blends into the KYC AML.

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side of things, especially with the context of the accountable wallet conversation that we just had.

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around it potentially being a vector or a layer for regulation and compliance if the wallet is that

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layer that becomes the focus of regulation

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how do we manage that? There's going to be wallets which are obviously non-compliant, emerging wallets, these sorts of things.

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So I kind of

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already have spoken about this slide because I just got ahead of myself, but we've noticed the convergence between the three, the financial governance and the identity and the call to participation

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is to try and receive as many unique perspectives on what they believe a wallet is to them.

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now and in the future.

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One other sort of fun angle and spin I want to add to this is

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there are going to be wallets that we permission

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However, we never use

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yet function on our behalf in

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when we see the advent of AI

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AI is going to be using blockchain-based wallets because it can't really

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Sign up to a bank account but it can

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manage a public private key pretty easily.

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what does that mean for these sorts of wallet governance systems and what we classify them as? Is there going to be a whole new category of

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wallets that get used by robots versus wallets that get used by people.

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and humans.

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So this is kind of a high level overview of

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part of the paper

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It's a fairly long paper now. It's around sort of 50 pages that goes through various different technologies to the types of wallets to explain and then the part that we're working on at the moment is these sort of use cases and

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identifying what the best use cases are so

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just really sort of high level because I'm more interested in the feedback from the

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the room, these are the primary categories. So custodial wallets, non-custodial hardware wallets

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multi-seqs and smart contract wallets that we discuss.

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And then I would like, I guess.

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If you have any interesting

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types of wallets that were used for a use case that

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hasn't really been listed here or is unique.

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To bring that up so then we can put it in the research.

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And then finally, the question here, how will reputation on chain impact the financial use cases of these wallets is an allusion to the accountable wallet.

03:22:39.000 --> 03:22:42.000
conversation that we just had.

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One thing that came up in the Tokyo

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begin layer two meetup, which I found quite interesting is that there was um

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like a pushback against wallets for governance as a category in of itself.

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A lot of the people in the room believed that it was binary. It was just either non-financial or financial use cases.

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for wallets and there wasn't this middle ground

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where you could potentially be using financial instruments, but for governance.

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use cases like Treasury, DAO governance, protocol governance, like these are moving funds in places in order to

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provide voting rights in particular systems.

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Which they're not spending the money that they're using that as leverage in

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like a plutocratic sense to vote on a blockchain network and a protocol and its path and future.

03:23:38.000 --> 03:23:46.000
Particularly in staking to contribute to the network. So should there be this third category of wallets for governance.

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Which then diverges away and

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could potentially create even more innovation in the wallet space when we get that specialization.

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around wallets which are fit for purpose for

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protocol governance, DAO governance, Oracle-based governance

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And then we can get potentially technologies in place that

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that really optimize for those outcomes.

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In the same way that, say, for example, an NPC, while it is optimized for an institutional

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Trading use case of

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cryptocurrency exchanges. And then finally in

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I think most importantly, while it's for identity, I think is the primary use case that we

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discuss and explore in the IKP working group more and more the sort of longer I've been on this planet, I've realized that identity is

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kind of like the thing, the social fabric

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That allows for us to interact and be a part of.

03:24:51.000 --> 03:24:55.000
society. So when we start to prescribe

03:24:55.000 --> 03:24:58.000
these things to wallets which are

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digital we should

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definitely think about the best type of wallet, the types of security, the types of technologies that we use in order to either

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secure or share.

03:25:13.000 --> 03:25:16.000
ourselves with the world.

03:25:16.000 --> 03:25:21.000
In the way that we choose, which is that privacy spin on identity.

03:25:21.000 --> 03:25:28.000
So our wallets for identity better left to focus only on identity or would

03:25:28.000 --> 03:25:33.000
adding that extra functionality, which we've seen with blockchain of being able to

03:25:33.000 --> 03:25:36.000
have NFTs in my wallet, which are a profile picture

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As well as you can see my DeFi trading at the same time.

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is that

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a positive

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or uh yeah nuanced conversation there.

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And then finally, the discussion piece of this paper is the wallet convergence piece.

03:25:55.000 --> 03:26:03.000
So I've been alluding to it throughout this whole presentation. But what is the ramifications and what are the ramifications of the

03:26:03.000 --> 03:26:09.000
convergence between financial and non-financial applications of wallets. Are they positive, negative?

03:26:09.000 --> 03:26:19.000
What does this mean for AI when they start needing more and more data? Is there a way that we can use wallet technology as a median

03:26:19.000 --> 03:26:22.000
for enabling better privacy, data dignity.

03:26:22.000 --> 03:26:25.000
when ai start

03:26:25.000 --> 03:26:28.000
making requests and sending me um

03:26:28.000 --> 03:26:31.000
information via this sort of public

03:26:31.000 --> 03:26:33.000
private key cryptography layout.

03:26:33.000 --> 03:26:36.000
that's building up.

03:26:36.000 --> 03:27:01.000
So I think I may actually leave it there for my presentation because it's better short and sweet and to get into the discussion. And then we get into the accountable wallet. But yeah, thank you.

03:27:01.000 --> 03:27:04.000
Awesome. Always a pleasure.

03:27:04.000 --> 03:27:15.000
So first off, mate, yeah, great presentation, great questions. If you go back a few slides, I'd really appreciate that.

03:27:15.000 --> 03:27:24.000
Because you had questions per slide, and I thought there was a few questions that we should definitely discuss.

03:27:24.000 --> 03:27:25.000
I don't know if someone else.

03:27:25.000 --> 03:27:27.000
Did you want to speak about the financial?

03:27:27.000 --> 03:27:28.000
Questions. Sorry, I…

03:27:28.000 --> 03:27:35.000

Oh, just going back a few of the slides to back further

03:27:35.000 --> 03:27:38.000
Oh, the…

03:27:38.000 --> 03:27:40.000
I think the Zoom is lagged a little bit.

03:27:40.000 --> 03:27:42.000
I was scrolling.

03:27:42.000 --> 03:27:46.000
While you were speaking, like our voice is coming.

03:27:46.000 --> 03:27:47.000
Three, before.

03:27:47.000 --> 03:27:50.000
A little bit of a lag.

03:27:50.000 --> 03:27:57.000
One of the truest truths that you mentioned before was around how AI won't be able to bank count, but it can definitely interact crypto.

03:27:57.000 --> 03:28:01.000
And I've never thought about that until just then. And that's, I think that's probably something you should put

03:28:01.000 --> 03:28:06.000
big bold letters, you know, 50 fonts on the front page because

03:28:06.000 --> 03:28:19.000
One of the questions I get asked a lot of times in working decentralized AI is like, well, what's the biggest use case? And I can go through a bunch of different use cases, right? But it always comes down to leveraging either data security through blockchain or having it as a kind of financial communication layer.

03:28:19.000 --> 03:28:21.000
And it's like, well, why is that useful?

03:28:21.000 --> 03:28:35.000
We look at the future of artificial intelligence, especially the next kind of three years when we see the explosion around AI agents, AI agent interactions, and then even AI agents to human to back to AI agent interactions. There's a possibility where you'll have things like

03:28:35.000 --> 03:28:47.000
an AI scientist where one solution may provide a

03:28:47.000 --> 03:28:48.000
Mm-hmm.

03:28:48.000 --> 03:28:56.000
I don't know, we'll go through the scientific method and then let's say it will pass it on to a peer review AI that will peer review that and then come back and then it'll keep going back and forth, back and forth right over and over again. So how do you kind of trans, you know, transact that, you know, with credits, probably with tokens makes sense.

03:28:56.000 --> 03:29:01.000
So I think that wallet governance and policies and like having the way that we have that kind of traction to have

03:29:01.000 --> 03:29:09.000
ingrained policies and equally identifies to allocate that to like an AI or an AGI, or obviously that's far future, or at least an agent

03:29:09.000 --> 03:29:12.000
And that's one thing we're really working hard with with like fetch.com.

03:29:12.000 --> 03:29:14.000
Sorry, fetch AI right now.

03:29:14.000 --> 03:29:21.000
around how do they use like Asian various in marketplaces with like AI services or an AI API all the way through to like an AI agent.

03:29:21.000 --> 03:29:27.000
So I'd be really keen to see exactly how, like from your paper, you start to talk about different policies

03:29:27.000 --> 03:29:47.000
That would be recommended to be implemented for, let's say, an AI agent provider that are looking to plug in with a wallet. And then I know that if I'm going to load this wallet up with, let's say a thousand US dollars, that it's not just going to go spend it like it's in a candy store, but it's actually going to be using it for a way that improves its purpose.

03:29:47.000 --> 03:30:02.000
So I'd be keen to see that in the paper. And moreover, exactly how can an individual load it up and allow for like private key infrastructure to be managed effectively and especially securely by what would be a very low intelligent entity.

03:30:02.000 --> 03:30:10.000
Which would just be saying, hey, I have money in account. Therefore, I do X, Y, Z things.

03:30:10.000 --> 03:30:11.000
Hmm.

03:30:11.000 --> 03:30:14.000
So just to point out there that I'd love to see those when you kind of get to it and equally message me about that. Could be keen to see that.

03:30:14.000 --> 03:30:19.000
Yeah, I like the comments, Joey. Thank you. And I think it's related to the

03:30:19.000 --> 03:30:22.000
the design space right of these wallets and

03:30:22.000 --> 03:30:27.000
I complete, and I'm glad that you picked up on that that

03:30:27.000 --> 03:30:31.000
Yeah, AI is going to be using blockchain while it's it's

03:30:31.000 --> 03:30:38.000
It's web3 data based economy there's there's no point them

03:30:38.000 --> 03:30:41.000
going backwards, if you know what i mean so

03:30:41.000 --> 03:30:44.000
that design space when we start thinking about

03:30:44.000 --> 03:30:46.000
how do we

03:30:46.000 --> 03:30:53.000
manage the accountability behind these AI agents with the money that we've

03:30:53.000 --> 03:30:55.000
And in the sort of

03:30:55.000 --> 03:31:00.000
economies.

03:31:00.000 --> 03:31:03.000
Are they just going to be able to go and

03:31:03.000 --> 03:31:16.000
potentially be corrupted and go spend all of my money on meme coins and then six months later, I realized that they've made a 10x but i didn't actually give them permission to that. Is that really what I wanted?

03:31:16.000 --> 03:31:19.000
Probably, but at the same token.

03:31:19.000 --> 03:31:21.000
that could become like a really slippery slope.

03:31:21.000 --> 03:31:28.000
And break a lot of systems. So I think that

03:31:28.000 --> 03:31:33.000
That's a really good suggestion for the paper. One, if I could just bounce this idea back to you.

03:31:33.000 --> 03:31:38.000
one idea that I had while you were speaking is that potentially we could come up with

03:31:38.000 --> 03:31:40.000
sort of best

03:31:40.000 --> 03:31:44.000
design of this wallet technology designs for

03:31:44.000 --> 03:31:49.000
the use cases of these AI agents. So when an AI agent wants to spend

03:31:49.000 --> 03:31:51.000
money um

03:31:51.000 --> 03:31:53.000
or transact value.

03:31:53.000 --> 03:32:02.000
then it should operate within a multi-sig structure or it should have some layer of

03:32:02.000 --> 03:32:05.000
smart contract wallets involved.

03:32:05.000 --> 03:32:07.000
and different

03:32:07.000 --> 03:32:20.000
sort of particles in place in order for it to even be permissioned to spend money. That could be one suggestion that we make in the paper.

03:32:20.000 --> 03:32:32.000
Yeah, we have thought about things similar to that in the last few months. And one of the problems we keep hitting is around like, okay, great. We want to put like a multi-sig or some kind of NPC or approval system. And we realize it's just not enough.

03:32:32.000 --> 03:32:37.000

Because we think about an agent to agent interaction, right? Oh, I want to interact a thousand times in a single day.

03:32:37.000 --> 03:32:38.000
Yeah.

03:32:38.000 --> 03:32:41.000
If I have someone that's me awake every single moment, 24 seven a day so

03:32:41.000 --> 03:32:48.000
you know, really how do you enable a AI agent to not have so many men in the middle when it comes to

03:32:48.000 --> 03:32:52.000
Interacting like fluidly.

03:32:52.000 --> 03:32:59.000
So that's why I'm saying the policies that you're kind of looking at for wall governance will probably be the most important piece to it because

03:32:59.000 --> 03:33:11.000
We either have to have a super manual and heavy process, right? We'll pretty much babysitting the AI is doing the right thing. Or we have a super fluid, potentially high risk situation. So we have this kind of dilemma here.

03:33:11.000 --> 03:33:16.000
And maybe the first case is that we can kind of use like supervised learning as we kind of help transact and allow

03:33:16.000 --> 03:33:23.000
the AI to actually learn by your preferences as building as opposed to an arbitrary policy system.

03:33:23.000 --> 03:33:28.000
And looking at kind of other like um

03:33:28.000 --> 03:33:52.000
looking at other ways to train AI on financial transactions, not for the sake of making a trade, but more so, hey, you know what, this is probably the most preferable transaction that I should do because the people that be will want to or have proven over history that this is probably a better transaction, right? Maybe the AI won't do any transaction between outside of work hours, for instance.

03:33:52.000 --> 03:33:53.000
Okay.

03:33:53.000 --> 03:34:03.000

But how do we kind of train those to have policy and build to it? And equally for us to teach us what policies we prefer, because a lot of times you create policies and transaction policies, which are super arbitrary, right? Because you want to be super high, like low risk.

03:34:03.000 --> 03:34:09.000
But maybe there is some kind of middle ground where there's a whole other region where we don't mind operating within.

03:34:09.000 --> 03:34:10.000
But yeah, I'm just…

03:34:10.000 --> 03:34:13.000
One thing that I wanted to sort of ask

03:34:13.000 --> 03:34:15.000
On this line of thought is

03:34:15.000 --> 03:34:22.000
So like we have plenty of systems which are just automated payment systems that could be

03:34:22.000 --> 03:34:29.000
considered through some eyes as like an AI payment routing system, but I don't really think it is

03:34:29.000 --> 03:34:30.000
It is routing, isn't it?

03:34:30.000 --> 03:34:34.000
Yeah, it's just setting up different

03:34:34.000 --> 03:34:36.000
articles and procedures which are

03:34:36.000 --> 03:34:38.000
just scripts that run.

03:34:38.000 --> 03:34:39.000
logic operations.

03:34:39.000 --> 03:34:45.000
Yeah, and have access. And I think that that's a misconception to call that AI. I think

03:34:45.000 --> 03:34:47.000
Which then we get into the problem of

03:34:47.000 --> 03:34:50.000
the complexity because

03:34:50.000 --> 03:34:57.000
Ideally, I want the AI to act in some level of autonomous, intelligent way where it

03:34:57.000 --> 03:35:06.000
does something that I wouldn't have considered as a human to optimize the experience and outcome, the goals that I give it.

03:35:06.000 --> 03:35:10.000
Which means that it can't fall down that sort of

03:35:10.000 --> 03:35:14.000
predetermined like deterministic path of

03:35:14.000 --> 03:35:27.000
majors choosing A to B as the agent's interaction. So I would say that these policies can't be fully arbitrary. They need to have some level of flexibility involved otherwise you don't get that

03:35:27.000 --> 03:35:31.000
sort of the bonus that

03:35:31.000 --> 03:35:34.000
having this external intelligence doing things for you.

03:35:34.000 --> 03:35:37.000
can actually bring.

03:35:37.000 --> 03:35:47.000
Agreed.

03:35:47.000 --> 03:35:51.000
We have only like two minutes left so

03:35:51.000 --> 03:35:55.000
Do you use in any other comments probably we can finish

03:35:55.000 --> 03:35:57.000
up this session.

03:35:57.000 --> 03:35:58.000
Any other comments?

03:35:58.000 --> 03:36:03.000
Okay, if I could hopefully can still share my screen.

03:36:03.000 --> 03:36:09.000
I'm just going to quickly put the wallet governance report

03:36:09.000 --> 03:36:10.000
Okay.

03:36:10.000 --> 03:36:15.000
in the chat, if anyone's welcome to come in and provide comments.

03:36:15.000 --> 03:36:18.000
And we host the IKP Working Group.

03:36:18.000 --> 03:36:26.000
Meeting fortnightly on Thursday. I think it's 1 p.m. UTC, but that would have to be confirmed.

03:36:26.000 --> 03:36:28.000
Currently discussing.

03:36:28.000 --> 03:36:32.000
this paper and would like to

03:36:32.000 --> 03:36:40.000
finalize it before the block 12 meeting ideally. So now's the time to contribute to the wallet governance policy in key.

03:36:40.000 --> 03:36:44.000
management study report. But thank you so much for your time.

03:36:44.000 --> 03:36:46.000
Thank you.

03:36:46.000 --> 03:37:02.000
Okay, so we have 15, 15.

03:37:02.000 --> 03:37:07.000
15 minutes.

03:37:07.000 --> 03:37:13.000
Yeah.

03:37:13.000 --> 03:37:37.000
I wasn't expecting that. I was going to come. I've just been a little bit like, I've been really busy today. Like, it's been huge for us.

03:37:37.000 --> 03:38:07.000
I don't know how to use it.

03:52:47.000 --> 03:53:00.000
Uh-oh.

03:53:00.000 --> 03:53:04.000

Yeah.

03:53:04.000 --> 03:53:10.000
So I don't know if you know this, but we did the one in tokyo

03:53:10.000 --> 03:53:14.000
a few weeks ago in the one in Sydney

03:53:14.000 --> 03:53:17.000
Yeah, last weekend. I think it was last week.

03:53:17.000 --> 03:53:21.000
Oh, yeah. And we're going to have…

03:53:21.000 --> 03:53:28.000
So it co-located with ISO, ISOTC 307, which is like…

03:53:28.000 --> 03:53:32.000
you know the singular is a member of ISO so

03:53:32.000 --> 03:53:36.000
We talked about cybersecurity with them.

03:53:36.000 --> 03:53:42.000
which is what we're going to talk about today as well but

03:53:42.000 --> 03:53:48.000
I don't remember it. I think the…

03:53:48.000 --> 03:53:51.000
He named it, I guess.

03:53:51.000 --> 03:54:01.000
Because the, you know, confidence that we have is like, you know, block conference, right? And we want to roll up all of the discussion points we made

03:54:01.000 --> 03:54:04.000
in layer two meetups to the block.

03:54:04.000 --> 03:54:06.000
you know?

03:54:06.000 --> 03:54:10.000
block meeting, yeah.

03:54:10.000 --> 03:54:14.000
Yeah, we're going to talk about the same thing too in block 11 and

03:54:14.000 --> 03:54:24.000
Yeah, yeah. So…

03:54:24.000 --> 03:54:29.000
Yes, yes, yes.

03:54:29.000 --> 03:54:34.000
So sometimes people misunderstand that, you know, we're going to talk only about l2

03:54:34.000 --> 03:54:43.000
in Leia to meet up, but it doesn't have anything to do with like a layer two so

03:54:43.000 --> 03:54:45.000
Yeah.

03:54:45.000 --> 03:54:51.000
Anyway, the next one is cybersecurity standards and

03:54:51.000 --> 03:54:55.000
We talked about this in like Tokyo and in Sydney as well.

03:54:55.000 --> 03:54:58.000
But I would have liked to…

03:54:58.000 --> 03:55:00.000
you know.

03:55:00.000 --> 03:55:04.000
give a brief introduction of like what we discussed and

03:55:04.000 --> 03:55:09.000
Yeah, this is a presentation I made in Tokyo and Sydney as well but

03:55:09.000 --> 03:55:15.000
So when it comes to thinking about cybersecurity information sharing alignment in the US,

03:55:15.000 --> 03:55:20.000
In general, it's not only, I'm not talking about crypto industry

Cybersecurity Information Sharing Alignment

03:55:20.000 --> 03:55:25.000
So this is an overview of how

03:55:25.000 --> 03:55:30.000
cybersecurity information sharing alignment works in general in any industry.

03:55:30.000 --> 03:55:34.000
So, um.

03:55:34.000 --> 03:55:40.000
So for example, like if there is a vulnerability in a product of vendors

03:55:40.000 --> 03:55:50.000
what happens is like researchers and security professionals discover and report vulnerabilities of the product

03:55:50.000 --> 03:55:52.000
of the product.

03:55:52.000 --> 03:56:00.000
directly to vendors and if you know vendors received information about like vulnerabilities they discover and report

03:56:00.000 --> 03:56:02.000
they're gonna like…

03:56:02.000 --> 03:56:05.000
fix the vulnerabilities and

03:56:05.000 --> 03:56:08.000
also create a security patch.

03:56:08.000 --> 03:56:11.000
information um

03:56:11.000 --> 03:56:16.000
of the information of the vulnerability, vulnerability information

03:56:16.000 --> 03:56:22.000
And they're going to put it into the CVE list, the common vulnerabilities and exposures program list.

03:56:22.000 --> 03:56:27.000
which is managed by MITA, which is a non-profit non

03:56:27.000 --> 03:56:30.000

or in addition based in the US, Washington, D.C,

03:56:30.000 --> 03:56:42.000
And researchers and security professionals sometimes discovering report vulnerabilities and they directly report the vulnerabilities information to the CBE list, CBE program.

03:56:42.000 --> 03:56:48.000
And the CBE program request to create security patch of the information

03:56:48.000 --> 03:56:51.000
and vulnerability information to the vendors.

03:56:51.000 --> 03:56:55.000
And the vendors need to deal with requests

03:56:55.000 --> 03:56:58.000
that are made by the CBE program.

03:56:58.000 --> 03:57:03.000
So that is how responsible disclosure practices work in general in any industry.

03:57:03.000 --> 03:57:16.000
And the responsibility of MITRE is to applying CBIDs to all of the vulnerabilities that are discovered and

03:57:16.000 --> 03:57:25.000
reported by researchers and security professionals. And my manages all of the vulnerability automations in the list.

03:57:25.000 --> 03:57:29.000
And once vulnerability information was

03:57:29.000 --> 03:57:33.000
has been put into the CVE list.

03:57:33.000 --> 03:57:36.000
they will be standardized with additional analysis

03:57:36.000 --> 03:57:42.000
in the national vulnerability Database, which is managed by NIST.

03:57:42.000 --> 03:57:45.000
Okay.

03:57:45.000 --> 03:57:48.000
these standardized database

03:57:48.000 --> 03:57:52.000

will be analyzed by

03:57:52.000 --> 03:57:58.000
industry specific ISAC, which stands for the information sharing

03:57:58.000 --> 03:58:01.000
And analysis center.

03:58:01.000 --> 03:58:04.000
And each industry has

03:58:04.000 --> 03:58:09.000
Isaacs, for example, like the financial industry has financed it.

03:58:09.000 --> 03:58:13.000
And also like…

03:58:13.000 --> 03:58:21.000
I can't come up with like any other industry, but for example um

03:58:21.000 --> 03:58:27.000
what is the good example of like other industry but you know any any industry has

03:58:27.000 --> 03:58:29.000
like Isaacs because

03:58:29.000 --> 03:58:34.000
national, like all of the vulnerability information that are put into

03:58:34.000 --> 03:58:41.000
the national database vulnerability database is like standardized and it's not industry specific

03:58:41.000 --> 03:58:49.000
So I analyze all of the vulnerability information in this database and share

03:58:49.000 --> 03:58:55.000
specific sector specific threat information and best practices

03:58:55.000 --> 03:58:58.000
to benders and they distribute

03:58:58.000 --> 03:59:00.000
all of the information that are you know

03:59:00.000 --> 03:59:06.000
I'm good for the vendors too so

03:59:06.000 --> 03:59:11.000

vulnerabilities and how they deal with these vulnerabilities and how

03:59:11.000 --> 03:59:14.000
They deal with like human threat?

03:59:14.000 --> 03:59:17.000
So this is how

03:59:17.000 --> 03:59:19.000
basically how

03:59:19.000 --> 03:59:25.000
cybersecurity information sharing alignment works in general in the US.

03:59:25.000 --> 03:59:28.000
But the question is…

03:59:28.000 --> 03:59:33.000
can this alignment, information sharing alignment work in the crypto industry?

03:59:33.000 --> 03:59:35.000
So…

03:59:35.000 --> 03:59:38.000
responsible disclosure practices

03:59:38.000 --> 03:59:43.000
do not seem to work in a crypto industry as of now.

03:59:43.000 --> 03:59:45.000
Because, you know, there are lightsab and Webstreet

03:59:45.000 --> 03:59:48.000
cybersecurity firms

03:59:48.000 --> 03:59:51.000
like, you know, smart contract audit auditors and like

03:59:51.000 --> 03:59:54.000
Web3 security firms.

03:59:54.000 --> 03:59:56.000
And they make a contract with like

03:59:56.000 --> 03:59:59.000
benders directly…

03:59:59.000 --> 04:00:03.000

And they are supposed to discover and report about the abilities of dependers.

04:00:03.000 --> 04:00:05.000
of the product of the vendors.

04:00:05.000 --> 04:00:08.000
But there isn't any…

04:00:08.000 --> 04:00:12.000
practices, um.

04:00:12.000 --> 04:00:13.000
like implemented in the whole crypto industry

04:00:13.000 --> 04:00:17.000
A whole bunch of different

04:00:17.000 --> 04:00:18.000
So this is um

04:00:18.000 --> 04:00:19.000
So this is gone, boom, boom, boom.

04:00:19.000 --> 04:00:23.000
Juan was a challenge is we have in the crypto industry

04:00:23.000 --> 04:00:26.000
and also

04:00:26.000 --> 04:00:28.000
vulnerability

04:00:28.000 --> 04:00:30.000
In the crypto industry should be

04:00:30.000 --> 04:00:33.000
like a little bit different from like

04:00:33.000 --> 04:00:35.000
As a bowling abilities.

04:00:35.000 --> 04:00:41.000
So we should have like a specific data format for the vulnerabilities in the crypto industry

04:00:41.000 --> 04:00:44.000
So…

04:00:44.000 --> 04:00:48.000

and but um

04:00:48.000 --> 04:00:52.000
We have CBE list in some of the

04:00:52.000 --> 04:00:55.000
crypto related vulnerability information

04:00:55.000 --> 04:01:00.000
are already put into the CBE list, but only a few

04:01:00.000 --> 04:01:02.000
So…

04:01:02.000 --> 04:01:07.000
I'm not really sure if MITA is going to be responsible for

04:01:07.000 --> 04:01:13.000
managing the inflammation, vulnerability, inflammation of the related to related

04:01:13.000 --> 04:01:16.000
to crypto and blockchain.

04:01:16.000 --> 04:01:20.000
So this is one of the questions we would like to

04:01:20.000 --> 04:01:23.000
ask.

04:01:23.000 --> 04:01:25.000
And so…

04:01:25.000 --> 04:01:28.000
In terms of Isaac in the U.S,

04:01:28.000 --> 04:01:33.000
Crypto Isaac was established recently, I think a few months ago.

04:01:33.000 --> 04:01:36.000
And yeah, they're gonna…

04:01:36.000 --> 04:01:38.000
being responsible for being responsible

04:01:38.000 --> 04:01:41.000
sharing threat intelligence

04:01:41.000 --> 04:01:45.000

And vulnerability information specific to

04:01:45.000 --> 04:01:47.000
on crypto and blockchain

04:01:47.000 --> 04:01:50.000
to vendors and companies

04:01:50.000 --> 04:01:52.000
the crypto related companies

04:01:52.000 --> 04:01:58.000
But crypto is like it's just a part of the, you know, whole framework

04:01:58.000 --> 04:02:02.000
on the cybersecurity information sharing alignment in the industry

04:02:02.000 --> 04:02:11.000
It's not like Crypto Isaac is going to solve all of the problem we have for the information sharing alignment.

04:02:11.000 --> 04:02:14.000
So…

04:02:14.000 --> 04:02:19.000
Yeah, it's great to have Crypto Isaac being established by

04:02:19.000 --> 04:02:23.000
It's just a part of it. So I think…

04:02:23.000 --> 04:02:26.000
We have to also think about how

04:02:26.000 --> 04:02:31.000
Crypto Isaac is what what kind of roles crypto isaac is playing in this

04:02:31.000 --> 04:02:35.000
alignment and also what are the missing pieces

04:02:35.000 --> 04:02:38.000
we have to think about

04:02:38.000 --> 04:02:45.000
Even after the establishment of crypto isaac.

04:02:45.000 --> 04:02:47.000
And…

04:02:47.000 --> 04:02:56.000
So we had a great conversation with people in the ISO TC307

04:02:56.000 --> 04:02:58.000
And also who is

04:02:58.000 --> 04:03:01.000
Bitcoin core developers about this

04:03:01.000 --> 04:03:05.000
And we also wanted to continue discussion

04:03:05.000 --> 04:03:09.000
around how what kind of roles begin can play

04:03:09.000 --> 04:03:14.000
in the cybersecurity information sharing alignment in the crypto industry.

04:03:14.000 --> 04:03:19.000
So yeah, we would like to talk about this

04:03:19.000 --> 04:03:23.000
In this session as well.

04:03:23.000 --> 04:03:31.000
And does anyone have any questions about this? If there isn't, I think I can…

04:03:31.000 --> 04:03:35.000
debrief recap the

04:03:35.000 --> 04:03:39.000
discussion we had in Tokyo and sydney

04:03:39.000 --> 04:03:44.000
But before that, if there's any question…

04:03:44.000 --> 04:03:48.000
these fields, feel free to

04:03:48.000 --> 04:03:58.000
do it to ask?

04:03:58.000 --> 04:04:07.000
any remote participant?

04:04:07.000 --> 04:04:09.000
Okay.

04:04:09.000 --> 04:04:12.000
So…

04:04:12.000 --> 04:04:25.000
I can share.

04:04:25.000 --> 04:04:36.000
This one, okay.

04:04:36.000 --> 04:04:38.000
All right. So…

04:04:38.000 --> 04:04:41.000
We're going to talk about this

04:04:41.000 --> 04:04:46.000
thing in Washington, D.C. next week as well and

04:04:46.000 --> 04:04:51.000
there will be people from Crypto Isaac and NIST and

04:04:51.000 --> 04:04:53.000
also mitre, I guess.

04:04:53.000 --> 04:05:00.000
But we already created an agenda for the session we're going to have

04:05:00.000 --> 04:05:02.000
about this topic.

04:05:02.000 --> 04:05:04.000
And yeah.

04:05:04.000 --> 04:05:08.000
This is an agenda for the

04:05:08.000 --> 04:05:15.000
session in Washington, D.C, but I have…

04:05:15.000 --> 04:05:20.000
at discussion points we made in layer two meetups in Tokyo and Sydney here.

04:05:20.000 --> 04:05:22.000
And yeah, I can…

04:05:22.000 --> 04:05:26.000
go through this.

04:05:26.000 --> 04:05:30.000
So when it comes to talking about vulnerabilities.

04:05:30.000 --> 04:05:35.000
So I think like pre-incident and post incidents is very

04:05:35.000 --> 04:05:41.000
different between each other like vulnerabilities and incidents should be treated separately

04:05:41.000 --> 04:05:47.000
So vulnerable abilities, maybe potential causes for incidents that haven't happened yet.

04:05:47.000 --> 04:05:53.000
while if once incidents happen, I think the

04:05:53.000 --> 04:05:57.000
you know process of dealing with it is very different from like

04:05:57.000 --> 04:06:04.000
you know, dealing with like vulnerabilities before incidents. So we have to separate the discussion

04:06:04.000 --> 04:06:06.000
Okay.

04:06:06.000 --> 04:06:13.000
And also we talk about a lack of talent in education

04:06:13.000 --> 04:06:15.000
So…

04:06:15.000 --> 04:06:18.000
even if

04:06:18.000 --> 04:06:26.000
Crypto Isaac has a lot of like security experts. If there isn't any expert cybersecurity expert in each country, each company.

04:06:26.000 --> 04:06:29.000
it doesn't really…

04:06:29.000 --> 04:06:32.000
it won't be really effective.

04:06:32.000 --> 04:06:36.000
So, um.

04:06:36.000 --> 04:06:45.000

But the general conversation we had, like the general concern we have around, you know, lack of talent and education is that like

04:06:45.000 --> 04:06:54.000
So there are a lot of like cybersecurity experts in the Web2 industry, but there isn't any, the number of cybersecurity experts is limited

04:06:54.000 --> 04:06:56.000
in the crypto industry.

04:06:56.000 --> 04:06:58.000
So this is a problem.

04:06:58.000 --> 04:07:00.000
In…

04:07:00.000 --> 04:07:02.000
um

04:07:02.000 --> 04:07:04.000
So there is an opinion

04:07:04.000 --> 04:07:16.000
that we should look at how other industries have succeeded in increasing the number, increasing the number of cybersecurity experts in these companies, in their companies.

04:07:16.000 --> 04:07:19.000
and also

04:07:19.000 --> 04:07:26.000
There was a discussion around

04:07:26.000 --> 04:07:32.000
So what kind of…

04:07:32.000 --> 04:07:38.000
So there was a discussion around the separation between product and infrastructure

04:07:38.000 --> 04:07:40.000
So…

04:07:40.000 --> 04:07:44.000
you know when it comes to thinking about the corporate security

04:07:44.000 --> 04:07:48.000
corporate part, the infrastructure

04:07:48.000 --> 04:07:58.000

fracture is no different from any other industry the other part is encrypting the only part that is cryptic, it's a product

04:07:58.000 --> 04:08:00.000
Somebody says this.

04:08:00.000 --> 04:08:05.000
But anyway, increasing the number of qualified

04:08:05.000 --> 04:08:07.000
young experts is critical

04:08:07.000 --> 04:08:12.000
and requires budgeting for educational initiatives.

04:08:12.000 --> 04:08:19.000
So, you know, I think one of the missions of the beginning is to create educational contents for

04:08:19.000 --> 04:08:21.000
um dumb.

04:08:21.000 --> 04:08:23.000
for the things

04:08:23.000 --> 04:08:26.000
that people do not

04:08:26.000 --> 04:08:28.000
Like, like…

04:08:28.000 --> 04:08:32.000
you know, people do not have a good understanding of so

04:08:32.000 --> 04:08:39.000
I think the beginning can play the role of creating educational contents for

04:08:39.000 --> 04:08:42.000
people to understand

04:08:42.000 --> 04:08:45.000
the cybersecurity more

04:08:45.000 --> 04:08:50.000
And yeah, so this is what was uh

04:08:50.000 --> 04:08:53.000
discussion points we made in Tokyo and Sydney.

04:08:53.000 --> 04:08:56.000
And also…

04:08:56.000 --> 04:09:02.000
In order for the responsible disclosure practice to work in

04:09:02.000 --> 04:09:08.000
the crypto industry and the question would be what is missing with

04:09:08.000 --> 04:09:11.000
current security practices.

04:09:11.000 --> 04:09:19.000
And yeah, so we already, as I mentioned, we already have like smart contract auditors and WebStreet security firms

04:09:19.000 --> 04:09:22.000
But…

04:09:22.000 --> 04:09:26.000
It's just a part of the cybersecurity information sharing alignment.

04:09:26.000 --> 04:09:29.000
in…

04:09:29.000 --> 04:09:35.000
Yeah, many blockchain projects have utilized the bug bounty schemes in recent years

04:09:35.000 --> 04:09:40.000
But it helps reduce the number of bags, but it's not perfect, as I mentioned.

04:09:40.000 --> 04:09:54.000
Like some projects claim that their product is secure because they use back bounties, but they might be misleading because it's just one of the tools to mitigate certain issues.

04:09:54.000 --> 04:09:56.000
And yeah, that's a

04:09:56.000 --> 04:10:00.000
as is written, a comprehensive cybersecurity information

04:10:00.000 --> 04:10:03.000
sharing framework should combine.

04:10:03.000 --> 04:10:05.000
different types of approaches including

04:10:05.000 --> 04:10:10.000
certification programs from celebrity testing labs

04:10:10.000 --> 04:10:15.000
And so on. This is one of the discussion points we

04:10:15.000 --> 04:10:19.000
made in Sydney and Tokyo

04:10:19.000 --> 04:10:23.000
And also we need crypto security standards

04:10:23.000 --> 04:10:31.000
As I mentioned when I talked about the slides I showed you all

04:10:31.000 --> 04:10:35.000
we do not have…

04:10:35.000 --> 04:10:39.000
specific date format for the, you know, threat intelligence and also

04:10:39.000 --> 04:10:42.000
vulnerability information.

04:10:42.000 --> 04:10:46.000
So we should have one poll specific to cybersecurity

04:10:46.000 --> 04:10:50.000
of crypto and blockchain

04:10:50.000 --> 04:10:52.000
And…

04:10:52.000 --> 04:10:56.000
Yes. So this is one of the

04:10:56.000 --> 04:10:59.000
things we discussed.

04:10:59.000 --> 04:11:05.000
But yeah, I heard that there is, you know, standard called CCS.

04:11:05.000 --> 04:11:08.000
But there is skepticism around it

04:11:08.000 --> 04:11:11.000
about its current effectiveness

04:11:11.000 --> 04:11:21.000
And yeah, while this is one of the few Web3 security standards, it may be somewhat static and not refractive of new or

04:11:21.000 --> 04:11:24.000
developments in the industry so we have to

04:11:24.000 --> 04:11:26.000
um

04:11:26.000 --> 04:11:29.000
crypto security standards.

04:11:29.000 --> 04:11:33.000
And obviously, like in um

04:11:33.000 --> 04:11:35.000
beginning is trying to be…

04:11:35.000 --> 04:11:37.000
standard setting body

04:11:37.000 --> 04:11:39.000
trying to aim

04:11:39.000 --> 04:11:44.000
aim to create standards and best practices for blockchain governance so

04:11:44.000 --> 04:11:46.000
this is what

04:11:46.000 --> 04:11:48.000
one of the things that

04:11:48.000 --> 04:11:52.000
big income can do.

04:11:52.000 --> 04:11:55.000
And also…

04:11:55.000 --> 04:11:58.000
Yeah, as I mentioned, we need um

04:11:58.000 --> 04:12:10.000
data format for threat intelligence and vulnerability information, but we should have a taxonomy of threat intelligence first because

04:12:10.000 --> 04:12:12.000

Do you even do any taxonomy?

04:12:12.000 --> 04:12:17.000
for threat intelligence and we can think about the data format

04:12:17.000 --> 04:12:20.000
of them right after

04:12:20.000 --> 04:12:24.000
defining a taxonomy of the certain intelligence.

04:12:24.000 --> 04:12:30.000
And yeah, somebody says that the blockchain and the crypto industry hasn't been

04:12:30.000 --> 04:12:33.000
hasn't matured yet enough.

04:12:33.000 --> 04:12:38.000
to identify specific items to be shared and exchanged by

04:12:38.000 --> 04:12:40.000
Yeah.

04:12:40.000 --> 04:12:45.000
Yeah. In the long term, we have to think about like how

04:12:45.000 --> 04:12:52.000
we can create the standards for the data format to abstract intelligence in

04:12:52.000 --> 04:12:56.000
vulnerability information, so…

04:12:56.000 --> 04:12:59.000
Okay.

04:12:59.000 --> 04:13:02.000
I have one common ccis is not static.

04:13:02.000 --> 04:13:07.000
version 8.2 will be released in a month or two.

04:13:07.000 --> 04:13:11.000
Okay, but…

04:13:11.000 --> 04:13:17.000
Okay, can you add any comments on AR8 on this?

04:13:17.000 --> 04:13:20.000

Mark, sorry, I can't

04:13:20.000 --> 04:13:25.000
Mark, if you can speak up.

04:13:25.000 --> 04:13:26.000
Oh, hello.

04:13:26.000 --> 04:13:28.000
Okay. Yeah, I can hear you.

04:13:28.000 --> 04:13:32.000
Yes, I'm a member of the um

04:13:32.000 --> 04:13:40.000
CCSS steering committee and I was also attending the TC307 plenary in Sydney

04:13:40.000 --> 04:13:42.000
and attended this meeting.

04:13:42.000 --> 04:13:47.000
Yes, there is a long history with CCSS it was designed

04:13:47.000 --> 04:13:50.000
and published in 2015. But over the last

04:13:50.000 --> 04:13:55.000
two years, there's been a massive input into

04:13:55.000 --> 04:13:57.000
bringing ccss

04:13:57.000 --> 04:14:02.000
online and reflective of the current attack vectors

04:14:02.000 --> 04:14:05.000
that are out there the um

04:14:05.000 --> 04:14:11.000
there's an auditors program now. There's over, well, I think about 20 ccs

04:14:11.000 --> 04:14:13.000
auditors around the world

04:14:13.000 --> 04:14:35.000
And there is currently that I know of as a committee member, five active or four to five active audits going on right now worldwide.

04:14:35.000 --> 04:14:37.000
Are you still talking?

04:14:37.000 --> 04:14:39.000
No, sorry, I've finished.

04:14:39.000 --> 04:14:41.000
Oh, okay. Great, great.

04:14:41.000 --> 04:14:42.000
Yeah.

04:14:42.000 --> 04:14:46.000
Okay.

04:14:46.000 --> 04:14:48.000
So the next…

04:14:48.000 --> 04:14:51.000
the new version will be released in a month.

04:14:51.000 --> 04:14:52.000
or two.

04:14:52.000 --> 04:14:56.000
Yeah, the new version's got 30% more requirements

04:14:56.000 --> 04:14:59.000
than the current version we've got

04:14:59.000 --> 04:15:03.000
There's risk, governance, service provider management.

04:15:03.000 --> 04:15:04.000
Okay.

04:15:04.000 --> 04:15:07.000
There's wallet monitoring requirements.

04:15:07.000 --> 04:15:11.000
there's support for single signature mechanisms.

04:15:11.000 --> 04:15:19.000
there's an enhancement on the existing requirements. It's gone for a substantial stakeholder feedback process as well so

04:15:19.000 --> 04:15:21.000
anyone in the world

04:15:21.000 --> 04:15:24.000
can submit their suggestions.

04:15:24.000 --> 04:15:28.000
for improving CCSS and that will be taken on board.

04:15:28.000 --> 04:15:32.000
You don't have to pay for membership. You don't have to

04:15:32.000 --> 04:15:35.000
You know, be on a particular

04:15:35.000 --> 04:15:39.000
committee to make suggestions for the changes so

04:15:39.000 --> 04:15:42.000
We conducted a stakeholders feedback

04:15:42.000 --> 04:15:48.000
process a couple of months ago and we received quite a lot of feedback from the community

04:15:48.000 --> 04:15:51.000
about how CCSCS can be improved

04:15:51.000 --> 04:15:56.000
And that's reflected in the fact that over 30% more requirements have been added to the standard.

04:15:56.000 --> 04:16:01.000
So we're quite pleased now. It's quite a well-rounded standard.

04:16:01.000 --> 04:16:04.000
That's it for the…

04:16:04.000 --> 04:16:06.000
the update on CCSS.

04:16:06.000 --> 04:16:08.000
Thank you.

04:16:08.000 --> 04:16:15.000
Okay, so sorry that I'm not really familiar with CCSS, but I

04:16:15.000 --> 04:16:19.000
Another question I wanted to ask is like what's

04:16:19.000 --> 04:16:21.000

Cobbers.

04:16:21.000 --> 04:16:24.000
what CCSS covers specifically like

04:16:24.000 --> 04:16:30.000
So when it comes to creating standards in the block in begin as well

04:16:30.000 --> 04:16:31.000
on what are the missing pieces

04:16:31.000 --> 04:16:34.000
In terms of standardization and

04:16:34.000 --> 04:16:38.000
We're not going to be like, you know.

04:16:38.000 --> 04:16:40.000
creating same standards

04:16:40.000 --> 04:16:44.000
all the specific problems for the same problem, right?

04:16:44.000 --> 04:16:48.000
So I want to have a more

04:16:48.000 --> 04:16:53.000
like better understanding of like what CCSS covers and like what are the

04:16:53.000 --> 04:16:56.000
missing pcs that block

04:16:56.000 --> 04:17:00.000
the beginning can play the role in.

04:17:00.000 --> 04:17:06.000
Absolutely. CCSCS is not a baseline information security standard.

04:17:06.000 --> 04:17:07.000
Mm-hmm.

04:17:07.000 --> 04:17:09.000
So it's not like ISO 27001

04:17:09.000 --> 04:17:14.000
where you have patch management, configuration management, deployment management.

04:17:14.000 --> 04:17:21.000

User access management, all the absolute basic fundamentals of information security

04:17:21.000 --> 04:17:25.000
CCSS takes the key management

04:17:25.000 --> 04:17:29.000
requirements from ISO 27001

04:17:29.000 --> 04:17:30.000
Mm-hmm.

04:17:30.000 --> 04:17:32.000
It adds further enhancement to those requirements.

04:17:32.000 --> 04:17:35.000
to basically

04:17:35.000 --> 04:17:41.000
tackle the unique properties of blockchain and DLT systems as well as cryptocurrency.

04:17:41.000 --> 04:17:46.000
For example, we've included wallet monitoring in the requirements.

04:17:46.000 --> 04:17:51.000
Which is not obviously a part of your ISO 27001

04:17:51.000 --> 04:17:54.000
standard controls.

04:17:54.000 --> 04:17:57.000
So it's always been classed as a bolt on

04:17:57.000 --> 04:18:00.000
to ISO 27001.

04:18:00.000 --> 04:18:07.000
But what we have noticed through the stakeholder feedback

04:18:07.000 --> 04:18:14.000
is that a lot of organizations are deciding whether to become ISO 27001 compliant

04:18:14.000 --> 04:18:16.000
or ccss

04:18:16.000 --> 04:18:23.000
compliant because a lot of the organisations feel that CCSS is more aligned to their

04:18:23.000 --> 04:18:25.000

architectural and design

04:18:25.000 --> 04:18:29.000
patterns, then ISO 27001 is.

04:18:29.000 --> 04:18:34.000
So we've found that because of this feedback, we've introduced

04:18:34.000 --> 04:18:40.000
some baseline information security controls such as service provider management

04:18:40.000 --> 04:18:45.000
which is obviously critical as service providers are a major attack vector.

04:18:45.000 --> 04:18:47.000
within organizations

04:18:47.000 --> 04:18:51.000
We've also added a risk management and governance such as

04:18:51.000 --> 04:18:54.000
an executive member

04:18:54.000 --> 04:18:58.000
must acknowledge in writing that they understand

04:18:58.000 --> 04:19:04.000
the risks that are currently attributed to information security within their organisation.

04:19:04.000 --> 04:19:08.000
And another one I forgot to mention that we've included in

04:19:08.000 --> 04:19:12.000
version 8.2 is smart contract audits

04:19:12.000 --> 04:19:13.000
Hmm.

04:19:13.000 --> 04:19:15.000
So we've addressed the

04:19:15.000 --> 04:19:21.000
One of the major issues that we see in the

04:19:21.000 --> 04:19:24.000
community currently is that

04:19:24.000 --> 04:19:29.000

one version of a smart contract will be audited by a third party external auditor

04:19:29.000 --> 04:19:31.000
say version one

04:19:31.000 --> 04:19:36.000
And then they make change and the audit is successful they pass

04:19:36.000 --> 04:19:43.000
and the reports generated and then say the next day or the next week, they add in some extra functionality

04:19:43.000 --> 04:19:47.000
into that smart contract and they call it version two.

04:19:47.000 --> 04:19:53.000
What we've found based on stakeholder feedback and also reading articles in the news is that

04:19:53.000 --> 04:19:59.000
In these situations when an additional function is added to the smart contract.

04:19:59.000 --> 04:20:06.000
the entity may decide that they won't do an audit because they believe that the update is not a significant change to the code base.

04:20:06.000 --> 04:20:09.000
So they released version two

04:20:09.000 --> 04:20:13.000
to the environments where stakeholders interact with the smart contracts.

04:20:13.000 --> 04:20:16.000
you know quote call it production

04:20:16.000 --> 04:20:22.000
And they released the audit report. However, the audit report is for version one.

04:20:22.000 --> 04:20:25.000
not version two because version two was never audited

04:20:25.000 --> 04:20:29.000
So the stakeholders look at the audit report

04:20:29.000 --> 04:20:32.000
and see that the code passed an audit.

04:20:32.000 --> 04:20:35.000

But they did not realize or they did not

04:20:35.000 --> 04:20:37.000
have the expertise to research

04:20:37.000 --> 04:20:43.000
The smart contract code and see that actually the smart contract code that they're currently interacting with

04:20:43.000 --> 04:20:45.000
is a different version.

04:20:45.000 --> 04:20:47.000
So in CCSS,

04:20:47.000 --> 04:20:50.000
we've reflected that situation

04:20:50.000 --> 04:20:52.000
that issue and um three

04:20:52.000 --> 04:20:59.000
smart contract audit requirements that are coming out in version 8.2.

04:20:59.000 --> 04:21:05.000
So that's basically the key difference between baseline information security standards

04:21:05.000 --> 04:21:10.000
in CCSS, it's a bolt-on to ISO, 27,001

04:21:10.000 --> 04:21:13.000
But at the same time.

04:21:13.000 --> 04:21:16.000
we've identified through the community that

04:21:16.000 --> 04:21:19.000
a lot of organizations or entities

04:21:19.000 --> 04:21:22.000
We'll look at CCSS.

04:21:22.000 --> 04:21:26.000
In totality, they won't look at ISO 27001 so we've

04:21:26.000 --> 04:21:30.000
looked at some of the you know the major attack vectors out there currently

04:21:30.000 --> 04:21:34.000
And we've put in baseline security controls that we believe

04:21:34.000 --> 04:21:39.000
will strengthen CCSS as an independent information security standard.

04:21:39.000 --> 04:21:42.000
such as service provider management, risk management.

04:21:42.000 --> 04:21:44.000
and other governance?

04:21:44.000 --> 04:21:47.000
I hope that's answered your question.

04:21:47.000 --> 04:21:50.000
Yes. Yeah. Thank you very much it's

04:21:50.000 --> 04:21:54.000
it quantifies um

04:21:54.000 --> 04:22:06.000
My questions. Okay.

04:22:06.000 --> 04:22:10.000
Hi, Mark. One question from central

04:22:10.000 --> 04:22:15.000
Okay, so please correct me if I'm wrong but

04:22:15.000 --> 04:22:18.000
So that work.

04:22:18.000 --> 04:22:21.000
at CCSS started with key management.

04:22:21.000 --> 04:22:24.000
So, you know, several years ago, that slow the

04:22:24.000 --> 04:22:28.000
target we need to think about cybersecurity exposure

04:22:28.000 --> 04:22:32.000
layer one, you know, simple blockchain technology

04:22:32.000 --> 04:22:36.000
Like, you know, under, so the key management at

04:22:36.000 --> 04:22:39.000
custody or cryptocurrency exchange.

04:22:39.000 --> 04:22:44.000
That's a not complicated. It's still complicated, but not complicated.

04:22:44.000 --> 04:22:48.000
But at this moment, so we have so many layer two

04:22:48.000 --> 04:22:51.000
protocols so which involved in that social

04:22:51.000 --> 04:22:53.000
So we see that it's crypto

04:22:53.000 --> 04:22:57.000
cryptographic protocols, including Xeronoge proof

04:22:57.000 --> 04:23:00.000
So that number of keys are increasing.

04:23:00.000 --> 04:23:04.000
from industry. So, you know, we need to

04:23:04.000 --> 04:23:10.000
think about that more complicated structure with technology and the key management

04:23:10.000 --> 04:23:13.000
And the number of kids increasing.

04:23:13.000 --> 04:23:15.000
key management life cycle is

04:23:15.000 --> 04:23:18.000
quite different from the you know eat

04:23:18.000 --> 04:23:21.000
of the PKIs. So

04:23:21.000 --> 04:23:23.000
what is our so future

04:23:23.000 --> 04:23:27.000
direction to handle that complicated decision

04:23:27.000 --> 04:23:31.000
They got new key management.

04:23:31.000 --> 04:23:34.000
That's an excellent question.

04:23:34.000 --> 04:23:36.000
Before, um.

04:23:36.000 --> 04:23:40.000
8.2 was started working on.

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there was some discussion in the community that CCSS was basically still in line with the general concepts of how Bitcoin works.

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For example, in the current standard version, you'll see it talk about private keys

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but it doesn't actually reflect

04:23:58.000 --> 04:24:00.000
protocols such as MPC,

04:24:00.000 --> 04:24:07.000
So what we've done in version 8.2 as the start of the journey to answer your question is that

04:24:07.000 --> 04:24:12.000
We've replaced the mention of

04:24:12.000 --> 04:24:17.000
Private keys in such a way that we talk about key material

04:24:17.000 --> 04:24:19.000
we talk about

04:24:19.000 --> 04:24:22.000
single sign-on mechanisms

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a multi-signature mechanisms instead of

04:24:26.000 --> 04:24:31.000
using terminology that provides more

04:24:31.000 --> 04:24:35.000
acceptance of new technologies that we don't even know exist yet.

04:24:35.000 --> 04:24:38.000
for example the use of mechanisms

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But again, this is a community driven standard and

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we accept and welcome any

04:24:46.000 --> 04:24:53.000
contributions from anyone around the world in terms of how CCSS can be improved

04:24:53.000 --> 04:24:55.000
And also, um.

04:24:55.000 --> 04:24:58.000
reflect new technologies in this space.

04:24:58.000 --> 04:25:01.000
And we do rely a lot on community input.

04:25:01.000 --> 04:25:03.000
to help us achieve this.

04:25:03.000 --> 04:25:07.000
we don't. It's a completely volunteer organization

04:25:07.000 --> 04:25:13.000
We rely on experts from around the world to freely submit their time and just help us achieve

04:25:13.000 --> 04:25:18.000
these new goals and make CCSS as relevant as it is today.

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CCSS is now with the steering committee in place very active

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in the community now it hadn't been prior we admit

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And that's where some of the comments that have come

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from and historically that is you know true

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But right now, there's an absolute core focus

04:25:39.000 --> 04:25:45.000
on keeping CCSS constantly updated to reflect these new technologies.

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So if there's anything that someone believes that CCSS is not covering or limiting

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based on the wording of the requirement we

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we've thoroughly encourage anyone to submit

04:25:59.000 --> 04:26:03.000
or just even just talk to the CCSS committee and say well

04:26:03.000 --> 04:26:11.000
you know this technology is not really taking account of in CCSS or the current requirements are limiting

04:26:11.000 --> 04:26:14.000
In some way, we will definitely take that on board

04:26:14.000 --> 04:26:18.000
And I'll turn around now for the standard can be

04:26:18.000 --> 04:26:21.000
In months, not, you know.

04:26:21.000 --> 04:26:25.000
in years for some standards.

04:26:25.000 --> 04:26:32.000
I hope that has helped you with your question.

04:26:32.000 --> 04:26:35.000
Thank you very much, Ando. So I believe…

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that we don't have to reinventing the inventing way

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So we don't have to create a new CCSS.

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So we should leverage that so existing CCS works.

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the future, you know, key management was a cyber security

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you know, operations.

04:26:53.000 --> 04:26:56.000
standard or something like that but uh so

04:26:56.000 --> 04:26:59.000
we need to involve

04:26:59.000 --> 04:27:01.000
that that you know

04:27:01.000 --> 04:27:04.000
knowledge or no ideas.

04:27:04.000 --> 04:27:08.000
for the newly invented technologies.

04:27:08.000 --> 04:27:14.000
for blockchain, I think that begin could be that so good you know

04:27:14.000 --> 04:27:16.000
good organization to

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Bring that to new

04:27:18.000 --> 04:27:23.000
new technology discussion around the new technology too

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to future revision of the CCS.

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standard or something like that. This is my

04:27:28.000 --> 04:27:31.000
opinion.

04:27:31.000 --> 04:27:34.000
Absolutely, completely agree and

04:27:34.000 --> 04:27:40.000
And this is why, you know, I was invited to this discussion in the plenary

04:27:40.000 --> 04:27:46.000
in Sydney and I identified that this is absolutely critical that

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CCSS steering committee

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and all the volunteers behind it.

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branch out more and see what other

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you know associations and nonprofits are doing in this space

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and work out how we can collaborate together.

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So we are definitely the steering committee is definitely 100%

04:28:05.000 --> 04:28:07.000
open to any type of

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collaboration just to improve

04:28:09.000 --> 04:28:14.000
the security and basically in this space

04:28:14.000 --> 04:28:16.000
Thank you.

04:28:16.000 --> 04:28:30.000
So the professor mentioned that we shouldn't reiterate. So we shouldn't recreate the standards we already have. They already existed. They exist.

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But like, so I'm not on a cybersecurity expert, so I was not really familiar with like what what was like ccss

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And also when I had a conversation with like people in the crypto isaac

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people, they mentioned another standard setting bodies like

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Security Council or something

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So…

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If there isn't any…

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like other organizations that is trying to create standards

04:28:59.000 --> 04:29:02.000
as we do as begin

04:29:02.000 --> 04:29:08.000
Like, and if there is anything, any organization that we should pay attention to and

04:29:08.000 --> 04:29:12.000
We are not going to be like you know

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competing with them like we are trying to

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Like feel the missing pieces we have in the industry so

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If there is any like other augmentation

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CC… Like…

04:29:26.000 --> 04:29:29.000
you know…

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like it's great to

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also involves them in this conversation as well. But is there any other orientations that is trying to create standards like ccss or like

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you know um as far as I know.

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There's only like CCSS and also like the Security Council, which is um

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Also.

04:29:56.000 --> 04:30:03.000

But if there is any other organization that we should pay attention to probably

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And it's great if you could

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Give us inputs on it.

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It's market yes so there's the

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blockchain Security Standards Council.

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Which is, I believe, a membership

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only organisation and

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there's not much information on their website, but I do believe that they are looking at introducing

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information security standards for

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the Web3 space.

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But that's all I'm aware of. I believe that they're going to reduce

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releasing standards by the end of this year.

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The only key difference I can see between CCSS

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And blockchain security standards council is that

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CCSS is not a membership only

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entity we

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We're volunteer based we accept

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anyone and anyone's submissions

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you're not restricted in terms of communication or the fact that

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you don't pay a membership fee, you can't contribute to the standard.

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That's only my opinion and my understanding based on the communication that the

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blockchain Security Councils has released but ultimately

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them coming into this

04:31:26.000 --> 04:31:30.000
environment where information security is

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is poorly represented is a positive in any way that you see it it's it's

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to be congratulated that

04:31:39.000 --> 04:31:43.000
the standard, this council is going to release their standards

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that is to be congratulated in terms of

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how you contribute to the

04:31:49.000 --> 04:31:51.000
Their standards versus ccss

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All I can say is that CCSS has obviously been around for a while now.

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And it's completely devoid of any membership requirements or fees to join.

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any type of restrictions whatsoever. It's completely community driven.

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I do not know.

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of any other security standards at this time apart from obviously TC307

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which is the ISO blockchain and DLT standards.

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I'm not aware.

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if there is a cryptocurrency

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standard being introduced into ISO.

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It could be covered by

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the other ISO TC307 standards that are currently in draft and that have been published as well.

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So I'm not 100% sure

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If there's anyone apart from CCSS,

04:32:46.000 --> 04:32:48.000
Blockchain Securities council

04:32:48.000 --> 04:32:52.000
by the end of the year and obviously the work that ISO is doing.

04:32:52.000 --> 04:32:54.000
Thank you.

04:32:54.000 --> 04:33:00.000
Great.

04:33:00.000 --> 04:33:03.000
It's my comment, but so

04:33:03.000 --> 04:33:07.000
Hey, so we have that participant who don't.

04:33:07.000 --> 04:33:09.000
protocol today.

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And so I would like to ask one question to you.

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So we are now discussing about cybersecurity issues as you know

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every day so we found

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so many vulnerability for protocols, products, software codes, something like that

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And so your company is creating the many products every day

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And so we are now talking about that so you know

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So this diagram shows that

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So general.

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You know, sad.

04:33:41.000 --> 04:33:44.000
mechanism for sharing cybersecurity

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information over attacks or information of vulnerability or

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you know incident handling or something like that so

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What kind of information

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useful to be shared to

04:33:58.000 --> 04:34:02.000
the company which created so you know blockchain product every day

04:34:02.000 --> 04:34:05.000
Because that, you know, information sharing

04:34:05.000 --> 04:34:09.000
scheme either or they exist for other industry

04:34:09.000 --> 04:34:12.000
But the whole crypto industry that you know

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the security or cybersecurity home is more complicated than other industry

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that we need to

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have them matching information to be shared

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to everyone but the the headache is also

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You know, to be honest so

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the major part of the crypto industry

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company don't care about that.

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So Barnabit information. Unfortunately, due to the lack of exports

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So we are now discussing that we could create that this similar kind of framework for

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You know, some auto startups. So, you know, what kind of information to be shared or what kind of support

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Supports are needed for the small startups

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So like the mini equipped industry companies.

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So I don't know who else is in here, but everybody. So just to answer your question, I think a lot of this information does exist.

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It's just not being shared in the right channels.

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So like currently I had to manually go to every single project in the near ecosystem

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And request audit reports.

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on an audit to get the report

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Do we get the permission to have the report in a public database. And now I have this public database of all these reports.

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Now it would be great for me to like submit the this database to

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this entity that is trying to create the standards just to kind of have you guys get a better sense of like what are the vulnerabilities that are being addressed before the product goes live.

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So I'd say like basic

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Basic, basic, basic would just be like get existing auto reports in a centralized place, which I think does exist in some other ecosystems, but it's not totally standardized. And so I think getting those reports would be great.

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could be super helpful for best practices is straightforward.

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We currently do have like certain blog posts that highlight this, but obviously it's like an information asymmetry issue. Like you don't always…

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The builders don't always know that these things exist. Someone needs to send them to them.

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I am normally that person, and that is not very scalable. And so having like just if you are a builder

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checking out like it needs to be, I guess…

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Discovery, I guess, is the thing that partially needs to be standardized as well.

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like making sure that if you guys are like running this consortium, checking these different documentation like near docs.

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to make sure that there are security best practices baked into that.

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And like they know where to also go to, you know.

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get advice if it's not from the near core team directly.

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So we do have office hours. We do have telegram support channels. We do have Discord channels to provide the support.

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But it's not necessarily standardized. We just do this for ourselves.

04:37:03.000 --> 04:37:07.000
I guess third thing is around, yeah, bug bounties. We do have a lot of bug bounties.

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That's great, but…

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Obviously, there's games that people play a lot um

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And that is…

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hard to navigate, especially if you're alone.

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It would be great, I think, to get the security professionals that are in the near ecosystem to engage more with like external professionals and other ecosystems. It's going to be more about how to navigate

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Crypto Bug Bounty bug bounty situations.

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Because I can say that there have been serious attempts on packing the Rainbow Bridge, for example. I think has…

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over a billion dollars stored in it, at least at one point.

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And people have to try to break that a bunch, like probably every month or so.

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So understanding how other teams sort of address those issues would be great.

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Another one, and this is something that I've personally tried to work on is insurance.

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Insurance is really hard. Yeah. Like I've talked to Lloyds of London. I've talked to these big brokers

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to see if they're willing to underwrite some of these really, really expensive contracts.

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And they're open to it for a very serious price. And so I'm

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like subsidy for key contracts to be insured or align with different

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auditing firms that maybe they would be willing to insure their audits. I don't think they would.

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But that's something that I have explored. Didn't get a ton of progress there.

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And then something that I've like we spoke very briefly about is formal verification, formal, you know, formal systems.

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That would obviously be great, but that's only for very specific types of contracts.

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But I'm just trying to think of like all of the different

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kind of types of security somebody can kind of pursue in crypto. But when it comes to information sharing, I would just say it's like

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Docs is probably the best place to standardize at first. Where do you see the information at first is the documentation, like docs.near.org.

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But…

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I don't know if I answered your question totally. That is kind of okay

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Thank you. So we are discussing how the beginning is

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We pray a law to…

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to English level security of this ecosystem

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So, you know, United States, Crypt Isaac was established early this year

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And yeah, this is Isaac is just for that information sharing

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I don't know that. So Quip Isaac is qualified or not but so they are they are starting that their activities for the information sharing

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But it's a part of the entire picture.

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And it says, so we

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I've already done that this kind of discussion

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In Tokyo, in Macedonia. So a few weeks ago

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That's the one thing we discuss is about education.

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You know, the number of the psychiat is very small

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in this area but so we have tons of information to be shared that's fine but there's there's no enough number of people who

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truly understand or translate to that that

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that each company. So we need to increase the number of that kind of security expert. This is the reason why we need to create some

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educational document, educational materials

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to share that all small startups

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that cannot have enough number of security experts.

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So in order to get those

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In order to get the leadership to hire the experts to create the market for experts, for more experts to come in, you need to scare them.

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You need to scare them saying, hey, you might be liable for the potential activities that might happen in your network or in your ecosystem.

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That's going to put a lot of pressure, at least like on your foundation or a similar foundation.

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to ensure that they have quality security experts to

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essentially participate in the system.

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And so I would say, and they do exist, like there is a direct, there are directors of security in here.

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At the foundation level.

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they're probably just not participating.

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And so why aren't they participating?

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Probably because they either don't know that it exists

04:41:08.000 --> 04:41:13.000
it might not be relevant for them because it might be, you know, we're WebAssembly based blockchain

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If you're just talking about solidity all day

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And slow to do smart contracts, they don't care.

04:41:18.000 --> 04:41:23.000
So I would say like you kind of need to really get focused on like either solidity or rust

04:41:23.000 --> 04:41:28.000
Maybe move. I know move is like kind of a growing ecosystem. It's still very small i

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I would say solidity and Rust are pretty much the two.

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And they share different problems.

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So, and then from, which I'm very excited to talk to you about is formal methods for Rust contracts.

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You know, Rust can be formally verified.

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And this to me is like

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the real future for smart contract development. And in perfect world, if we are able to formally verify Rust efficiently.

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then we can train agents to develop contracts or at least

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Before we go all the way to development, at least do like certain security benchmarks automatically.

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So if we do build these automatic like benchmark

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essentially frameworks and tools uh

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that is going to be amazing public good.

04:42:08.000 --> 04:42:11.000
to provide these smart contract developers to ensure security.

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I'm pretty sure every foundation, every large

04:42:14.000 --> 04:42:16.000
Treasury would be willing to pay.

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to subsidize access to these tools for their ecosystems.

04:42:22.000 --> 04:42:24.000
If it is cheaper than hiring security experts.

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So it depends on, I think it's just providing the developers and founders the tools that they need because it's really expensive to create an audit.

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Like to get an audit. Like I've gotten many smart contract audits and it's ridiculous. It's like not feasible.

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for an independent team to pay security experts. They're already not making money a lot of times.

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So like to pay $50,000 for an audit

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is a huge ask and actually limits the barrier of entry to the space.

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more than I think anything else in the industry.

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So I'd say if we could solve that problem, that would make Web3 much more successful.

04:42:59.000 --> 04:43:01.000
Thank you very much. That's useful.

04:43:01.000 --> 04:43:03.000
Yeah, information.

04:43:03.000 --> 04:43:07.000
So you mentioned the different languages you know like each

04:43:07.000 --> 04:43:11.000
protocol niche like you know blockchain you know use right

04:43:11.000 --> 04:43:13.000
And I'm not really sure like

04:43:13.000 --> 04:43:19.000
if crypto is like he's thinking about this because like, you know, they're just talking about, you know, blockchain

04:43:19.000 --> 04:43:21.000
as a whole

04:43:21.000 --> 04:43:32.000
I think we should start from like talking about what are the main languages we should deal with. I think, yeah, this is a great discussion point, I think. Thank you. Thank you very much for it.

04:43:32.000 --> 04:43:37.000
Yeah.

04:43:37.000 --> 04:43:42.000
based on the comments he made, does any…

04:43:42.000 --> 04:43:51.000
Remote participants have any opinion about this?

04:43:51.000 --> 04:44:00.000
What time is it now?

04:44:00.000 --> 04:44:02.000
Mark, do you have any…

04:44:02.000 --> 04:44:04.000
comments?

04:44:04.000 --> 04:44:06.000
Oh, sorry. I was…

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I was just thinking, I was actually doing a presentation

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to a country's regulators

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this morning on the importance of

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standards when they're doing their policy

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policymaking and the regulations around

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Introducing RESPs into

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into their country. And one of the key things I was thinking of as the gentleman before

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This is speaking about

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You know, where do we put all this type of information and and then I reflected back on the factors that another

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point was made is that there's no big stick in this sector yet.

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for example as mentioned before with PCI DSS the

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And the credit card security standard, there's a very big stick there and that's called the card schemes. It's the visa and the MasterCards that actually created the standard and

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If an entity is not PCI DSS compliant, then there's a very good chance they could be fined or have their credit card merchant services taken away from them, meaning they can't accept credit cards from their customers.

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And that's an extremely powerful big stick that they have. And then I was reflecting back on the meeting I had this morning

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with the regulators and it seems that there's almost

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a possibility that we could introduce a big stick by the way of regulations

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I believe a lot of jurisdictions out there are attempting to implement

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requirements around information security controls for VEFs when they come in to be licensed.

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Some of them are quite vague some of them

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appear to be

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you know missing key controls so i was just wondering if we could

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work with the regulators

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and the policymakers around the world to actually educate them on the importance of having

04:46:05.000 --> 04:46:07.000
information security control

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regulations for the vast

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And as one of those controls would be

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some type of vulnerability

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management.

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And from there, something in the industry could be, this could be a tipping point

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for the factors now we need to look at vulnerability management in order to get a license to operate in a jurisdiction.

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And, you know, somehow this big stick forces vulnerability management

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to come up to a head where there's more discussion

04:46:41.000 --> 04:46:46.000
in the marketplace about how do we tackle vulnerability management

04:46:46.000 --> 04:46:49.000
Thank you.

04:46:49.000 --> 04:46:51.000
Thank you.

04:46:51.000 --> 04:46:57.000
Um.

04:46:57.000 --> 04:46:59.000
We still have time left so

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I don't know if we could also…

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give a brief introduction will be like

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wallet security thing you

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discussed recently.

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I don't have any materials to

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display today but

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So regarding the cybersecurity, this is a just

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you know that incident handling or barnabity handling or something like that

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But the other aspect is how we satisfy the level of security of product.

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Two.

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Especially for those who are it, but so we have different types of you know software called the product how do we have software

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code a product, something like that.

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So I'm an expert on the cryptography.

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And so…

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In other industry than script uh blockchain, we already have the standards

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for the you know certified you know

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proving or certifying that quality of protocol specification

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was, you know, operations, you know, or implementation of the security

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for the implementation. So we already have the ISO standards

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I don't know, the 27,000 series

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So as Mark mentioned before.

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It's a so-called common criteria

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So this is a gorgeous international standard.

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Of course, the process

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to have certification of the product.

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aligning the ISO 27000 series of British

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very heavy for small startups.

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It's not too easy with most startups to align that.

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You know, ISM standard.

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or common criteria standard.

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So especially that for crypto industry that some companies say that

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you know that that warrant, how do I word align the ISO standard or, you know, NISTA standard, like a 140-3?

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But there's, you know.

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It takes a few years, three or four or five years.

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to have you know cm with fibs 100

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what it does through the certificate from East.

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So if you Google that list

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website was a you know the list of certified products for the 123

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doesn't know how to list it in that list.

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Of course, some companies say that it's a you know 140 DAC3 compliant but

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They don't care about that extra certification process by third party

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So, you know.

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certificate authorities

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But anyway, so we already have that existing standard for the you know

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keeping the level of security of the implementation of software hardware, something like that

04:49:58.000 --> 04:50:08.000
And I'm not sure that it's feasible for smaller startup, but we need to reconsider that how we compribe is that existing standard.

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If that process is very

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heavy for you know crypto industry or small startup.

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We need to create that some modified version of it.

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But, you know, that heavy standard.

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to make that alignment to the standard easy for small startups.

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And I think as a facial step you know you know that

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you know common criteria.

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request to create uh you know

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some specification, there's common specifications for water security or something like that

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So protection profile or you know security target or something like that this is a

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you know, standard document for

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you know security specification or something like that

04:50:53.000 --> 04:50:59.000
But there is no protection profile or security target document

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or cryptocurrency world.

04:51:02.000 --> 04:51:07.000
You know, that is a good puzzle step to create that kind of things

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Anyway, that is my observation, but let's start with learning that

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current international standard for

04:51:16.000 --> 04:51:21.000
the security implementation security of the software and hardware thing

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creating some uh you know

04:51:23.000 --> 04:51:26.000
modified version of this standard for

04:51:26.000 --> 04:51:28.000
crypto industry.

04:51:28.000 --> 04:51:31.000
And it's just mark here again.

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I understand exactly what you're saying. And yes, ISO 27001 certification is quite heavy for

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startups, especially when the budget is limited and the resources are limited

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CCSS actually has three levels of certification.

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Level one is for

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particularly designed for startups where

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budget and resource is limited and that

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only what we consider what the community considers to be controls that if implemented will

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you know greatly reduce the risk of a breach

04:52:09.000 --> 04:52:12.000
of the most common attack vectors.

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And that's at level one. And then level two is you add additional controls to strengthen

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your environment more if you've got more functionality or services provided. And then there's CCSS level three which is

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all of the CCSS requirements are applied and that's generally

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expected of larger enterprises that

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have the money in the budget to really enforce

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very, very strict processes

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and information security controls.

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In terms of adding to the requirements or

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adding more levels and so forth then the CCSS steering committee is again open

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to any suggestions for additional levels or additional requirements

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Even at CCSS level one, which is considered the startup

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level. So we're definitely open to collaboration and

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using our framework and the standard

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to accommodate this type of technologies as well.

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Thank you.

04:53:19.000 --> 04:53:28.000
Yeah, two questions. First question is on the, correct me if I'm wrong, CCS standard three.

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or the level three.

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How many institutional exchanges, custodians uh

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you know, people who are custodying a lot of assets on behalf of users, how many of them are complying with this?

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And are you meeting them where they are or are you expecting them to meet you where you are?

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Right now, there's three. There's liminal fire blocks and folios

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All three are wallet infrastructure providers. So there's a designation in CCCS called the qualified service provider

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So they are termed as service providers. They're not full system. So that's another designation in CCSS.

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a full system would be, for example, a custody provider that uses fire blocks for wallet management.

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that entity could become a full system and use fire blocks

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CCSS certification.

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The audit program for CCSS wasn't released until 2021.

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there was a stagnation in terms of the overall

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focus on

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propelling the standard and making it more publicly aware

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And that's obviously, as I mentioned before, that's changed

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There are currently five

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four to five audits

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active that we know because being on the steering committee

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an auditor must submit an intent to audit form

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So we're aware of that an audit is starting so

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There's four confirmed and there's another one that's currently just in the signing process and all four i believe

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of those new audits of what we call

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full system so they are custody providers

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that are using

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possibly service providers or other wallet infrastructure services.

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So it started. There's a great deal of momentum starting up now, but right now, to answer your question, there's three service providers who provide wallet infrastructure.

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Fire blocks, liminal and tholios.

04:55:30.000 --> 04:55:37.000
Very cool. Thank you. And then for the level one, this is always an interesting, uh.

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dynamic that I navigate is

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Would you prefer contracts that are upgradable and not knowing who the contract builders are?

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or contracts are not non-agreeable

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And not knowing who the builders are.

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like what is what is more ideal?

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like an upgradable contract or a non-upgradable contract, assuming you don't know the people who deployed it and can't trust them.

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For the CCSS certification, one of the core principles of being certified is accountability

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by way of not full transparency.

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So as I mentioned.

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you know it's fire blocks, liminal and folios

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there wouldn't be a certified entity that

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would get through the process without actually knowing

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who they are, their company, a contact name that's publicly available on the websites.

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Got it.

04:56:33.000 --> 04:56:40.000
there's no anonymity in CCSS in terms of certification and going through the audit process.

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Okay. Yeah. I guess like when people are engaging with contracts online

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They just don't always know who the builders are. And so when they don't know who the builders are

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I'm generally more of a fan of those contracts not being upgradable despite potential vulnerabilities that might exist in the contract.

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Because if it is upgradable, then they can just run off with the money.

04:56:59.000 --> 04:57:00.000
So, yeah. Okay, cool. Thanks.

04:57:00.000 --> 04:57:04.000
Oh, completely agree. And, you know.

04:57:04.000 --> 04:57:09.000
We have released only three smart contract requirements

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This time in this version release 8.2 which is related to

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smart contract audits

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However, we are looking at and also seeking help from anyone in the community

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on what other smart contract requirements we should be putting into CCSS. So it's basically a call to for help in terms of

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you know what else do we need to add to the standard

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for smart contracts to make this environment more safer for people.

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Thank you.

04:57:42.000 --> 04:57:44.000
How do you audit the auditors?

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Like, you know, verify that they're actually

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doing good work because all too often people pay a lot of money for an audit and then

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they're still vulnerabilities and obviously it's not a surefire thing just to get an audit and everything's safe.

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But who is actually keeping track of the success of these auditors despite the market?

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There is another excellent question.

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So when an auditor does their first audit for CCSS, they are

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They are basically mentored and their report on compliance, their ROC,

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There's peer review by a CCSS steering committee member.

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If they don't do that, then they can't have their systems that they've audited certified.

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So they go through a thorough and rigorous auditing of their initial rock, the peer review, making sure that it's up to the CCSS standards

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there is and once they've done the initial audit and it's peer reviewed by a committee member

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then they can go and basically get their

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audit report

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peer reviewed by another external auditor

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So the rule here in CCSS is that when you produce a report

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there must be another CCSA

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person who peer reviews that report

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And that CCSSA

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is chosen from a list

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of randomly generated selected CCSAs

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and sent to the CCSA, sorry for these acronyms but basically

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Every single rock that is produced

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must be peer reviewed by another auditor that is completely independent

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of the entity being audited

04:59:36.000 --> 04:59:42.000
in the auditor who audited the entity.

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Got it. Thank you.

04:59:44.000 --> 04:59:52.000
Thank you for the question. It's very good.

04:59:52.000 --> 04:59:57.000
Okay.

04:59:57.000 --> 05:00:01.000
Right. So…

05:00:01.000 --> 05:00:04.000
I think we can finish up this session.

05:00:04.000 --> 05:00:15.000
Oh, yeah? Okay. Go ahead. Okay, great. So curious about like, there's a bunch of these new startups have been coming out creating essentially smart contract auditing

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AI auditing tools, one being like octane.security.

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I know them just because I actually really respect their security team.

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And it seems like this is going to be the fastest approach to ensuring

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developers actually have some sort of checks

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Instead of having to pay insane amounts of money for third party audits. I'm curious if that like these sort of automated checkers would be a better

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sort of approach to provide a bit more of a bottom up

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adoption of the

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I'm sorry for the acronyms. CCCSA, if I'm getting it right. And that's a bit more on the bottom up side of things. On the top down side of things, just making sure that exchanges actually support these things and ensure that the

05:01:03.000 --> 05:01:08.000
the teams that their list, the projects that they're listing also comply.

05:01:08.000 --> 05:01:10.000
Maybe not with like the full…

05:01:10.000 --> 05:01:16.000
number, you know, three standard, but like, you know, not standard, but like three level of uh

05:01:16.000 --> 05:01:26.000
rigor, but maybe something a little less. So like how is sort of you navigating this like top down versus bottom approach of adoption of these standards?

05:01:26.000 --> 05:01:29.000
Another good question the uh

05:01:29.000 --> 05:01:34.000
The three requirements that are currently going to be released in version 8.2

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they're very generic. They're pretty much like an ISO standard in that

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the wording is generic enough

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for the auditor to decide what is the most appropriate for the entity that they're auditing.

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So when we mention

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a smart contract audit report

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we don't specify

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exactly who does that report

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what qualifications are needed

05:02:02.000 --> 05:02:06.000
And what frameworks are used.

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We leave that up to the auditor

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to decide what is the most appropriate

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or the entity. So if the audit report is generated by a tool.

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then the auditor

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must gain sufficient assurance from that report and that tool

05:02:21.000 --> 05:02:24.000
that it's performing its function

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correctly and the auditor has enough assurance that

05:02:28.000 --> 05:02:31.000
they accept that audit report.

05:02:31.000 --> 05:02:34.000
Got it. You know, something that I've been

05:02:34.000 --> 05:02:39.000
I've been in the space for about eight years, and so I've seen a lot of hacks and bad things happen.

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And I would love to see some sort of insurance fund

05:02:44.000 --> 05:02:54.000
for teams in order to qualify for it, you have to meet certain standards. And because the main thing about these standards is like you need carrots or sticks.

05:02:54.000 --> 05:03:05.000
like you need to either incentivize people to use them or like, you know, like hurt people in some way or like maybe they'll just hurt themselves by bad security by not using them. And so…

05:03:05.000 --> 05:03:10.000
Curious, like, what are potential incentive mechanisms to get people to adopt these?

05:03:10.000 --> 05:03:14.000
And what have you explored there?

05:03:14.000 --> 05:03:17.000
We've explored the insurance industry so

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We've contacted insurance

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underwriters and so forth in the Web3 space. And we've basically talked to them about how CCSS can provide

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the type of assurance that they need

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to look at the risk profile of an entity and work out whether their information security program is robust enough

05:03:38.000 --> 05:03:44.000
You know, things like SOC 2 and ISA 27001, they're very good baseline standards

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But CCSS adds the further level of requirements that are

05:03:50.000 --> 05:03:57.000
you know unique properties of blockchain and DLT systems. So we've looked at the insurance industry. They seem to be quite interested with this.

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We have now started to approach the regulators around the world

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We have also introduced certain CCSS requirements into the Commonwealth's

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virtual asset model law framework

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So the Commonwealth is a set of 157 countries

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And in the Commonwealth and what they do is this body

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will create

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policy frameworks

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to help these countries implement regulations.

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And laws and regulations. And one of these frameworks is the one I've just mentioned is the virtual asset model framework. And in there, CCSS is mentioned as possible

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Information security controls that could be

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required from

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an entity seeking compliance in a jurisdiction

05:04:56.000 --> 05:05:01.000
So we've looked at that. We've obviously joined

05:05:01.000 --> 05:05:09.000
as many associations and communities as we can to promote CCSS. So the more people who talk about CCSS,

05:05:09.000 --> 05:05:16.000
the more the larger exchanges and so forth start thinking maybe you know we need to have CCS.

05:05:16.000 --> 05:05:27.000
We are aware that some requests for information going to vendors in this space have had questions related to CCSS as well.

05:05:27.000 --> 05:05:35.000
So we may not have a big stick right now, but what we're trying to do is to promote the standard as much as possible.

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through channels where we believe questions will be asked whether they're CCSS certified or not, but we really

05:05:45.000 --> 05:05:47.000
hoping on

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providing guidance to the regulators around the world about what should be

05:05:52.000 --> 05:05:55.000
placed in terms of information security

05:05:55.000 --> 05:05:59.000
control requirements and their regulation.

05:05:59.000 --> 05:06:08.000
Got it. Thank you.

05:06:08.000 --> 05:06:11.000
Okay, that's it.

05:06:11.000 --> 05:06:17.000
So thank you very much for the participation and this is the end of the program.

05:06:17.000 --> 05:06:21.000
So thank you very much.

05:06:21.000 --> 05:06:27.000
Okay, so…

05:06:27.000 --> 05:06:29.000
Yes. Yeah. We will have

05:06:29.000 --> 05:06:32.000
So general meeting.

05:06:32.000 --> 05:06:35.000
Next week in Washington, D.C.

05:06:35.000 --> 05:06:39.000
And we have a remote option as well so

05:06:39.000 --> 05:06:42.000
Please sign up.

05:06:42.000 --> 05:06:45.000
Are you coming? Can you…

05:06:45.000 --> 05:06:48.000
participate remotely? Oh, yeah.

05:06:48.000 --> 05:06:54.000
It's like we got to have the AI session as well. So yeah.

05:06:54.000 --> 05:06:56.000
Okay, okay, perfect, perfect.

05:06:56.000 --> 05:06:58.000
It's like Chloe is coming as well.

05:06:58.000 --> 05:07:04.000
Yeah, no, she's going to be remote. Yeah.

05:07:04.000 --> 05:07:07.000
Oh, yes. I'm asking him now.

05:07:07.000 --> 05:07:13.000
Yes, yes. Yeah.

05:07:13.000 --> 05:07:22.000
Yeah, Jessica will not be available, but anyway yeah

05:07:22.000 --> 05:07:25.000
Okay, thank you very much, everybody.

05:07:25.000 --> 05:07:34.000
Yep.

05:07:34.000 --> 05:07:39.000
Yes. Great.

05:07:39.000 --> 05:07:43.000
dinner.

05:07:43.000 --> 05:07:46.000
What did you say? Dinner.

05:07:46.000 --> 05:07:48.000
I don't know, like…

05:07:48.000 --> 05:07:50.000
We're gonna…

05:07:50.000 --> 05:07:53.000
he…

05:07:53.000 --> 05:07:56.000
He made a reservation, but, like, I don't know if

05:07:56.000 --> 05:08:00.000
Yeah, I can ask him, Neda.